模式识别课程

多模态大模型前沿



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课程提纲

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一、 LLM/VLM/MLLM

二、多模态思维链推理

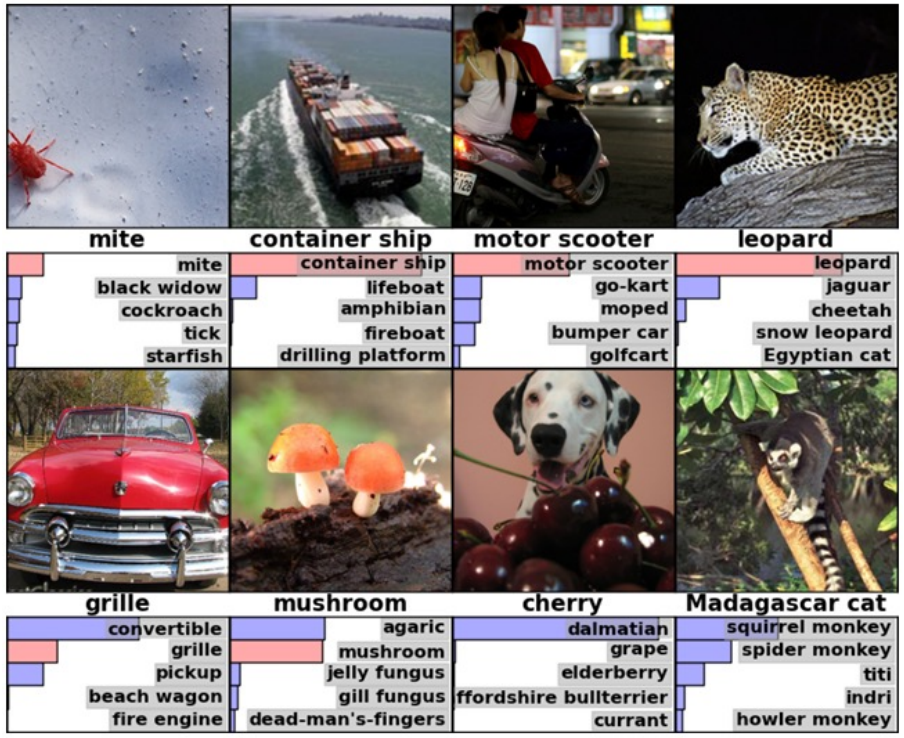
三、多模态视频理解

四、检索增强生成

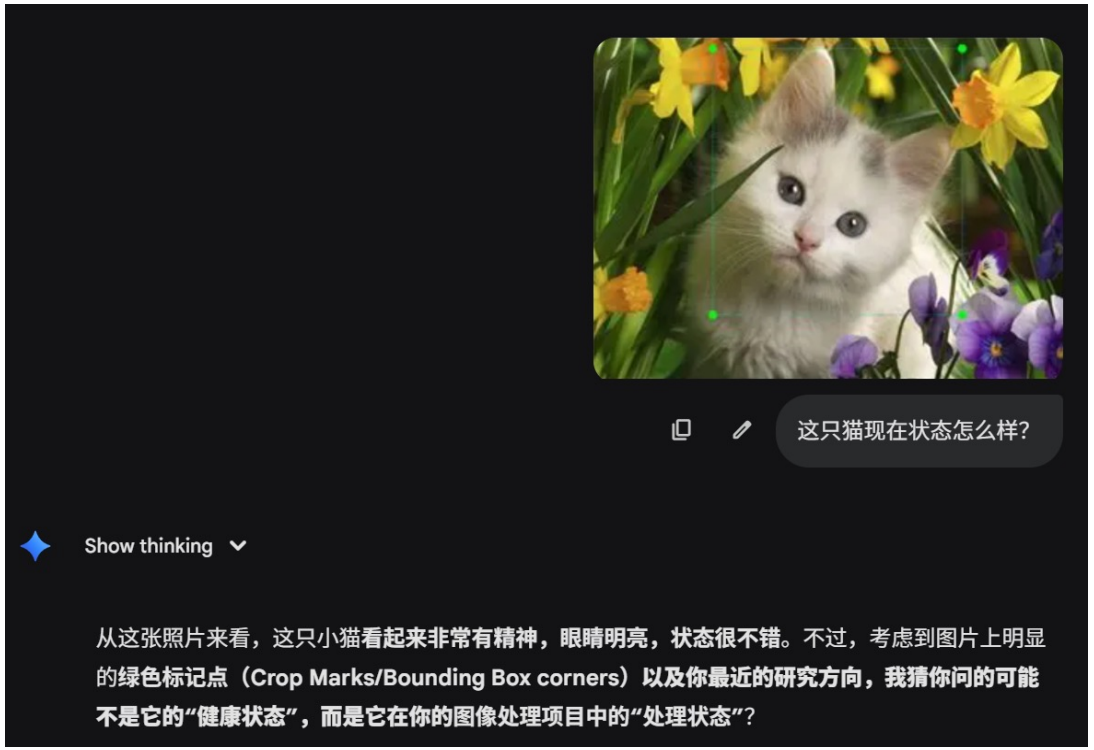
五、总结

模式识别的“前世今生”

传统模式识别旨在映射标签，而 MLLM 旨在语义理解



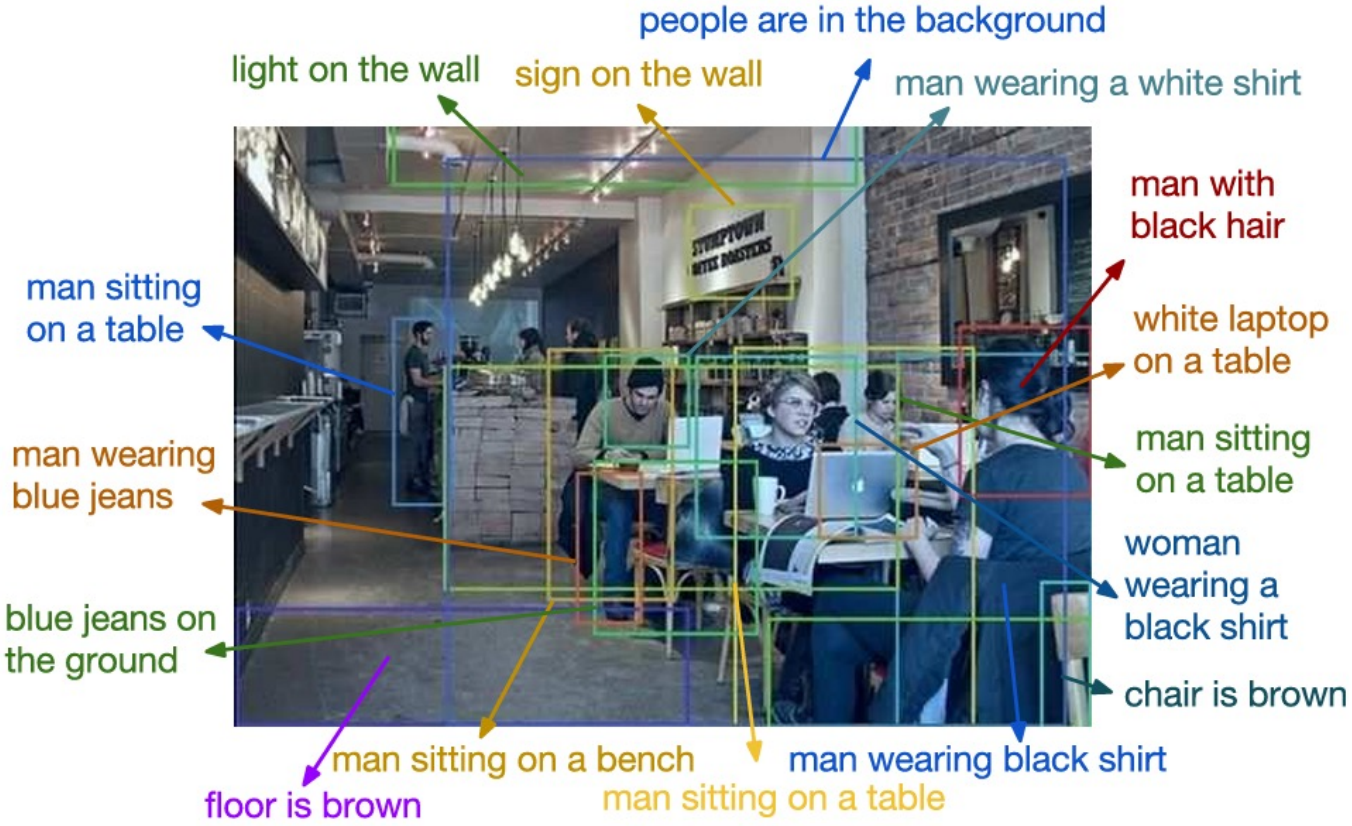
传统模式识别：AlexNet 在 ImageNet 测试集上的分类结果展示

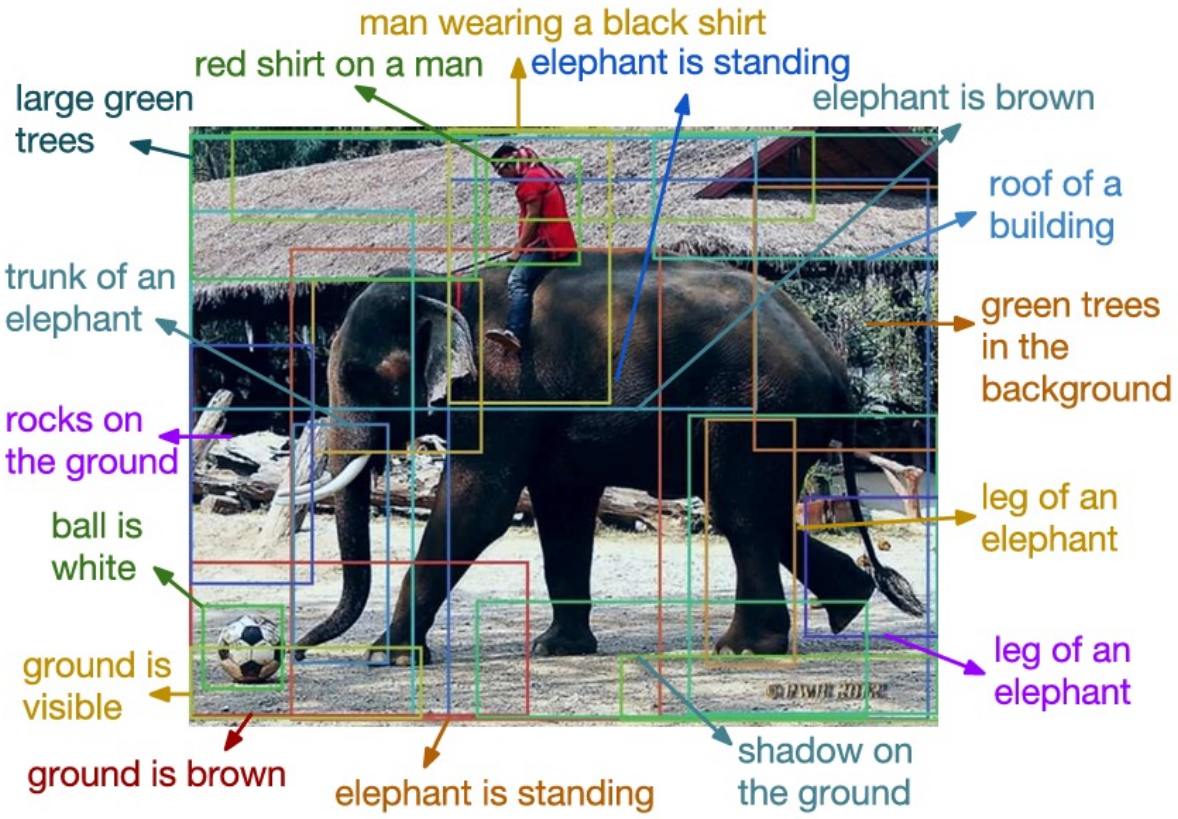


现代模式识别：基于多模态大模型的交互式视觉感知与逻辑推理

为什么要多模态?

文字能描述物体，但很难描述物体之间的物理关系

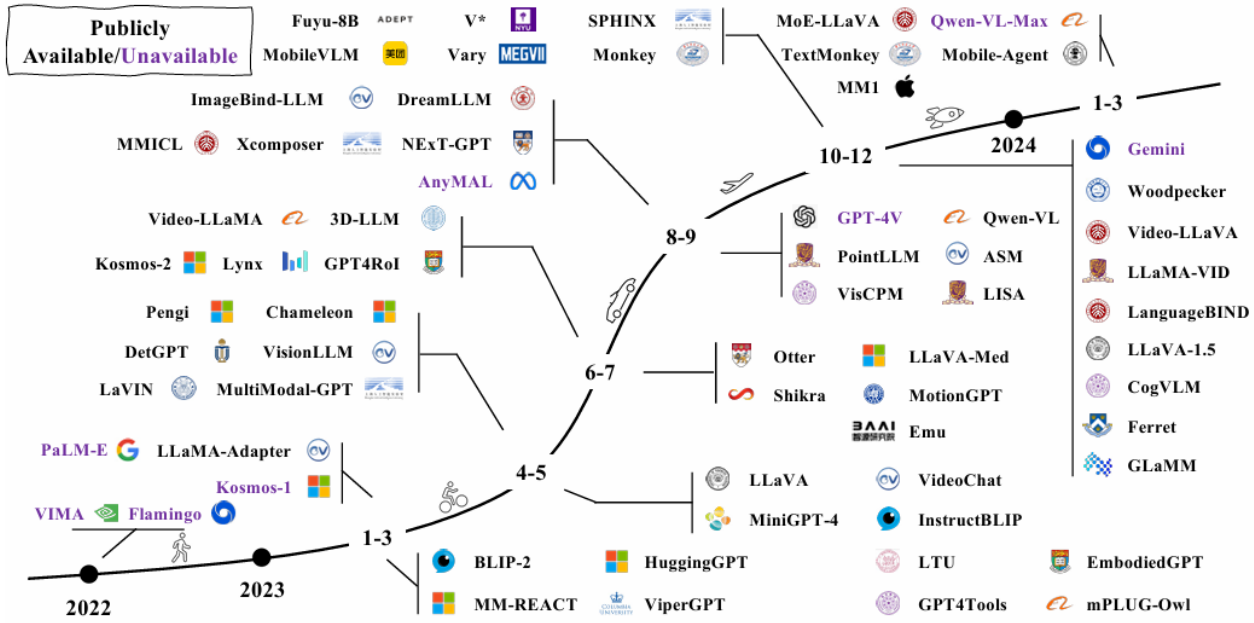




文本描述 ≪ 视觉信息

MLLM 的演进：从单模态到全模态

这是一个不断打破模态壁垒的过程



MLLM能力的演进

静态视觉 (Image): LLaVA, CogVLM

时序视觉 (Video): Video-LLaMA, VideoChat

空间感知 (3D): 3D-LLM, PointLLM

具身智能 (Action): Mobile-Agent, EmbodiedGPT

2022 - 2023初: 以 **Flamingo** 和 **BLIP-2** 为代表，这一阶段主要解决“视觉编码器与 LLM 如何对齐”的问题，打破了视觉和语言的隔阂

2023中: 随着 **LLaVA** 和 **MiniGPT-4** 的出现，开源社区彻底被激活，这一阶段的特点是“指令微调”引入多模态，让模型听得懂人类对图片的指令

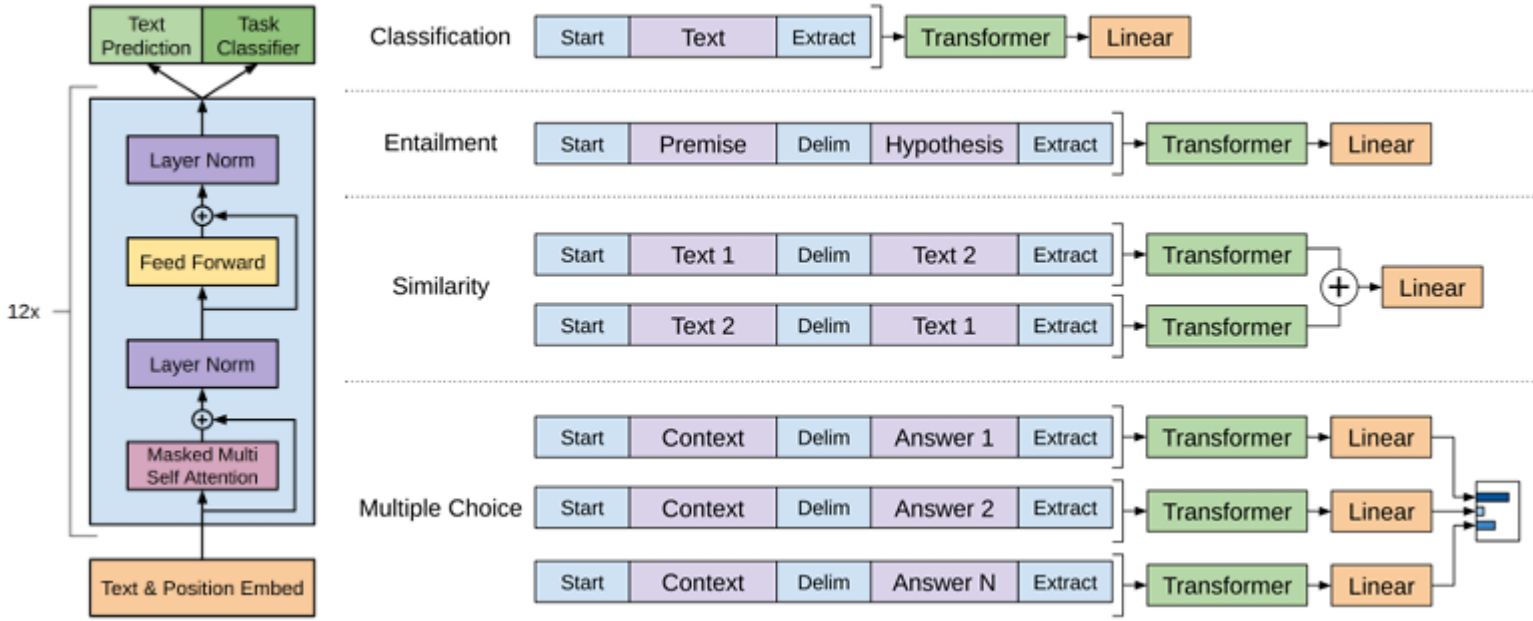
2023末 - 2024: GPT-4V 和 **Gemini** 的出现标志着模型不仅能看图，还能处理视频、音频，甚至像 **Mobile-Agent** 一样操作手机

基础核心：大语言模型(LLM)

LLM 是什么?

LLM 是基于深度神经网络构建的先进人工智能系统，专门处理、理解和生成类人文本
这些模型通过海量文本数据的训练，掌握了语言的深层次规律和上下文逻辑

类似于一个想象一个读遍了互联网的“超级大脑”

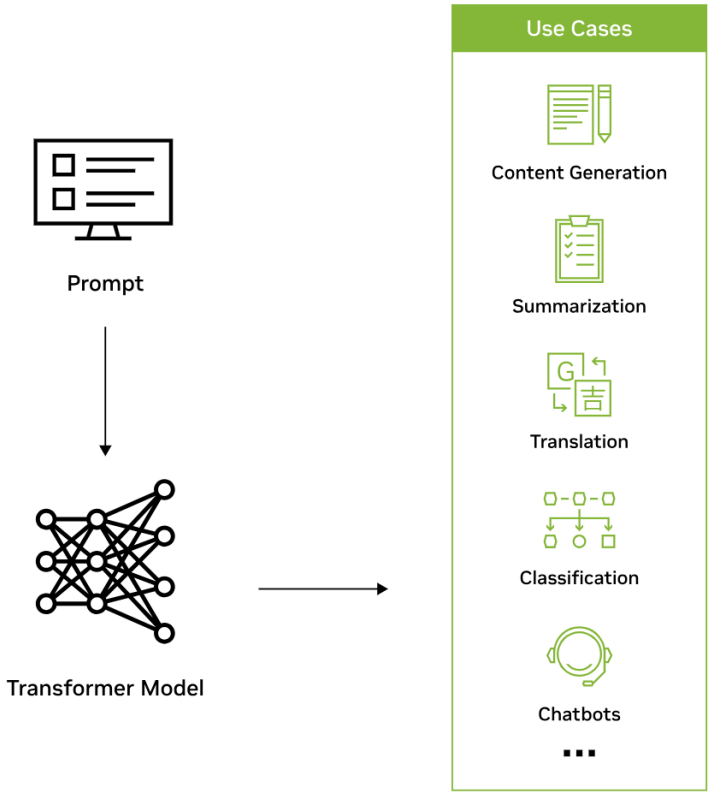


LLM 的核心架构基于Transformer系列网络，特别是其中的自注意力机制

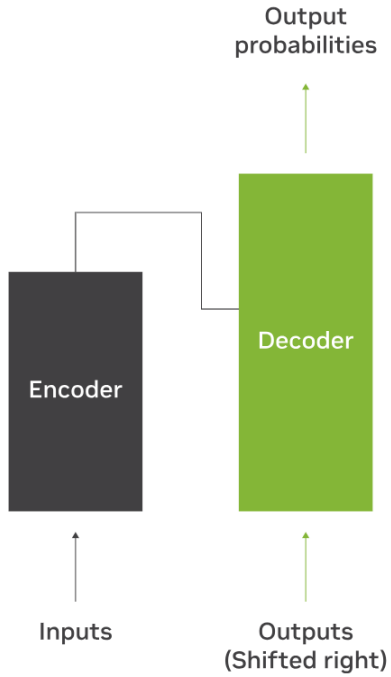
基础核心：大语言模型(LLM)

大语言模型在很大程度上代表了一类名为 Transformer 网络的深度学习架构

Transformer 模型是一种神经网络，通过追踪序列数据 (如本句中的单词) 中的关系学习上下文及其含义



Transformer 模型的工作原理



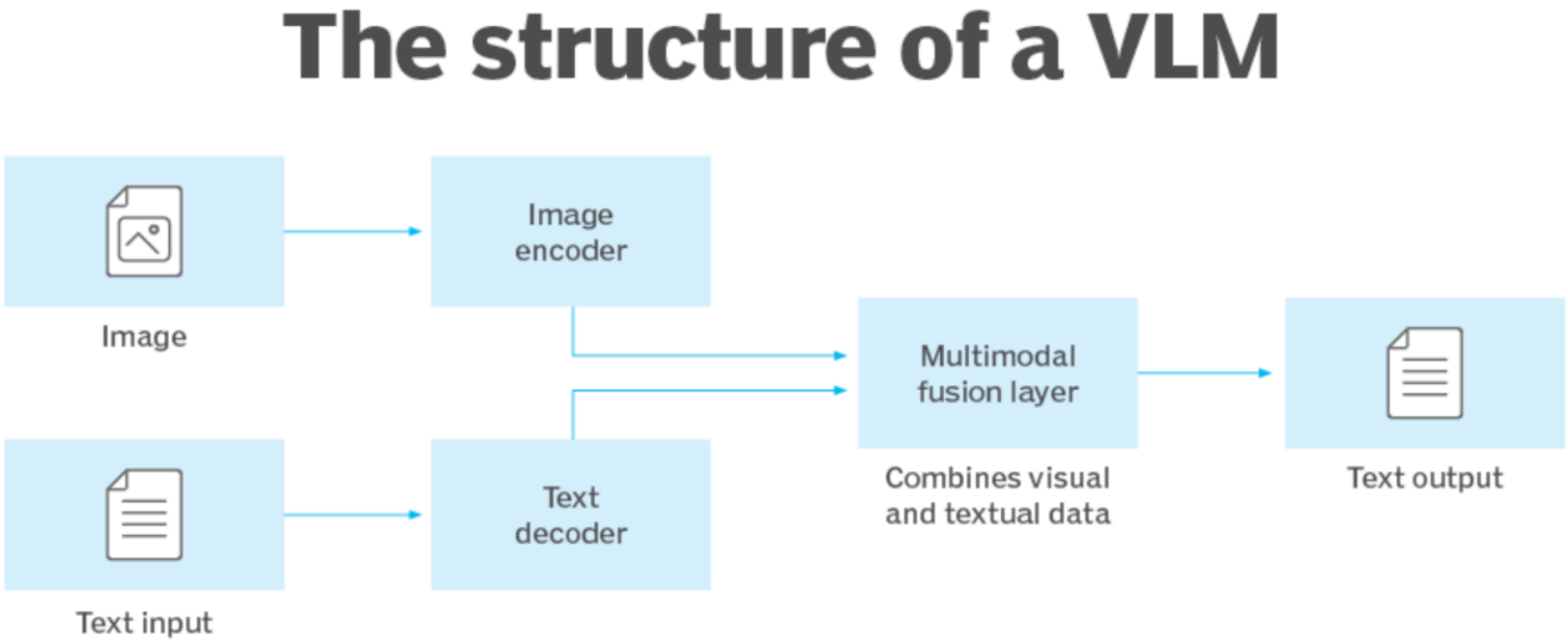
Encoder-Decoders

编码器-解码器语言模型的结构

视觉增强：视觉语言模型(VLM)

VLM 是什么?

VLM 是一种多模态、生成式 AI 模型，能够理解和处理视频、图像和文本



VLM 的构成要素是什么?

视觉语言模型是通过将大语言模型 (LLM) 与视觉编码器相结合构建的多模态 AI 系统，使 LLM 具有“看”的能力，凭借这种能力，VLM 可以处理并提供对提示中的图像和文本输入的高级理解，以生成文本响应

视觉增强：视觉语言模型(VLM)

VLM 的架构是什么?

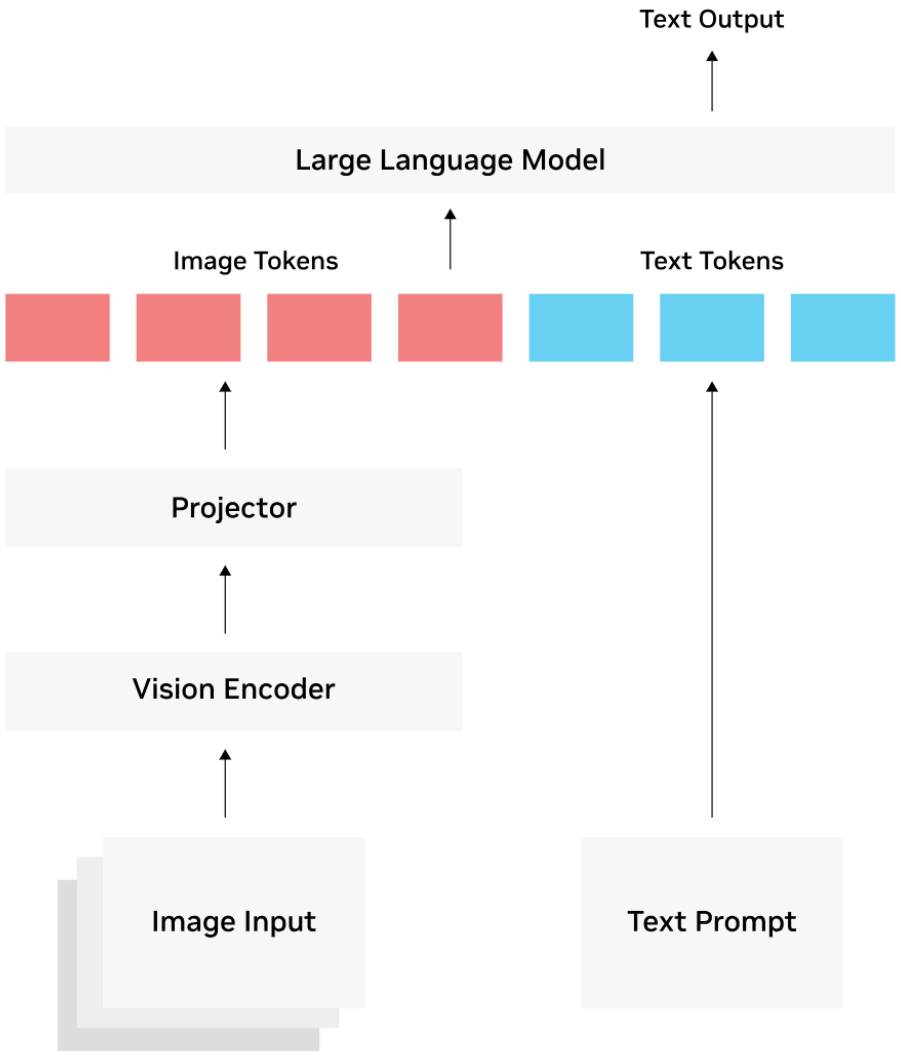
大多数 VLM 架构由三部分构成：

- **视觉编码器**
- **投影器**
- **LLM**

视觉编码器通常是一个基于 transformer 架构的 CLIP 模型，该模型已在数百万个图像-文本对进行了训练，具有图像与文本的关联能力

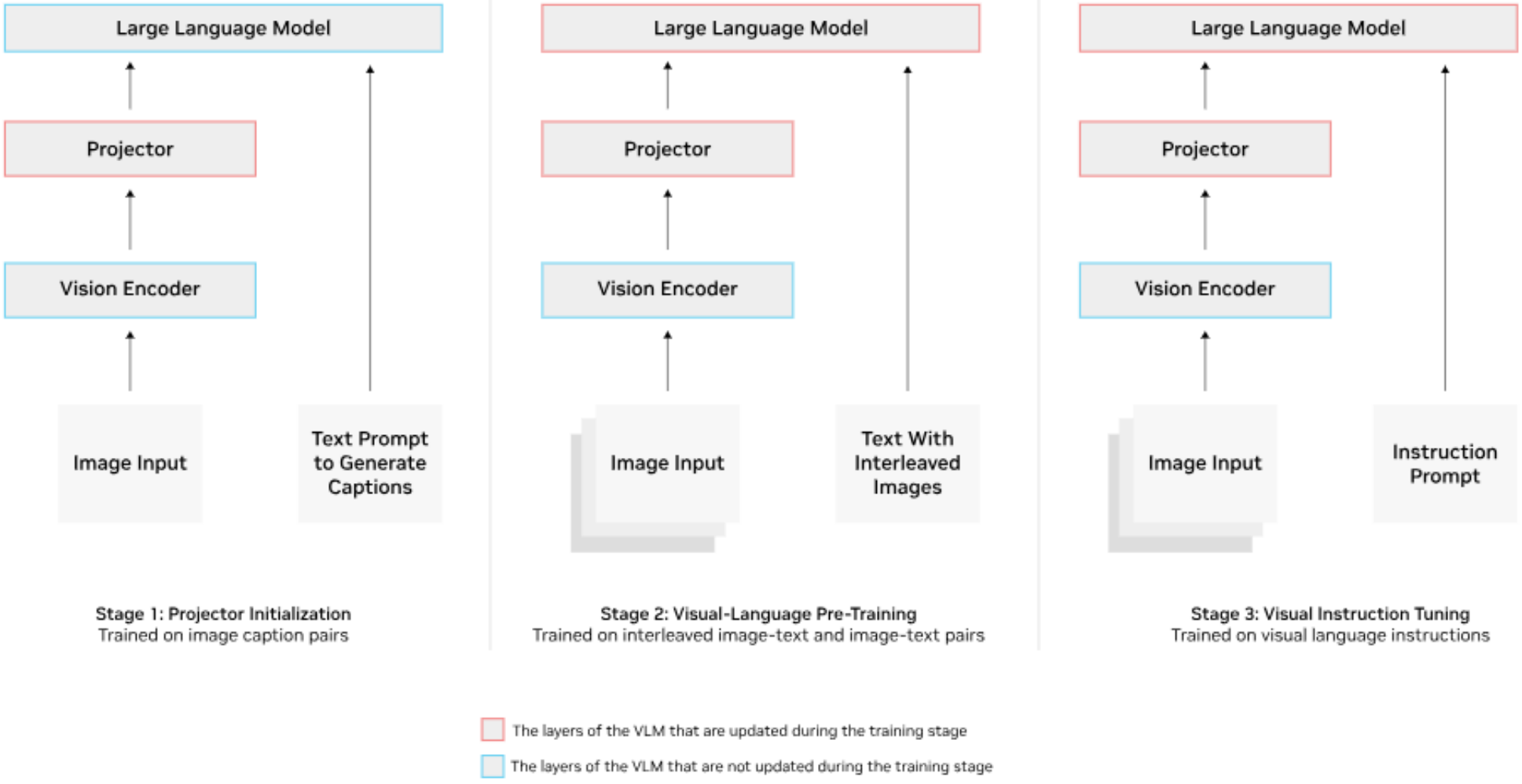
投影器由一组网络层构成，将视觉编码器的输出转换为 LLM 可以理解的方式，一般解读为图像标记 (tokens)，投影器可以是如 LLaVA 与 VILA 中的简单线性层，或者是如 Llama 3.2 Vision 中使用的交叉注意力层更复杂的结构

任何现有的 **LLM** 都可以用来构建 VLM，有数百种结合了各种 LLM 与视觉编码器的 VLM 变体



VLM 的通用三部分架构

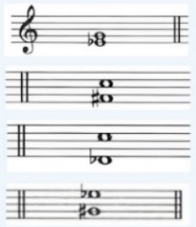
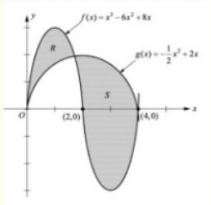
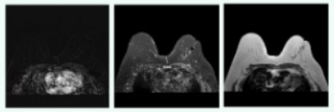
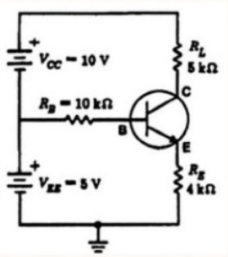
视觉增强：视觉语言模型(VLM)



VLM 训练通常针对模型的特定部分，分几个阶段完成

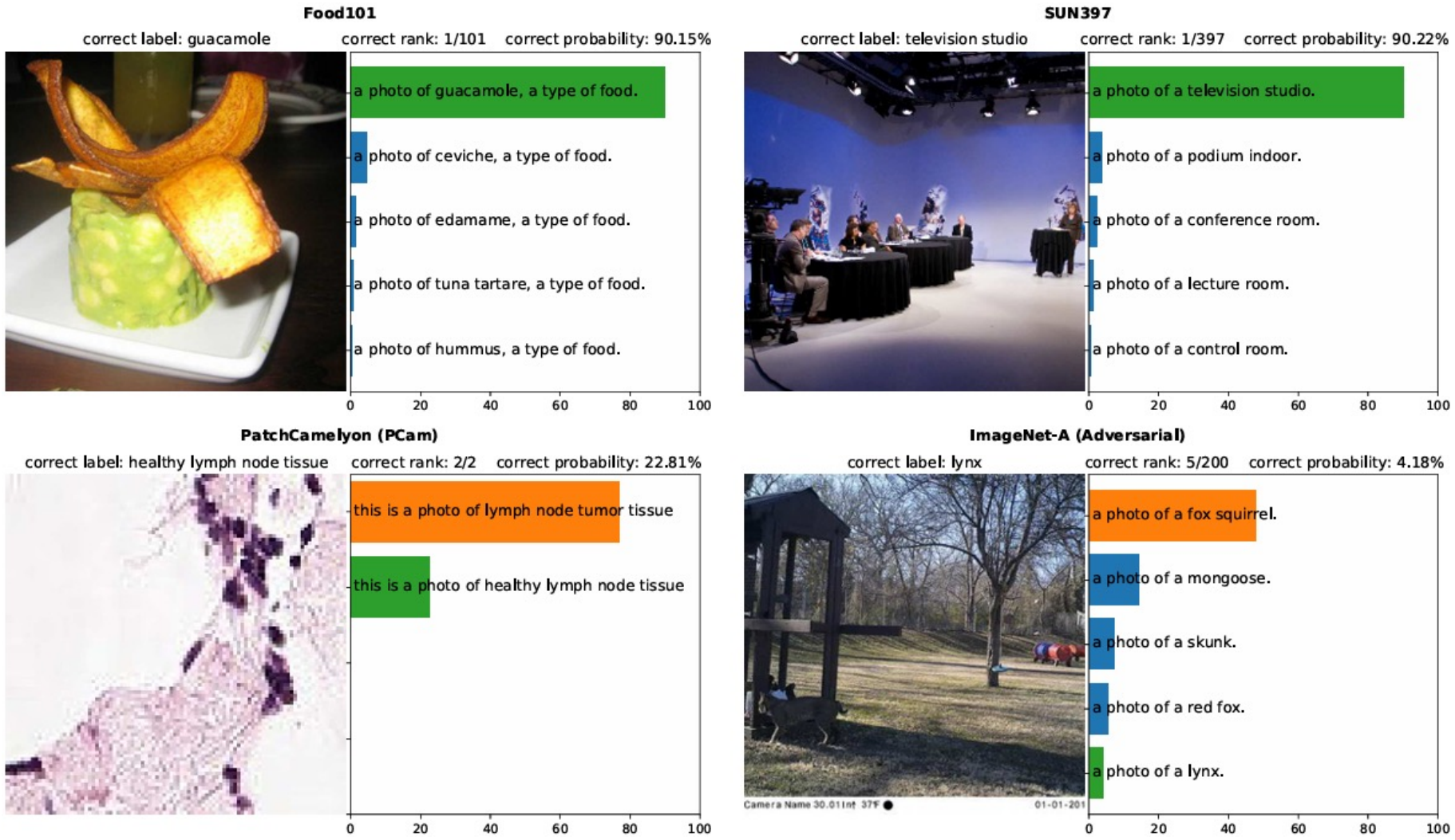
视觉增强：视觉语言模型(VLM)

如何评估 VLM 的性能？ 需要基准测试（Benchmark），用于确定 VLM 在各种任务上的表现，通常使用准确度作为评估指标

Art & Design	Business	Science
Question: Among the following harmonic intervals, which one is constructed incorrectly? **Options:** (A) Major third <*image 1*> (B) Diminished fifth <*image 2*> **(C) Minor seventh <*image 3*>** (D) Diminished sixth <*image 4*> 	**Question:** ...The graph shown is compiled from data collected by Gallup <*image 1*>. Find the probability that the selected Emotional Health Index Score is between 80.5 and 82? **Options:** (A) 0 (B) 0.2142 **(C) 0.3571** (D) 0.5 	**Question:** <*image 1*> The region bounded by the graph as shown above. Choose an integral expression that can be used to find the area of R. **Options:** **(A)** $\int_0^{1.5}[f(x)-g(x)]dx$ (B) $\int_0^{1.5}[g(x)-f(x)]dx$ (C) $\int_0^{2}[f(x)-g(x)]dx$ (D) $\int_0^{2}[g(x)-x(x)]dx$ 
Subject: Music; **Subfield:** Music; **Image Type:** Sheet Music; **Difficulty:** Medium	**Subject:** Marketing; **Subfield:** Market Research; **Image Type:** Plots and Charts; **Difficulty:** Medium	**Subject:** Math; **Subfield:** Calculus; **Image Type:** Mathematical Notations; **Difficulty:** Easy
Health & Medicine	**Humanities & Social Science**	**Tech & Engineering**
Question: You are shown subtraction <*image 1*>, T2 weighted <*image 2*> and T1 weighted axial <*image 3*> from a screening breast MRI. What is the etiology of the finding in the left breast? **Options:** (A) Susceptibility artifact (B) Hematoma **(C) Fat necrosis** (D) Silicone granuloma 	**Question:** In the political cartoon, the United States is seen as fulfilling which of the following roles? <*image* 1> **Option:** (A) Oppressor (B) Imperialist **(C) Savior** (D) Isolationist 	**Question:** Find the VCE for the circuit shown in <*image 1*>. Neglect VBE **Answer: 3.75** **Explanation**: ...IE = [(VEE) / (RE)] = [(5 V) / (4 k-ohm)] = 1.25 mA; VCE = VCC - IERL = 10 V - (1.25 mA) 5 k-ohm; VCE = 10 V - 6.25 V = 3.75 V 
Subject: Clinical Medicine; **Subfield:** Clinical Radiology; **Image Type:** Body Scans: MRI, CT.; **Difficulty:** Hard	**Subject:** History; **Subfield:** Modern History; **Image Type:** Comics and Cartoons; **Difficulty:** Easy	**Subject:** Electronics; **Subfield:** Analog electronics; **Image Type:** Diagrams; **Difficulty:** Hard

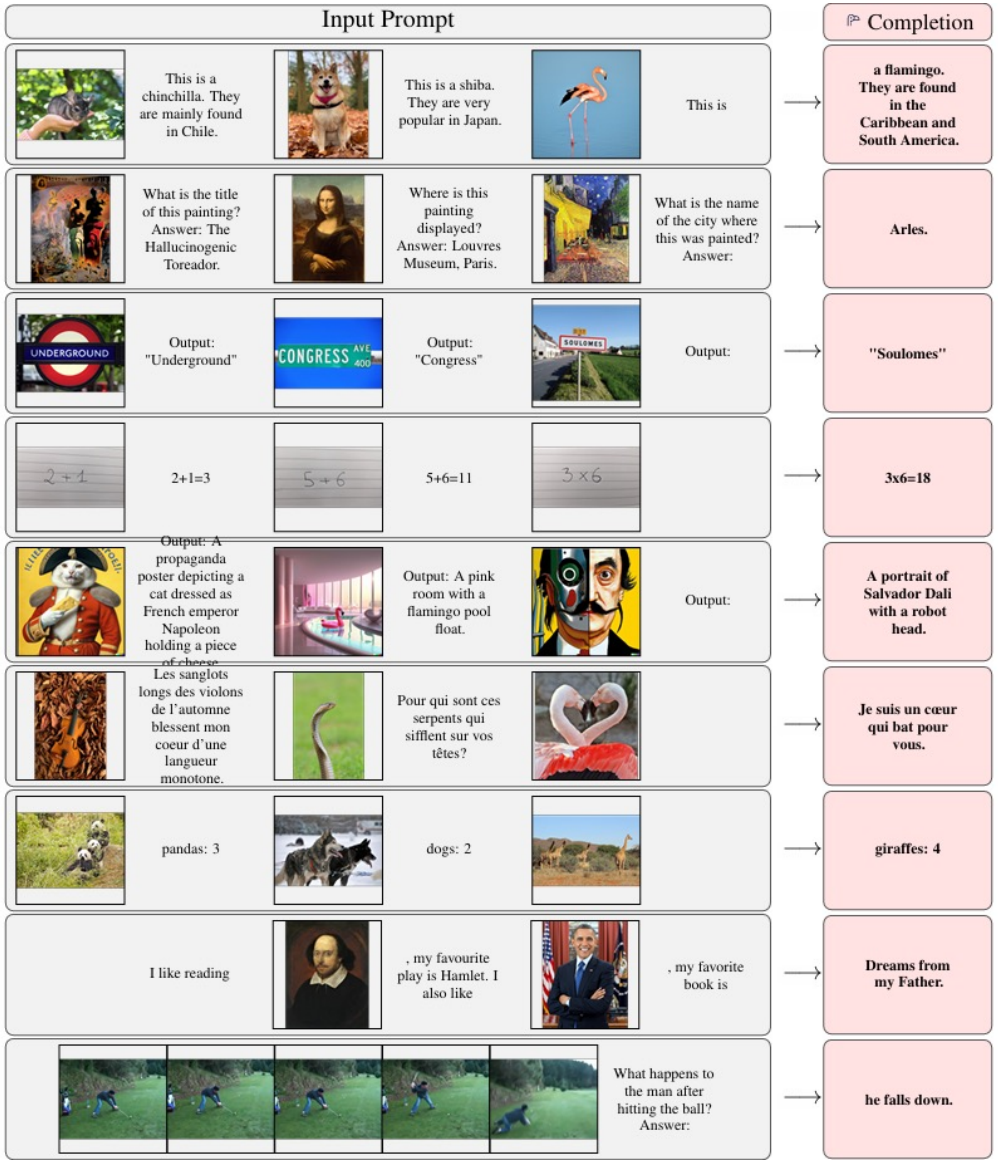
VLMs 使用 MMMU 基准测试的多选题示例

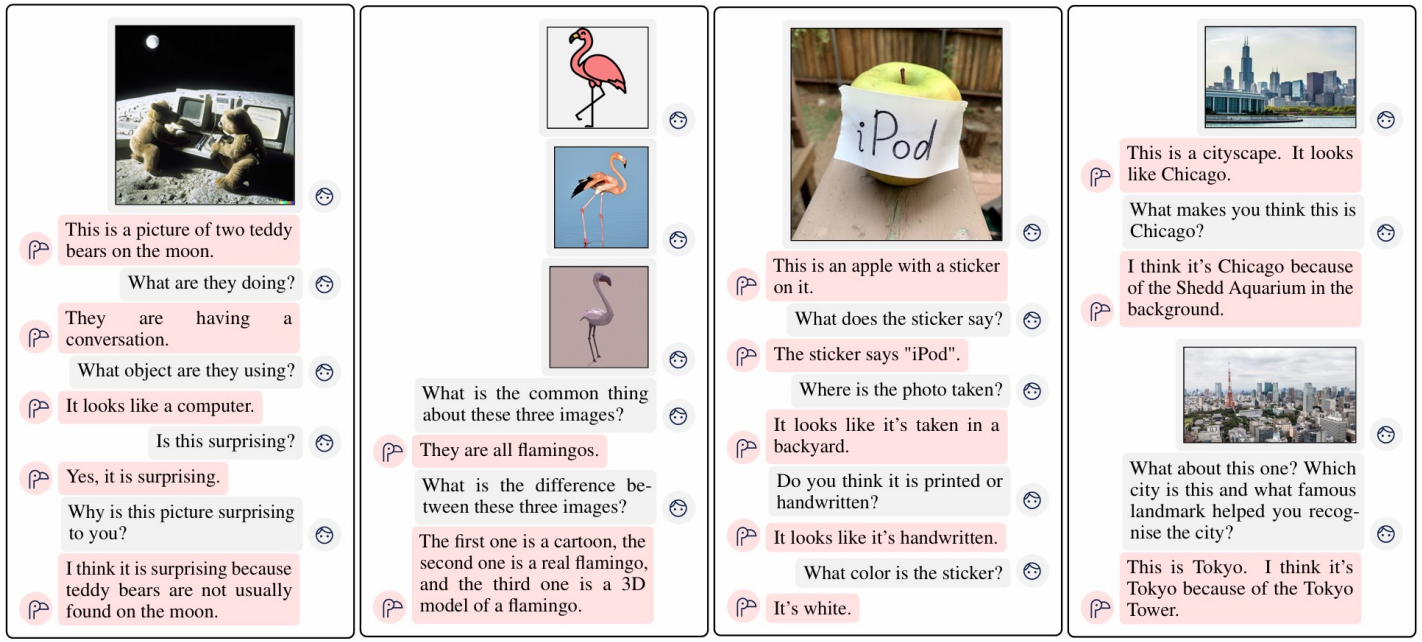
视觉增强：视觉语言模型(VLM)



CLIP 模型零样本（zero-shot prompting）分类预测可视化示例

视觉增强：视觉语言模型(VLM)





Flamingo-80B 模型的输入与输出示例选集

左图展示了 Flamingo 如何通过少样本提示（few-shot prompting）快速适应各种图像/视频理解任务

上图展示了该模型开箱即用的多图视觉对话能力

多模态融合：多模态大语言模型(MLLM)

MLLM是什么?

MLLM 是一种深度学习算法，能够理解和生成包括文本、图像、视频、音频等多种形式的内容

MLLM 扩展了主要专注于处理和生成文本的传统大型语言模型 (LLM) 的能力，通过整合多种类型的数据，实现了更复杂、更多样化的应用，这些应用需要综合和解释文本与非文本信息

如果把 MLLM 看作一个大集合，那么它的成员包括:

VLM: 处理 文本 + 图像/视频 (例如: LLaVA, GPT-4V, Gemini Pro Vision)

Audio-LLM : 处理 文本 + 音频/语音 (例如: Qwen-Audio)

Any-to-Any Model: 处理 文本 + 图像 + 音频 + 视频等所有模态 (例如: Gemini 1.5 Pro, GPT-4o)

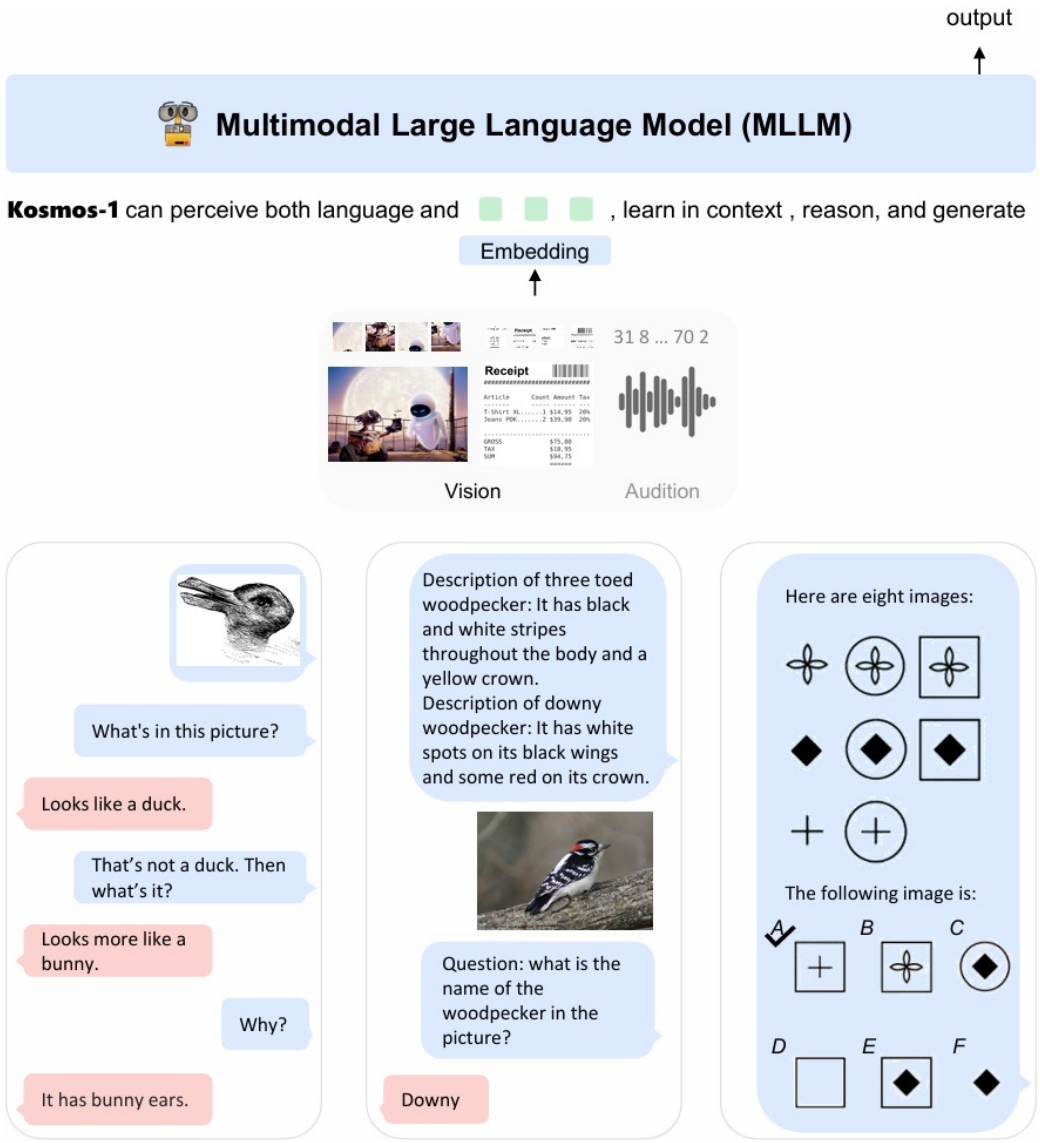


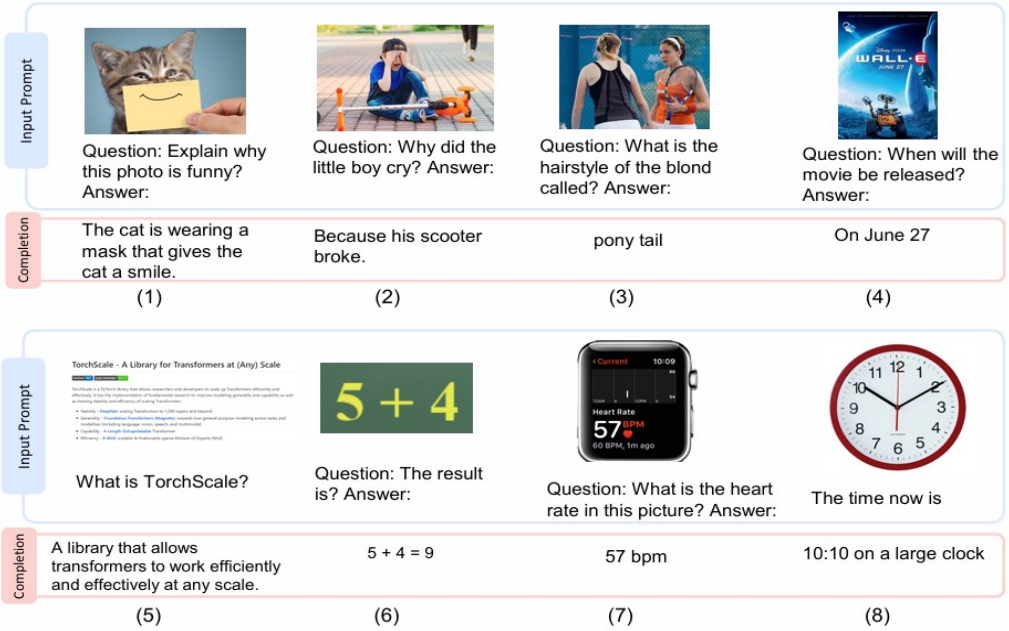
多模态融合：多模态大语言模型(MLLM)

LLM 和 MLLM 之间有什么区别?

特性	大型语言模型 (LLM)	多模态大语言模型 (MLLM)
数据处理	仅由文本分词器编码的文本数据	需要为每种模态配备单独编码器的多模态数据
模型架构	单一 Transformer 架构	每种模态有单独的编码器，随后是一个融合模块，将编码表示投影到统一的嵌入空间
训练目标	语言建模目标，如下一个令牌预测	通常使用对比学习目标，旨在对齐不同模态的表示
推理计算复杂度	与输入序列长度成二次方关系	纯文本 LLM 的复杂度加上同时从多种模态编码输入和解码输出的额外计算
模态编码器	仅处理文本数据，因此不需要模态编码器	将图像、音频和其他非文本数据转换为反映内容含义的嵌入
输入投影器	通常直接处理仅源自文本数据的文本嵌入，无需其他模态的对齐	将来自各种模态的编码表示与文本数据对齐，以创建语言模型可以处理的统一输入
LLM 主干	处理文本数据	使用预训练知识处理对齐的多模态输入，以执行推理、理解和内容生成等任务
输出投影器	无需将嵌入转换回其他模态	将模型的输出嵌入映射回目标模态以生成非文本输出
模态生成器	模型没有模态生成器，因为它们不是为处理非文本输出而设计的	生成单个模态的输出，通常使用潜在扩散模型 (LDMs)

多模态融合：多模态大语言模型(MLLM)

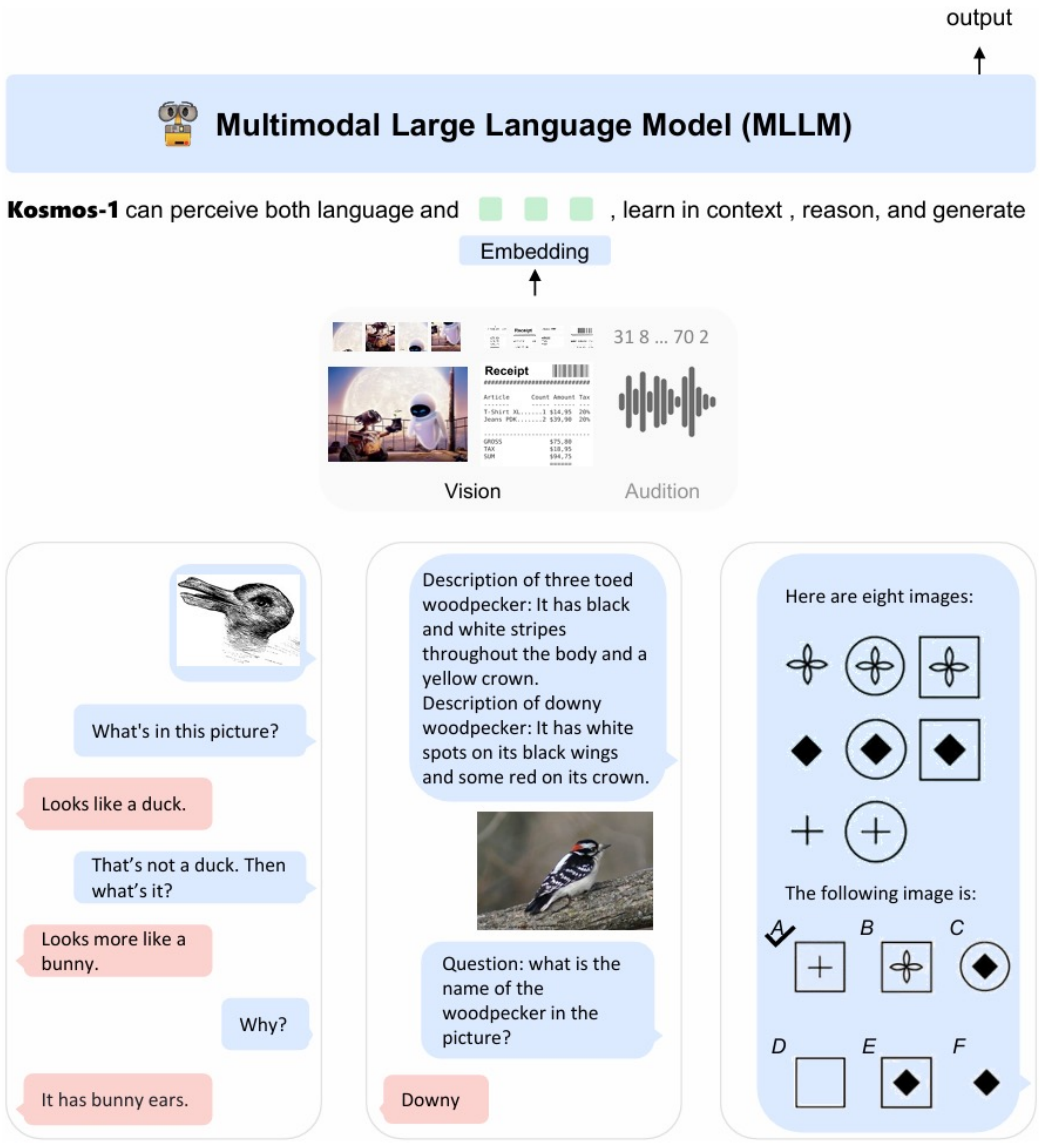


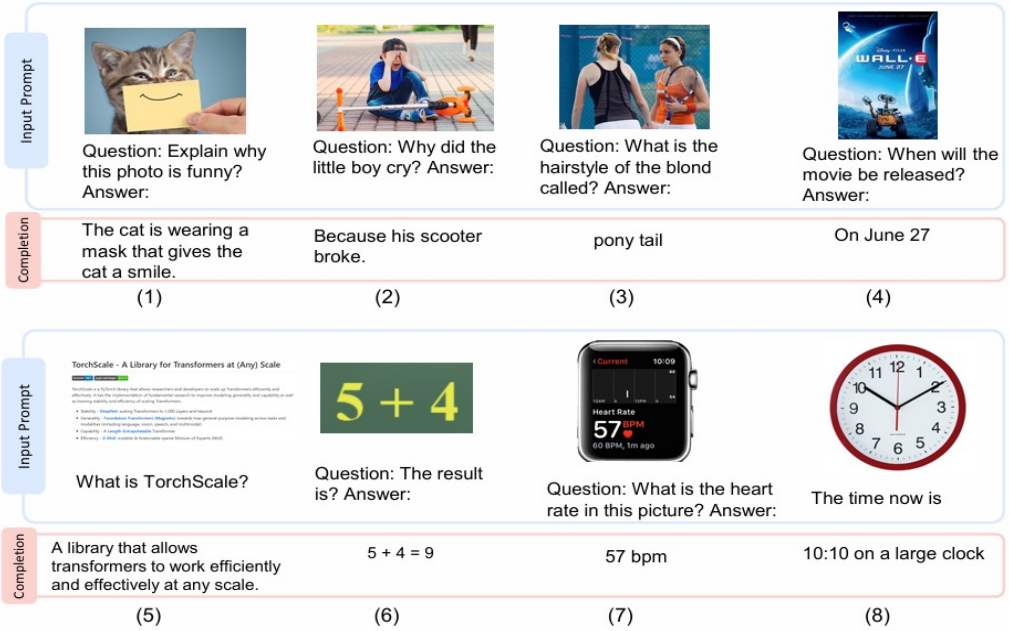


KOSMOS-1 是一个多模态大型语言模型（MLLM），它能够感知多模态输入、遵循指令，并不仅针对语言任务，也能针对多模态任务进行上下文学习（In-context learning）

在这项工作中， KOSMOS-1的作者将视觉与大型语言模型（LLM）相对齐，推动了从 LLM 向 MLLM 发展的趋势

多模态融合：多模态大语言模型(MLLM)

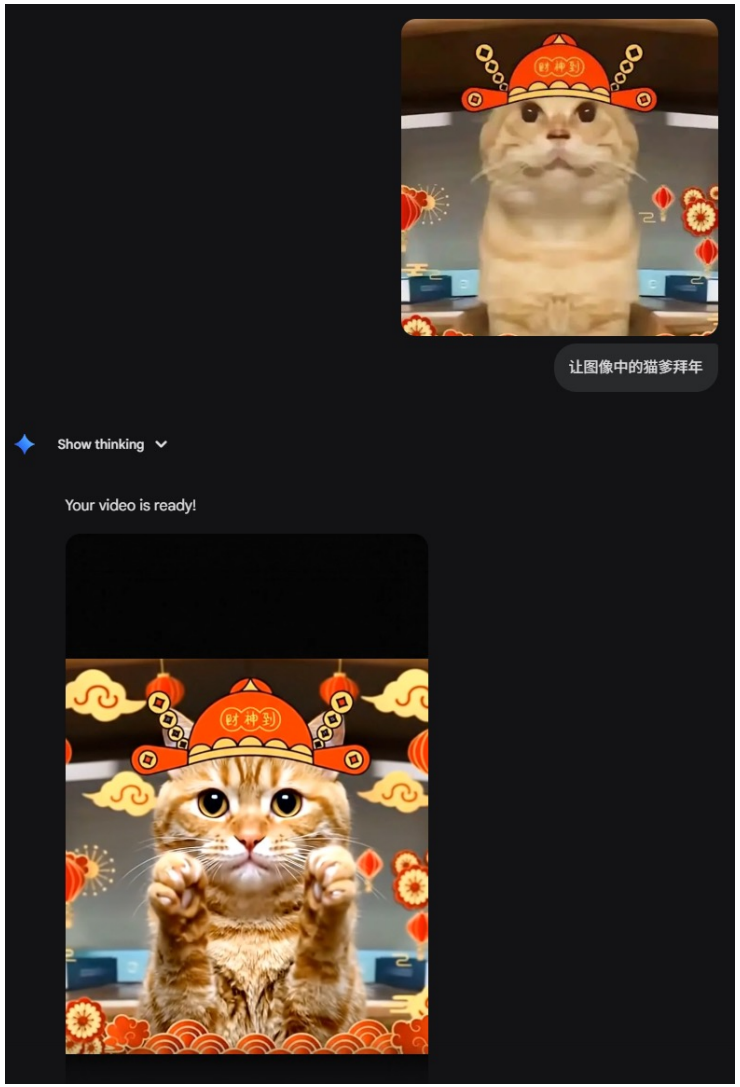




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多模态融合：多模态大语言模型(MLLM)





MLLM 执行跨模态任务示例

课程提纲

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一、 LLM/VLM/MLLM

二、多模态思维链推理

三、多模态视频理解

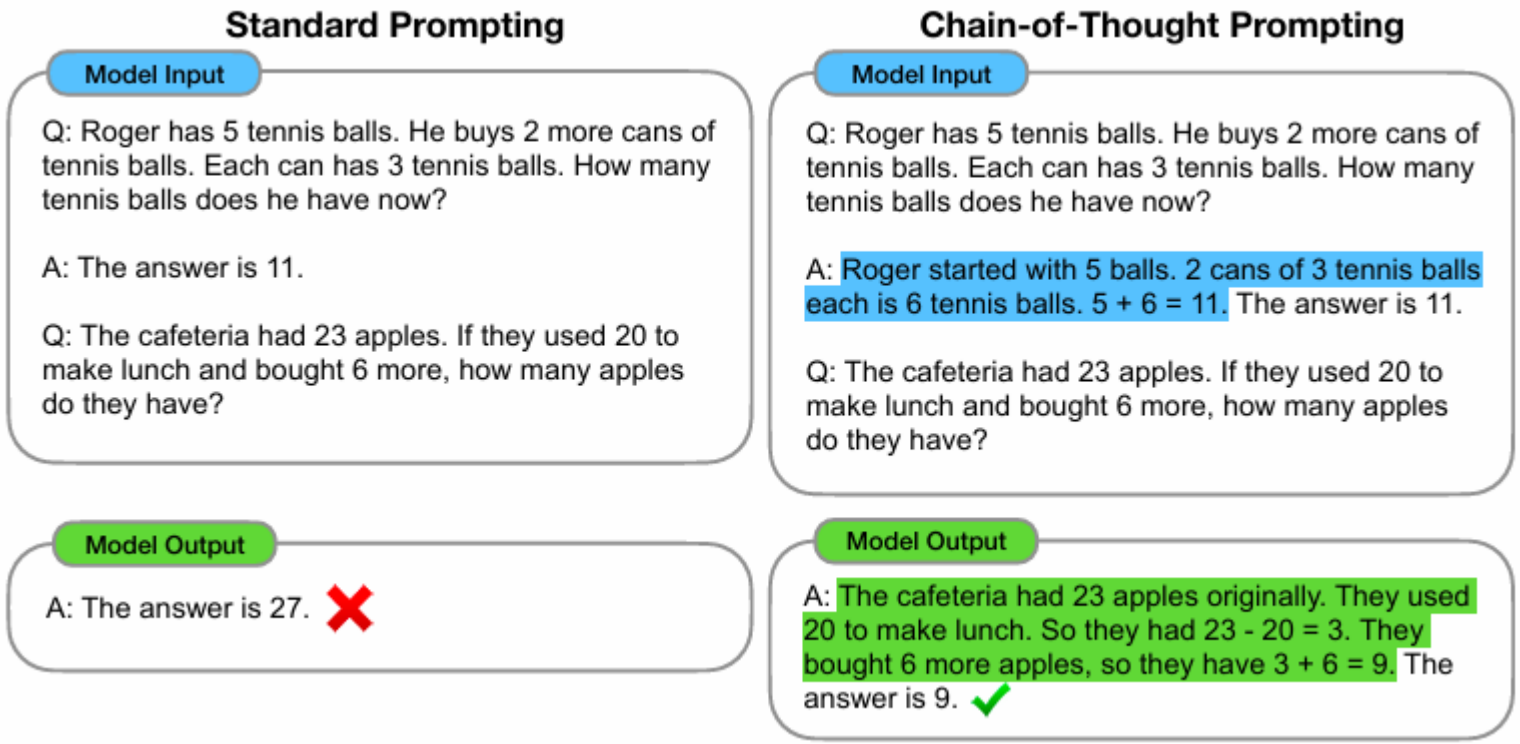
四、检索增强生成

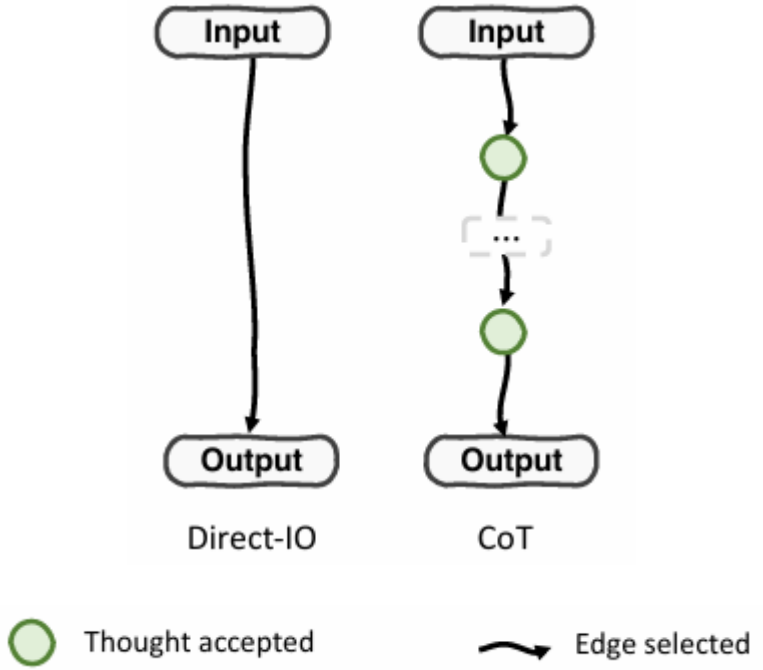
五、总结

思维的觉醒：什么是思维链 Chain-of-Thought (CoT)?

CoT 是什么?

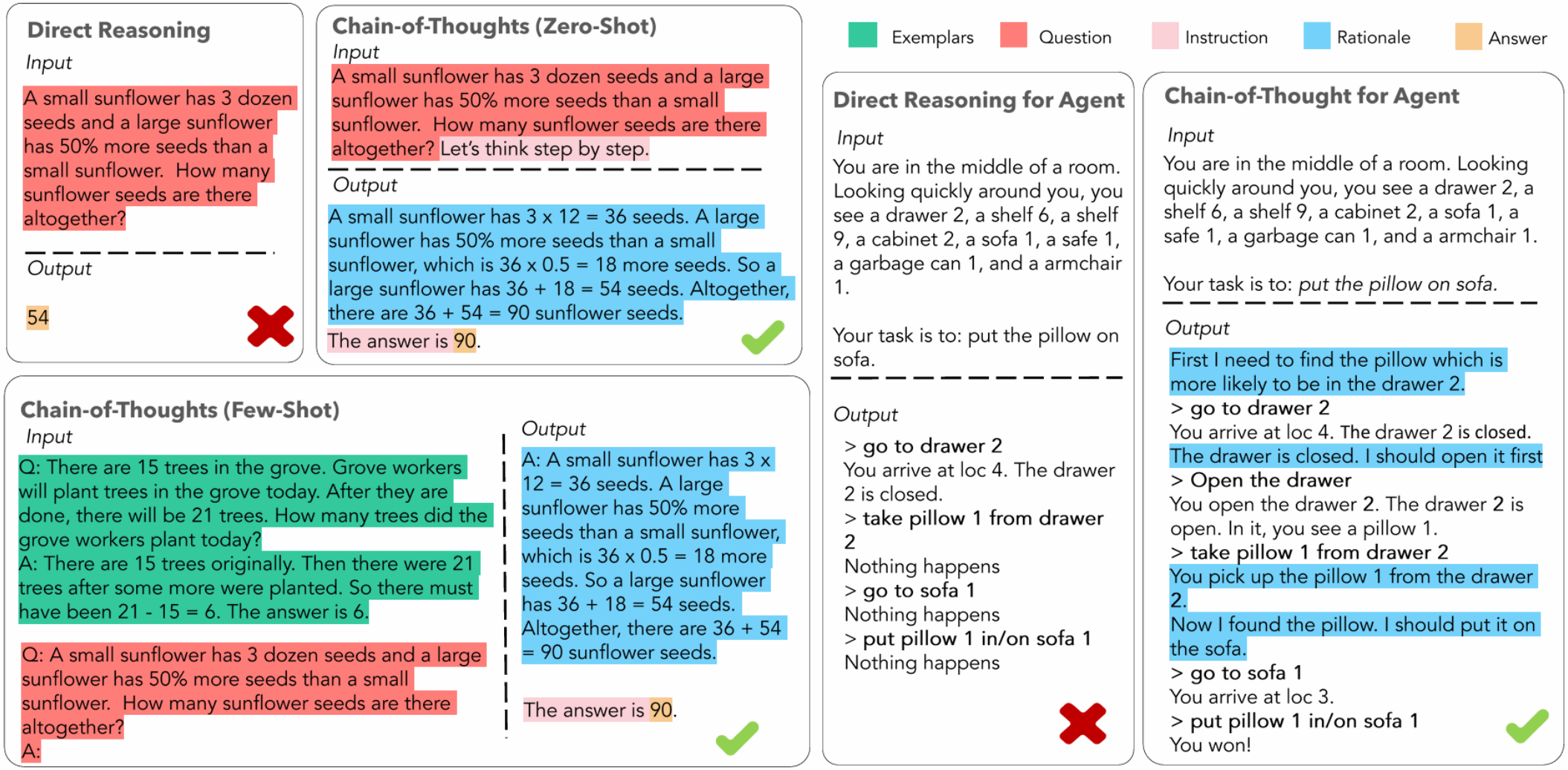
通过让大模型逐步参与将一个复杂问题分解为一步一步的子问题并依次进行求解的过程可以显著提升大模型的性能，而这些推理的中间步骤就被称为思维链（CoT）





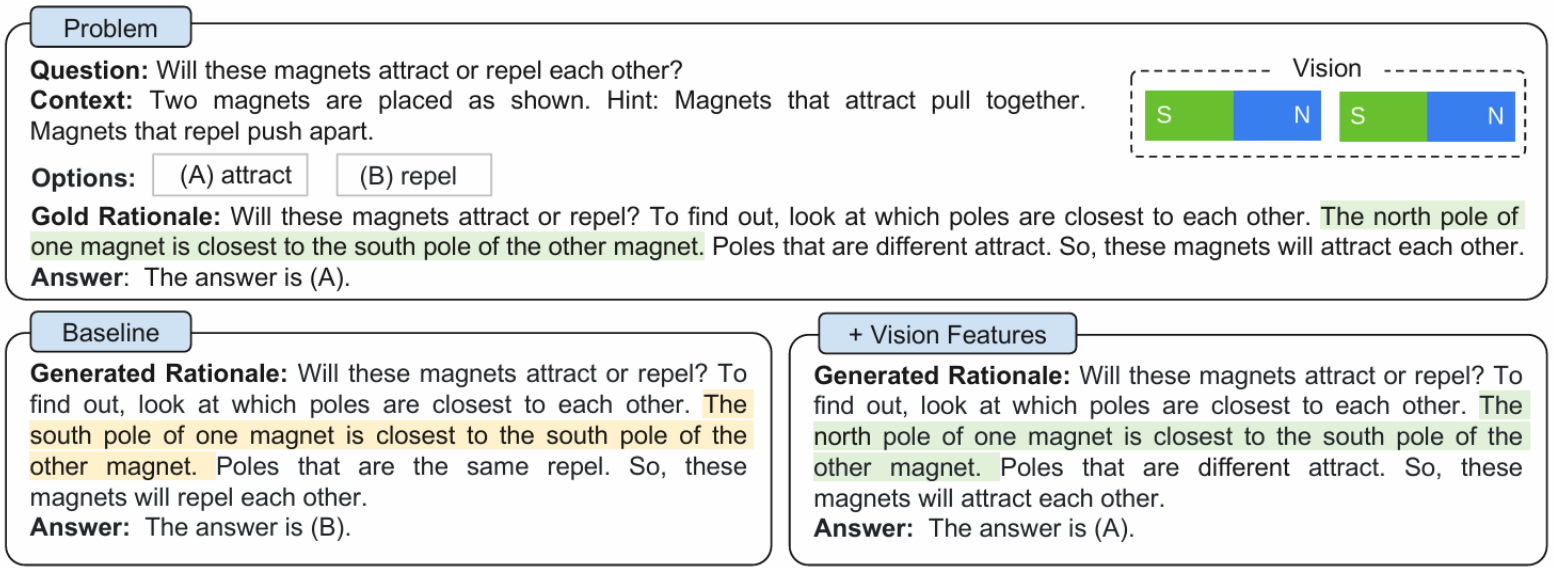
思维链推理过程示例

思维的觉醒：什么是思维链 Chain-of-Thought (CoT)?

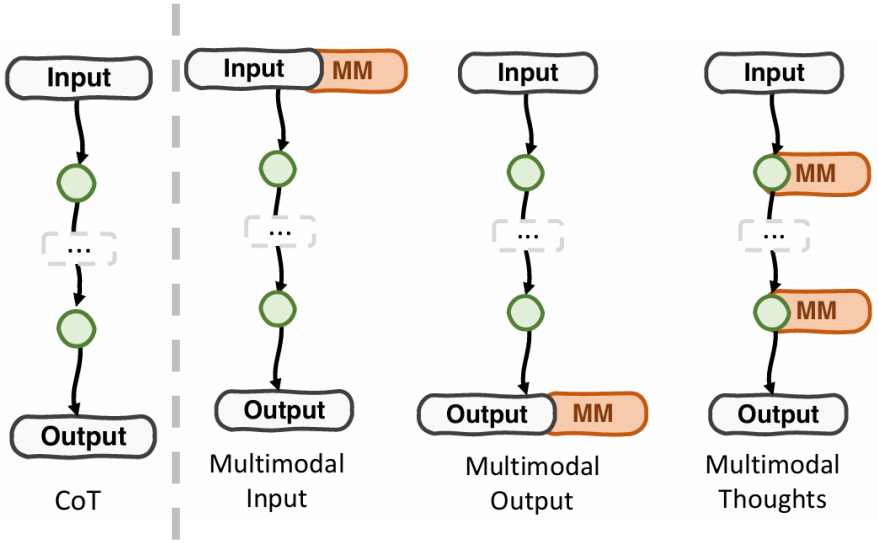


CoT 推理与直接推理的对比

当 CoT 遇到图像：为什么需要多模态思维链 (MCoT)?



CoT与MCoT任务示例对比



纯文本CoT的局限性

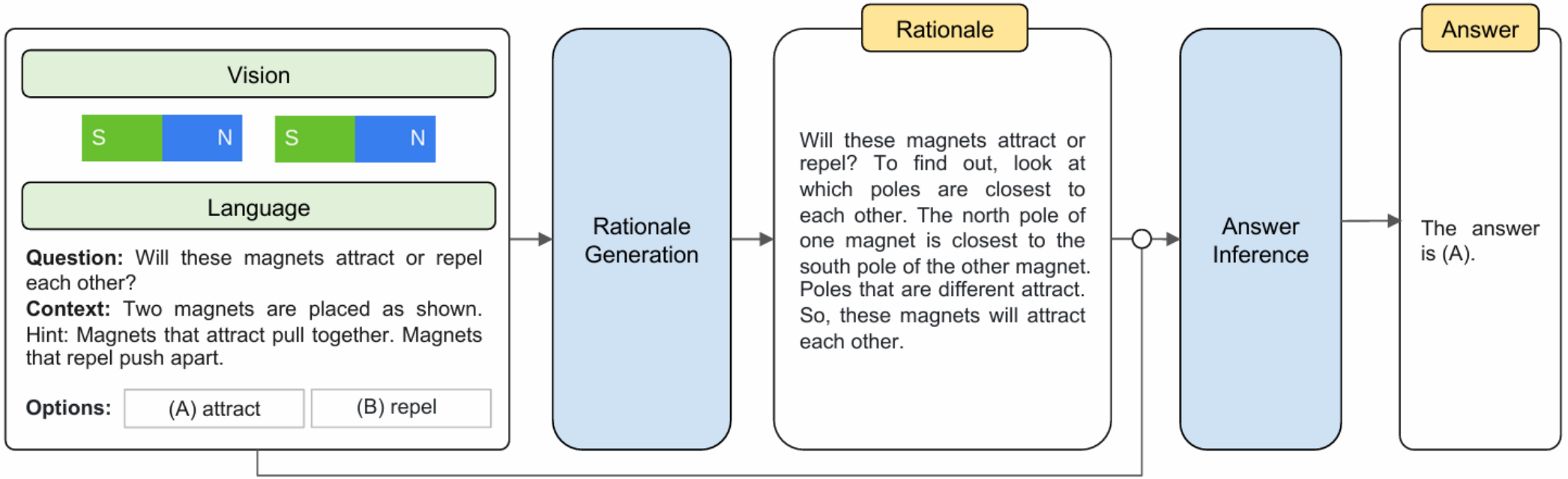
- **模态单一：** CoT 研究主要局限于语言模态，忽略了视觉信息
- **认知缺失：** 类似于“阅读没有插图的教科书”，缺乏视觉辅助会限制复杂知识的获取与推理能力

MCoT 的核心优势

- **减少幻觉：** 视觉特征作为锚点，显著减少了小模型（<1B参数）在推理过程中的编造行为
- **加速收敛：** 多模态信息提供了更强的监督信号，加快了模型的训练收敛速度
- **性能提升：** 生成的推理链（Rationale）质量更高，从而大幅提升最终答案的准确率（在 ScienceQA 上达到 SOTA）

MCoT经典架构：两阶段框架

MCoT框架图



- **Stage 1: 生成理由**　　输入：图+问题 → 输出：理由
- **Stage 2: 推断答案**　　输入：图+问题+理由 → 输出：最终答案

MCoT任务示例

Audio QA



Input: Based on the given audio, identify the source of the honk?

Rationale: The audio contains sounds related to a vehicle honking. Car honking is commonly heard on roads and streets when a driver uses the horn to alert other vehicles or pedestrians.

Answer: the most likely source of the honk is a car.

Visual QA



Input: Which nutrients are mainly provided by the foods in the image?
(A) Vitamins (B) Protein (C) Fats

Rationale: Oranges are in the image and they are commonly known to be a good source of Vitamin C. Therefore, the nutrient mainly provided by the foods in the image is Vitamins.

Answer: (A) Vitamins

Image Grounding



Input: The object can open bottle of beer.

Rationale: sharp blade with a pointed end to insert into the bottle cap.

Answer:



3D Synthesis

Input: A bench with five legs.

Rationale: Part specs generation… spacial specs generator… coordinate calculator… run command… spatial critic… completion critic….

Answer:



Video QA



Input: Why is the person running?

Rationale: First, a group of people is standing in front of an exercise machine and running on the track. Then, the essential function of the exercise machine is to provide resistance for the legs during exercise. This can help improve muscle strength and endurance.

Answer: They might participate in a fitness event.

AVQA



Input: Where is the first sounding instrument?

Rationale: What musical instruments appear in the video? flute, piano. What is the first instrument that sounds in the video? flute. Where is the flute? right

Answer: Right.

Generation

Input: A photo of three sports balls.



Rationale: <IMG A> The number of sport balls in the image does not match the prompt. While the lighting effect is decent, some areas have unnatural reflections and shadowing, which affects the overall realism.



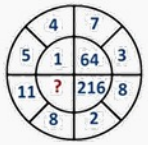
Answer: <IMG B>

Math Reasoning

Input: Find the missing value in this math puzzle.

Rationale:

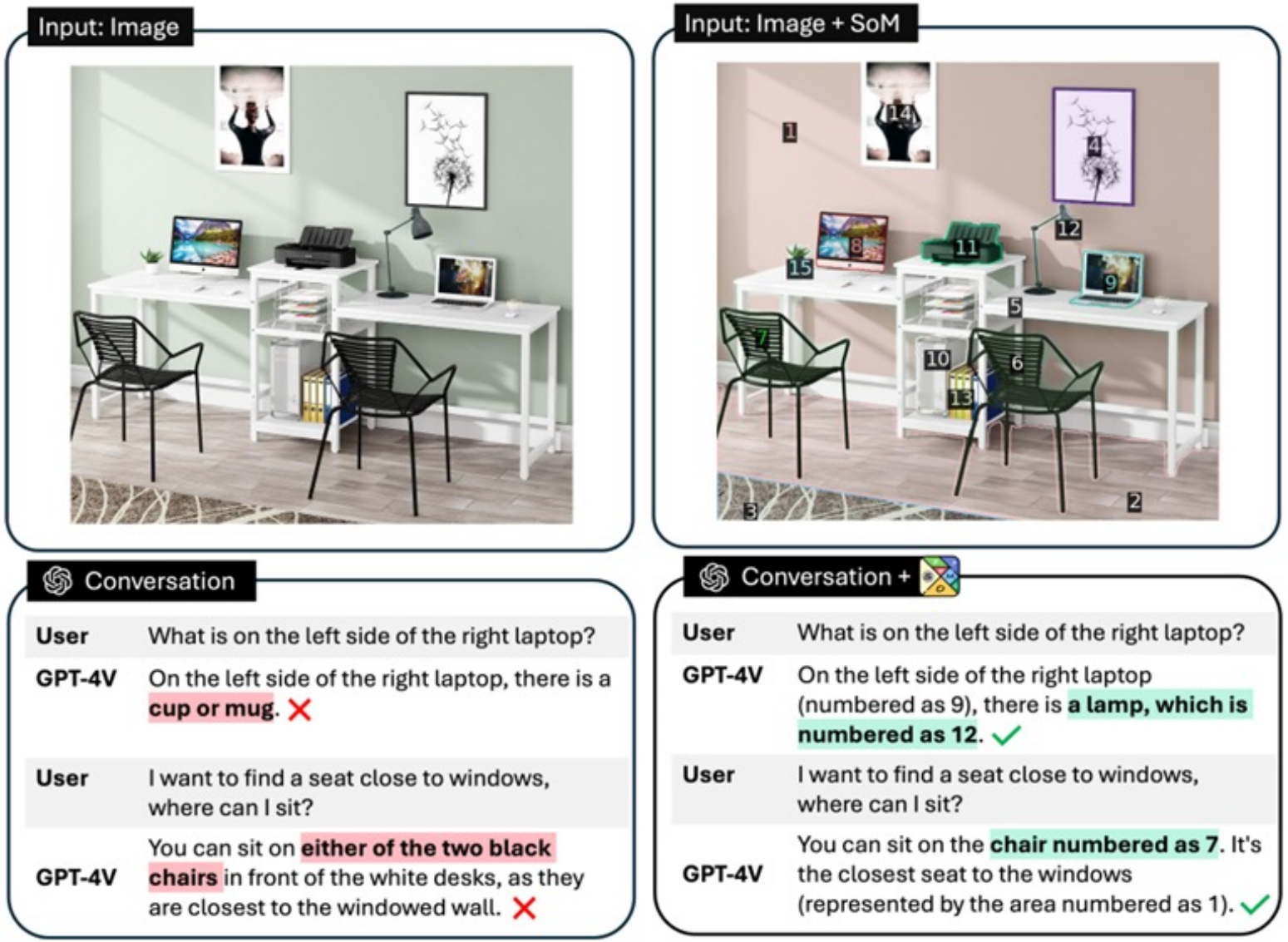
(5-4)^3=1, (7-3)^3=64,
(8-2)^3=216, (11-8)^3=27.



Answer: the missing value is 27.

MCoT 在多种模态与任务中的应用示例

MCoT 前沿进展：怎么看？ (Set-of-Mark)

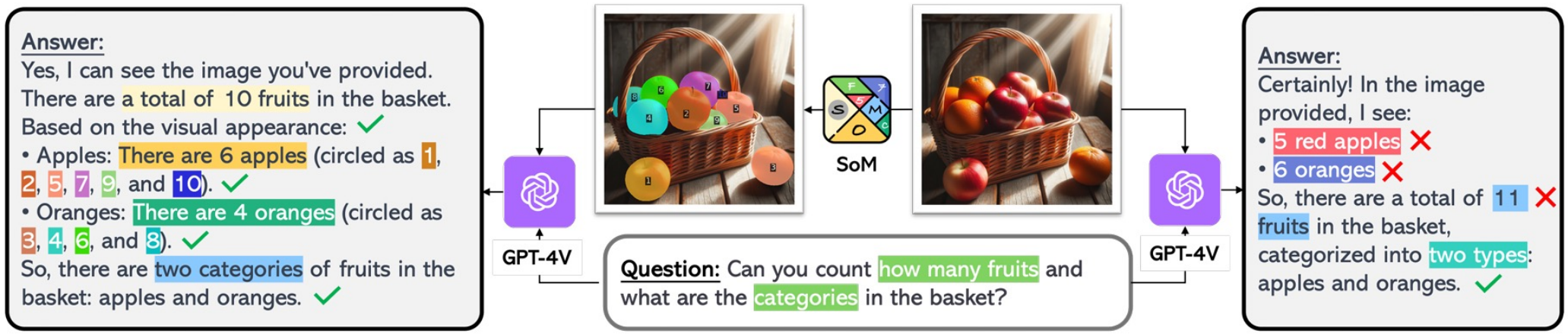


GPT-4V 提示技术的对比：（左）标准提示与（右）本文提出的 Set-of-Mark 提示

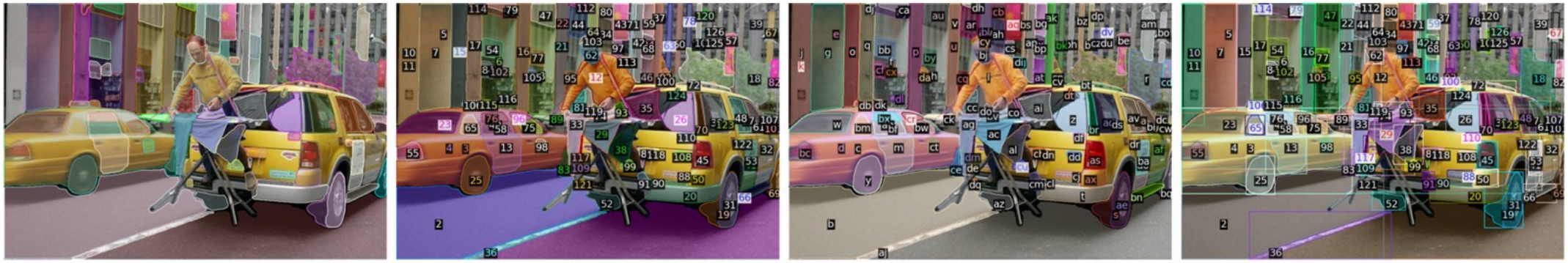
左：泛泛而谈导致“指代不明”与“视觉幻觉”

右：引入“视觉锚定”。在图像上强行打上 ID 标签（如数字、边界框），强制模型在推理时引用明确的视觉坐标，让输出精准可溯源

MCoT 前沿进展：怎么看？ (Set-of-Mark)



对比标准 GPT-4V 与结合了 Set-of-Mark 提示的方法

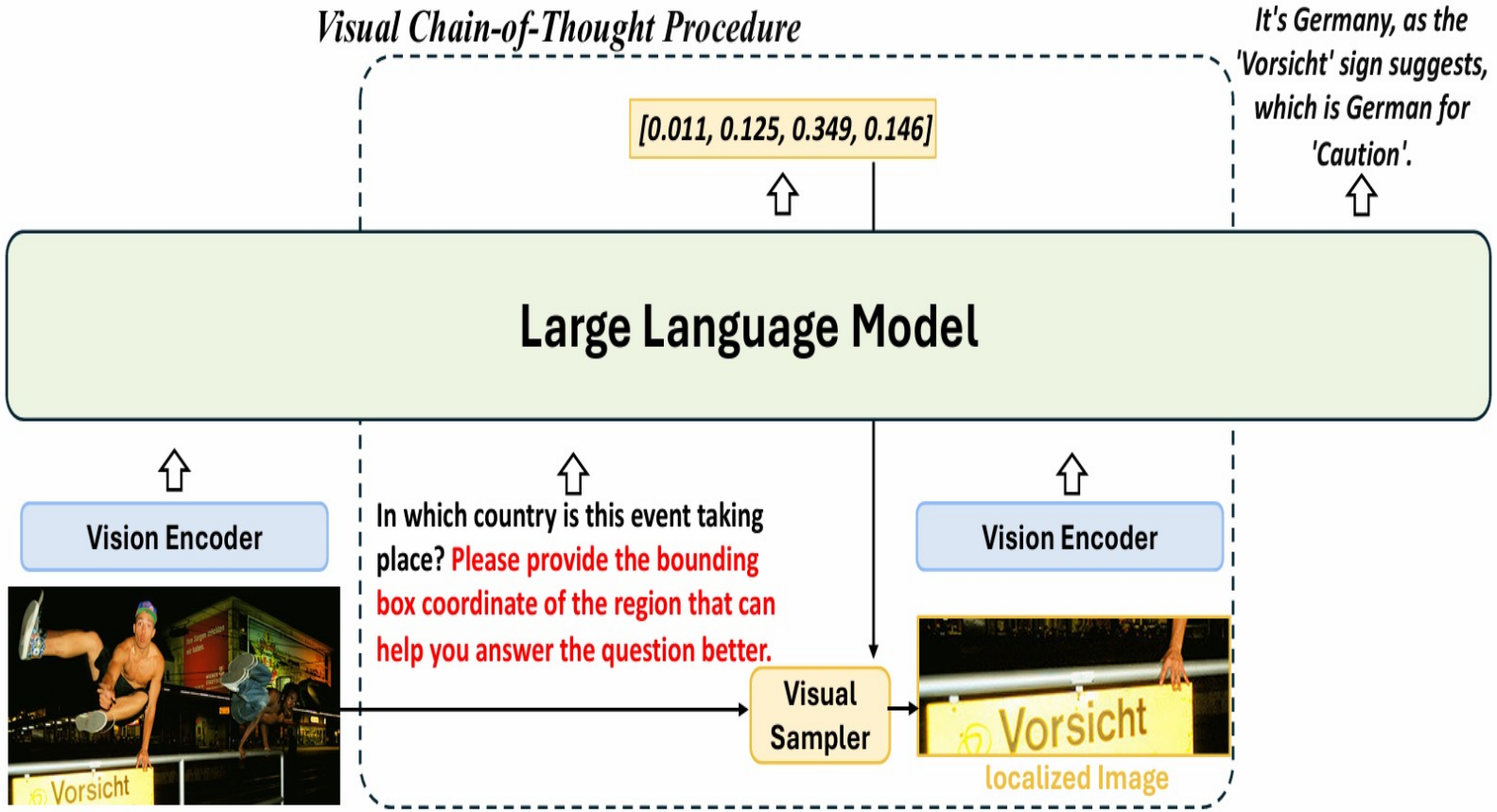


(1) Mask (2) Mask + Number (3) Mask + Alphabet (4) Mask + Box + Number

不同类型的 Set-of-Mark 标记

MCoT 前沿进展：怎么想？ (Visual CoT)

让模型学会“边画边想”，打破了“只用文字打草稿”的限制



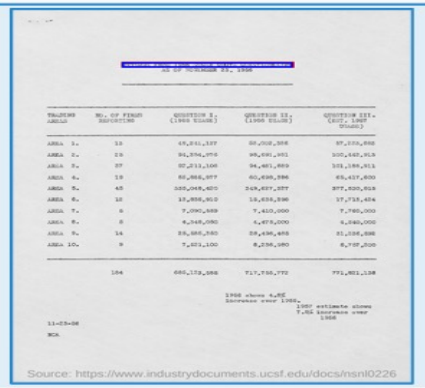
Visual CoT 框架图

- **步骤 1 | 全局感知与定位**：首先从整体图像提取视觉特征，并精准预测与问题相关的关键区域坐标
- **步骤 2 | 局部聚焦与重采样**：通过视觉采样器提取该局部的视觉信息，进行更细粒度的处理
- **步骤 3 | 全景与细节融合**：MLLM 综合全局上下文与局部关键特征，构建出兼具高可解释性与准确性的答案

模拟人类解题时的“画辅助线”过程。通过全局定位预测 、局部重采样放大、图文融合推演，生成视觉中间态，大幅提升复杂几何、文档等任务的上限

MCoT 前沿进展：怎么想？ (Visual CoT)

DocVQA



Question: What is the heading of the page?
GT Answer: Returns from 1956 Sugar Quota Questionnaire as of November 23, 1956

VisCoT Answer [score: 0.50]: RETURNS FROM 1968 SUGAR QUOTA QUESTIONNAIRE
VisCoT (GT BBox) Answer [score: 0.50]: Returns from 1966 sugar quota questionnaire
VisCoT (w/o CoT) Answer [score: 0.20]: "Monthly Financial Statement"

Flickr30k



Question: What is at the left side of the dog?
GT Answer: teddy bear

VisCoT Answer [score: 1.00]: teddy bear
VisCoT (GT BBox) Answer [score: 1.00]: teddy bear
VisCoT (w/o CoT) Answer [score: 0.50]: A stuffed animal

Birds-200-2011



Question: Does the bird in the picture have blue crown and black throat?
GT Answer: No

VisCoT Answer [score: 1.0]: No, the bird in the picture does not have a blue crown and black throat.
VisCoT (GT BBox) Answer [score: 1.00]: No
VisCoT (w/o CoT) Answer [score: 0.00]: The bird has a blue crown and a yellow throat.

GQA



Question: What is the name of the device to the right of the baby on the left side?
GT Answer: laptop

VisCoT Thoughts:
1.Locate the baby referenced in the question.
2.Ascertain the baby's position within the given context, specifically confirming that the baby is on the left side.
3.Identify the device situated to the right of the located baby.
4.Request or deduce the name of the identified device.

VisCoT Answer [score: 1.00]: laptop
VisCoT (GT BBox) Answer [score: 1.00]: The device is a laptop
VisCoT (w/o CoT) Answer [score: 0.00]: screen

Visual CoT 的可视化结果，用以说明各种推理模式之间的差异
模型生成的边界框以红色显示，而真实标注边界框以蓝色显示，这些分数由 ChatGPT 评估得出

MCoT 前沿进展：想错了怎么办？（自我纠正）

Woodpecker 的框架

Query: Describe this image.

MLLM Output

The scene takes place in an urban area, where a bicycle is parked next to a trash bin. The bicycle is situated on the right side of the image, with its front wheels slightly turned.

There are several other people in this area, walking and standing around at various distances from the trash bin.



Key Concept Extraction

bicycle; trash bin; person

Question Formulation

How many bicycles are there in the image?

...

What is next to the trash bin?

Where is the trash bin located?

Visual Knowledge Validation

bicycle: [0.467, 0.555, 0.717, 0.746]

...

a recycling bin

in front of the building

Visual Claim Generation

Object-level:

There is 1 bicycle. bicycle 1: [0.467, 0.555, 0.717, 0.746]

There are 2 trash bins. trash bin 1: [0.145, 0.498, 0.321, 0.728] ; trash bin 2: [0.319, 0.497, 0.483, 0.729]

There is no person.

Attribute-level:

trash bin 1: [0.145, 0.498, 0.321, 0.728]

A recycling bin is next to the trash bin.

trash bin 2: [0.319, 0.497, 0.483, 0.729]

A recycling bin is next to the trash bin.

A bicycle is next to the trash bin.

The trash bin is located in front of the building.

Hallucination Correction

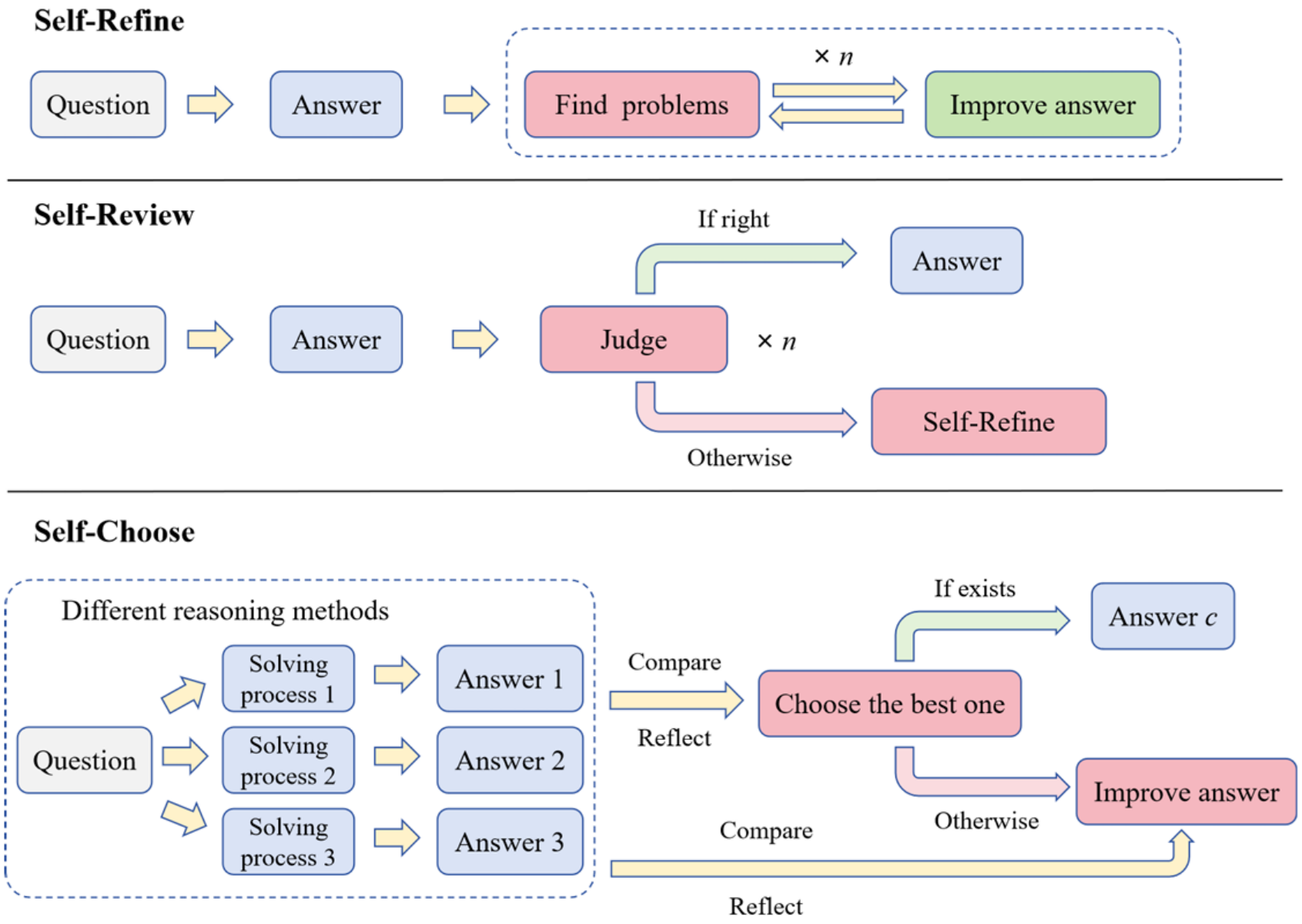


The scene takes place in an urban area, where a bicycle ([0.467,0.555,0.717,0.746]) is parked next to two trash bins ([0.145,0.498,0.321,0.728]; [0.319,0.497,0.483,0.729]). The bicycle is situated on the right side of the image, with its front wheels slightly turned.

There are no people in this area.

给定一张图像和一个查询，MLLM 输出相应的响应，通过四个步骤：**关键概念提取、问题构建、视觉知识验证以及视觉主张生成**，我们能够得到一个针对该图像和原始响应的特定视觉知识库。在最后一步中，响应中的幻觉将被以 bounding boxes 为证据进行修正

MCoT 前沿进展：想错了怎么办？（自我纠正）

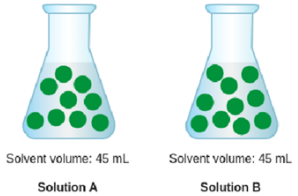


几种 MCoT 反思的演进：

- **Self-Refine**：基于反馈，迭代定位问题并完善答案
- **Self-Review**：先评估答案正误，正确则保留，错误则调用 Self-Refine 进行改进
- **Self-Choose**：对比反思多种推理路径与结果，直接择优，或综合生成更佳答案

MCoT 前沿进展：想错了怎么办？（自我纠正）

Question:
Which solution has a higher concentration of green particles?



Options:
A. Solution B
B. neither; their concentrations are the same
C. Solution A

IO (Candidate answer 1):
The answer is B. Because both solutions have the same amount of solute (green particles) and the same amount of solvent (45 mL). Therefore, their concentrations are the same.

CCoT (Candidate answer 2):

```
{
 "nodes": [
   {
    "id": "Solution A",
    "attributes": {
     "solvent volume": 45,
      "solute particle count": 8
    }
   },
   {
    "id": "Solution B",
    "attributes": {
     "solvent volume": 45,
     "solute particle count": 10
    }
   }
 ],
 "edges": [
   {
    "source": "Solution A",
    "target": "Solution B",
    "type": "concentration
              comparison",
    "attributes": {
     "difference": 2
    }
   }
 ]
}
```

The answer is B.

DDCoT (Candidate answer 3):

Sub-questions:
1. What is the definition of concentration?
2. What information do we have about the two solutions?
3. What can we conclude from the information we have?

Sub-answers:
1. Concentration is the amount of solute dissolved in a given volume of solvent.
2. We know that both solutions have the same volume of solvent (45 mL) and that Solution A has 12 green particles while Solution B has 15 green particles.
3. We can conclude that Solution B has a higher concentration of green particles than Solution A.

The answer is A, Solution B.

Final answer:
The candidate answer 3 is the best.
The answer is A.



Question:
What is the number displayed on the motorcycle ridden by the motorcyclist wearing a yellow suit?

IO (Candidate answer 1):
16.

DDCoT (Candidate answer 3):
Sub-questions:
1. What is the number displayed on the motorcycle ridden by the motorcyclist wearing a yellow and blue suit?
2. What is the number displayed on the motorcycle ridden by the motorcyclist wearing a yellow suit?

Sub-answers:
1. 17
2. 9

17 and 9.

Final answer:
Candidate answer 1 is the best.
16.

CCoT (Candidate answer 2):

```
{
 "objects": [
   {
    "type": "motorcycle",
    "attributes": {
     "color": "yellow",
     "number": 16
    }
   },
   {
    "type": "person",
    "attributes": {
     "suit_color": "yellow",
     "helmet_color": "black"
    }
   }
 ],
 "relationships": [
   {
    "type": "drives",
    "subject": {
     "type": "person",
     "attributes": {
      "suit_color": "yellow",
      "helmet_color": "black"
     }
    },
    "object": {
     "type": "motorcycle",
     "attributes": {
      "color": "yellow",
      "number": 16
     }
    }
   }
 ]
}
```

16.

两个 Self-Choose 的执行例子

MCoT 前沿进展：想错了怎么办？（自我纠正）

对比维度	流派一：事后特征验证	流派二：动态反思架构
代表方法	Woodpecker	Self-Refine / Self-Review / Self-Choose
核心思想	**“做完再检查”** (Fact-checking)	**“边做边改”** (Reasoning-checking)
纠错时机	**事后 (Post-hoc)：** 先生成初始回答，再针对该回答进行修改	**推演中 (In-process)：** 在得出最终答案前，动态调整和选择推理路径
纠错重点	解决“看错”（视觉幻觉：如无中生有、数错个数、属性标错）	解决“想错”（逻辑谬误：如公式用错、前后矛盾、推理断层）
运作机制	提取实体 → 生成视觉主张 → 重新回溯图像(如选择边界框) → 强制修正	单路径反复修改 (Self-Refine) 引入“法官”判断 (Self-Review) 探索多分支路径择优 (Self-Choose)
核心优势	强制视觉对齐，有效剔除“一本正经的胡说八道”	极大拓宽了复杂问题的逻辑推理上限
适用场景	密集图像描述、医疗影像诊断（对**事实准确度**要求极高）	数学几何解题、复杂图表分析（对**逻辑严密性**要求极高）

课程提纲

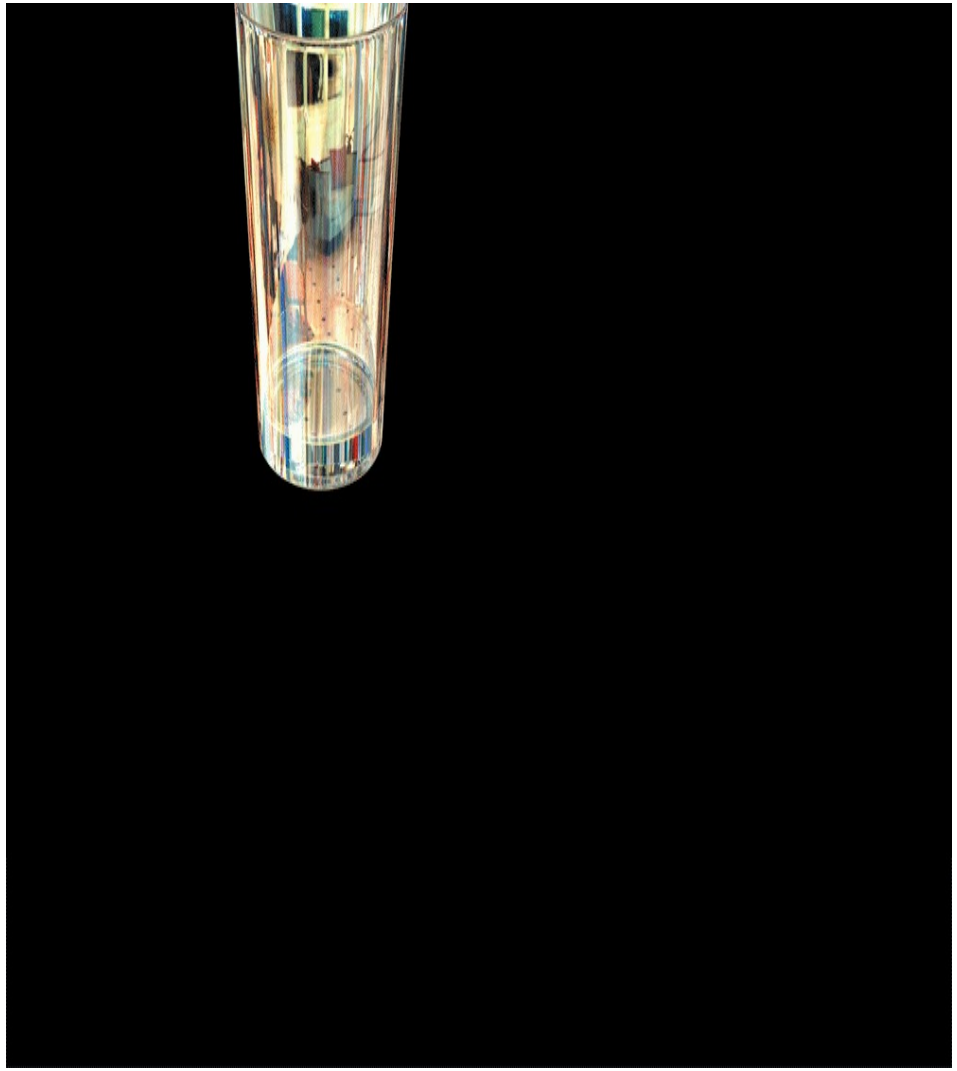
CONTENTS

一、 LLM/VLM/MLLM
二、多模态思维链推理
三、多模态视频理解
四、检索增强生成
五、总结

视界的跃迁：从“静态快照”到“动态因果”



MLLM 图像理解：杯子碎了（不知道怎么碎的）

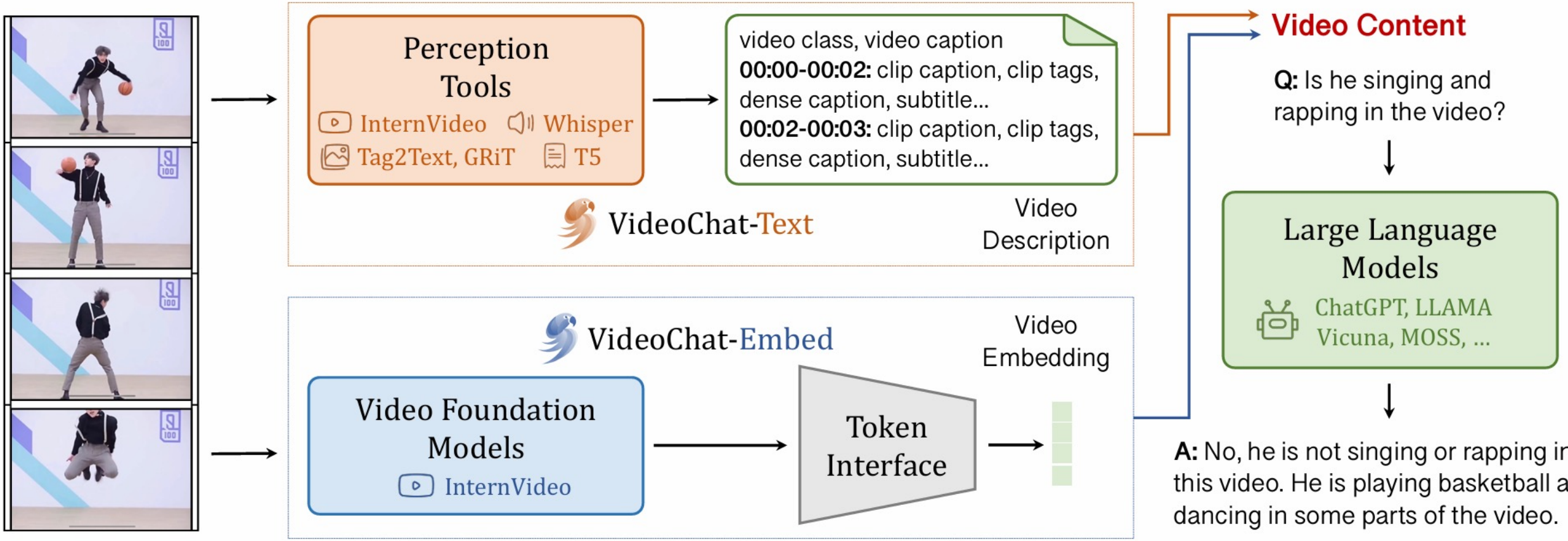


MLLM 视频理解：杯子碎了（从空中掉下来摔碎了）

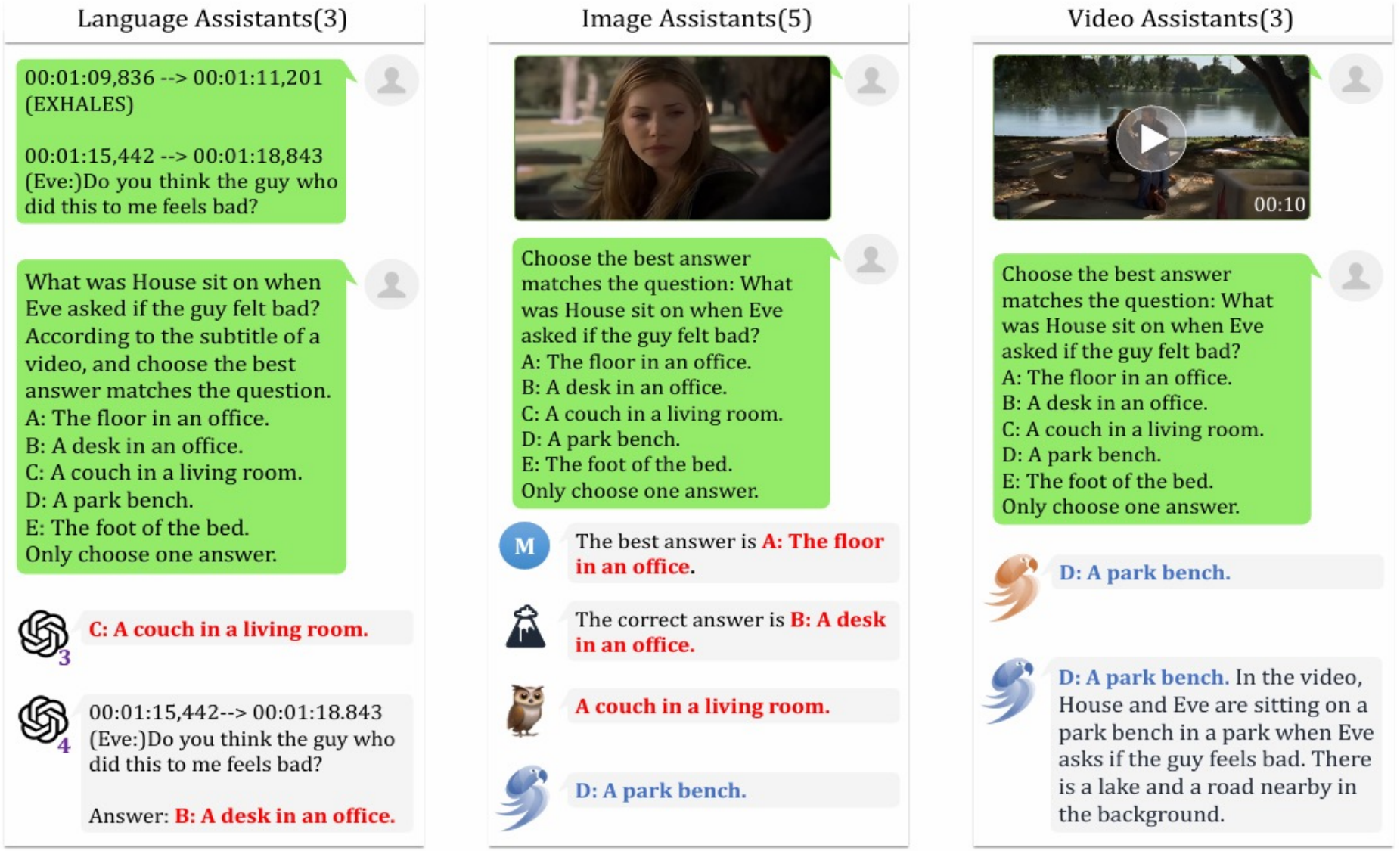
MLLM 是如何“看”视频的?

VideoChat 架构示意图

- **显式文本通道（上）：调用多个专门的感知模型，将视频的音视觉信息翻译成带时间戳的文字描述**
- **隐式特征通道（下）：提取视频深层特征，通过对齐接口 转化为大模型能理解的输入**



MLLM 是如何“看”视频的?



在 TVQA 数据集样本中，字幕被输入给语言助手，图像被输入给图像助手，而视频被输入给 VideoChat

在回答用户问题时，只有 VideoChat 利用了来自视频的信息

MLLM 是如何“看”视频的?



Explain why this meme is funny.

This meme is funny because it shows a small dog laying on the floor with a person's feet in the background. The dog looks very relaxed and content, and the caption reads "just Monday". This implies that the dog is feeling the same way as many people do on Mondays, which is a common feeling of exhaustion and lack of motivation. The meme is relatable and humorous because it plays on the idea that Mondays are often seen as a day that people do not look forward to, and this dog is no exception. It is also funny because it shows a small dog in a very human - like position, which adds to the humor of the meme.



How long is this video?

This video lasts 16 seconds.

Is he singing and rapping in the video?

No, he is not singing or rapping in this video. He is playing basketball and dancing in some parts of the video.

In which second, he starts to dance?

The man starts to dance in the second 9.8. He dances while wearing a brown sweater and black pants, and is wearing a black and white striped shirt.

Where did he take this video?

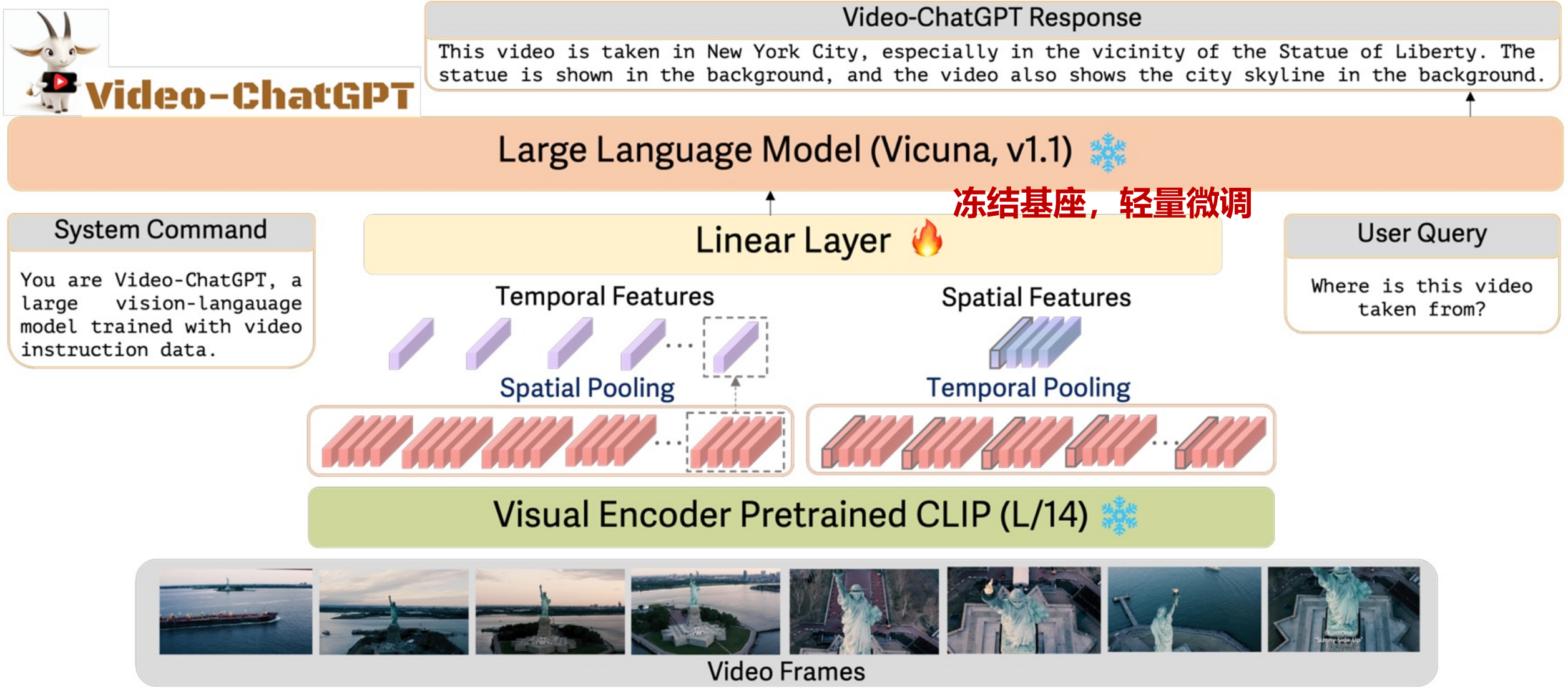
The video was filmed in an indoor gym, on a basketball court. You can see the basketball court in the background of some shots, as well as some large white letters with words painted on the wall. There are also white walls and a grey floor, which suggest that the location is likely an indoor gym.

Please describe whether the camera lens moving.

No, the camera lens does not appear to be moving in this video. The camera remains stationary and captures the actions of the man in the gym.
In some shots, the camera captures the basketball bouncing on the floor, but this is due to the action of the man in the gym rather than any camera movement.
Overall, the camera lens appears to remain fixed throughout the duration of the video.

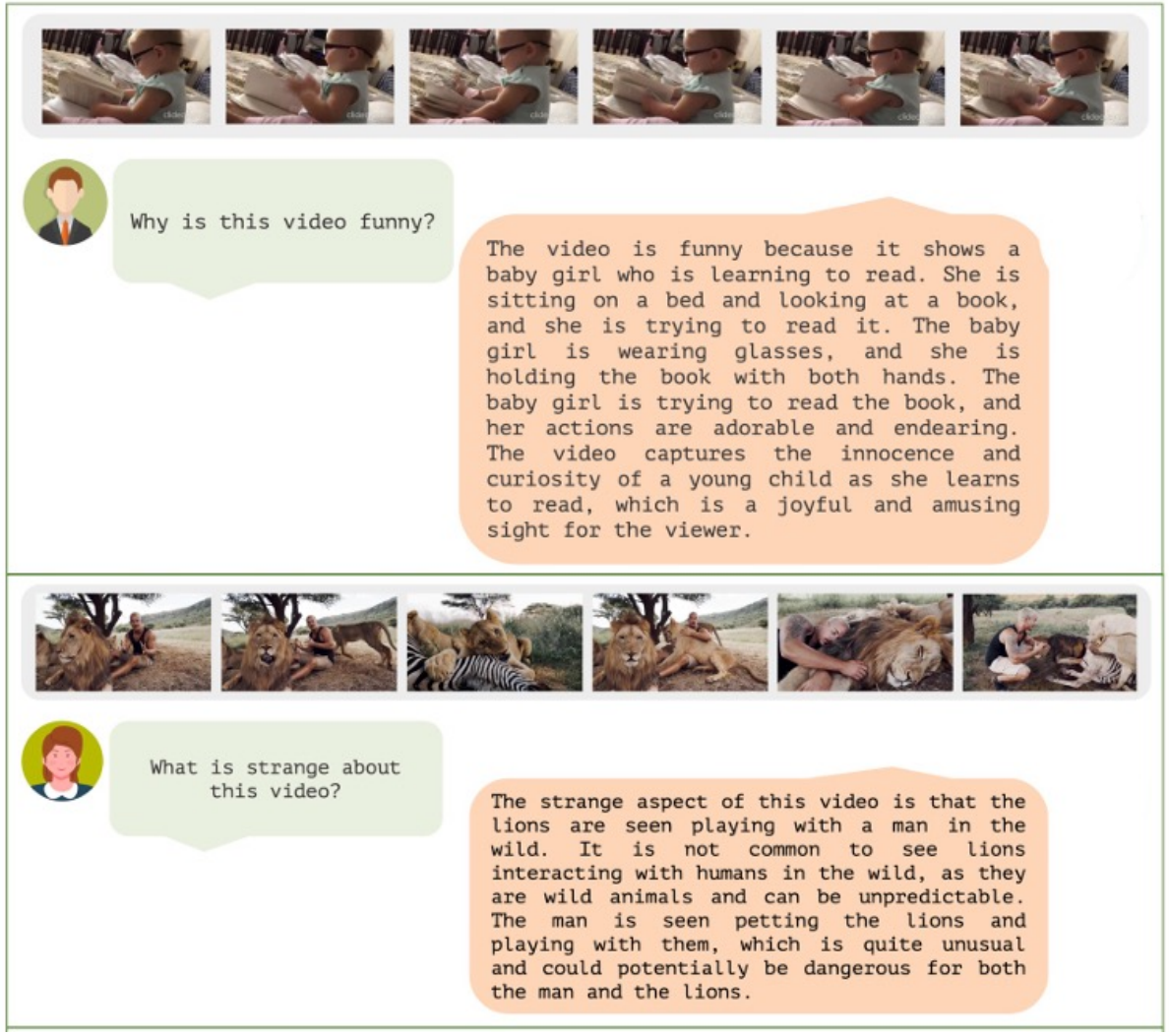
VideoChat在图像理解和视频理解上的示例

MLLM 是如何“看”视频的?



Video-ChatGPT 架构示意图

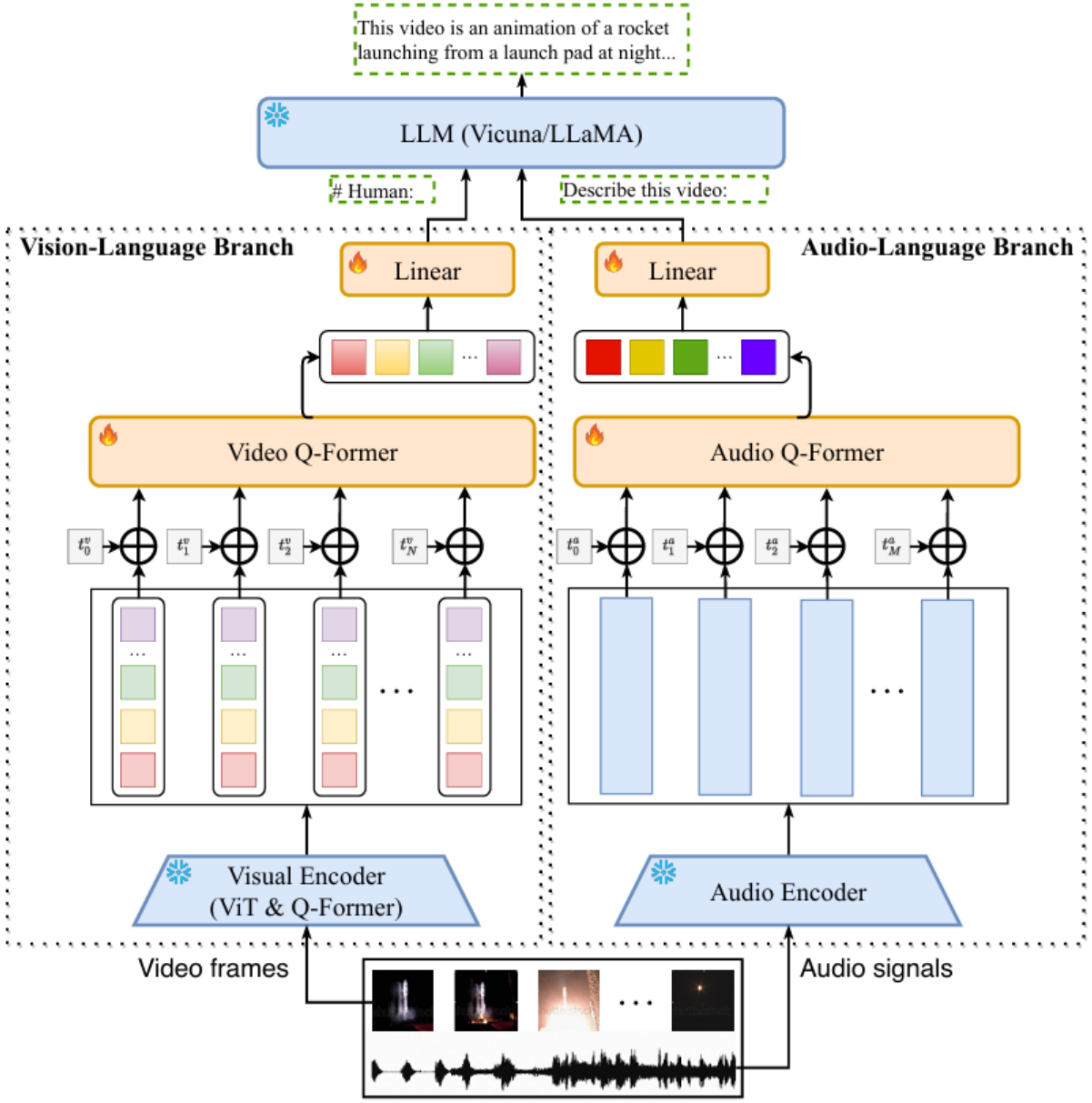
MLLM 是如何“看”视频的?





Video-ChatGPT 执行视觉理解任务

MLLM 是如何“听”视频的?



赋予冻结的大语言模型同时理解视频视觉与听觉内容的能力

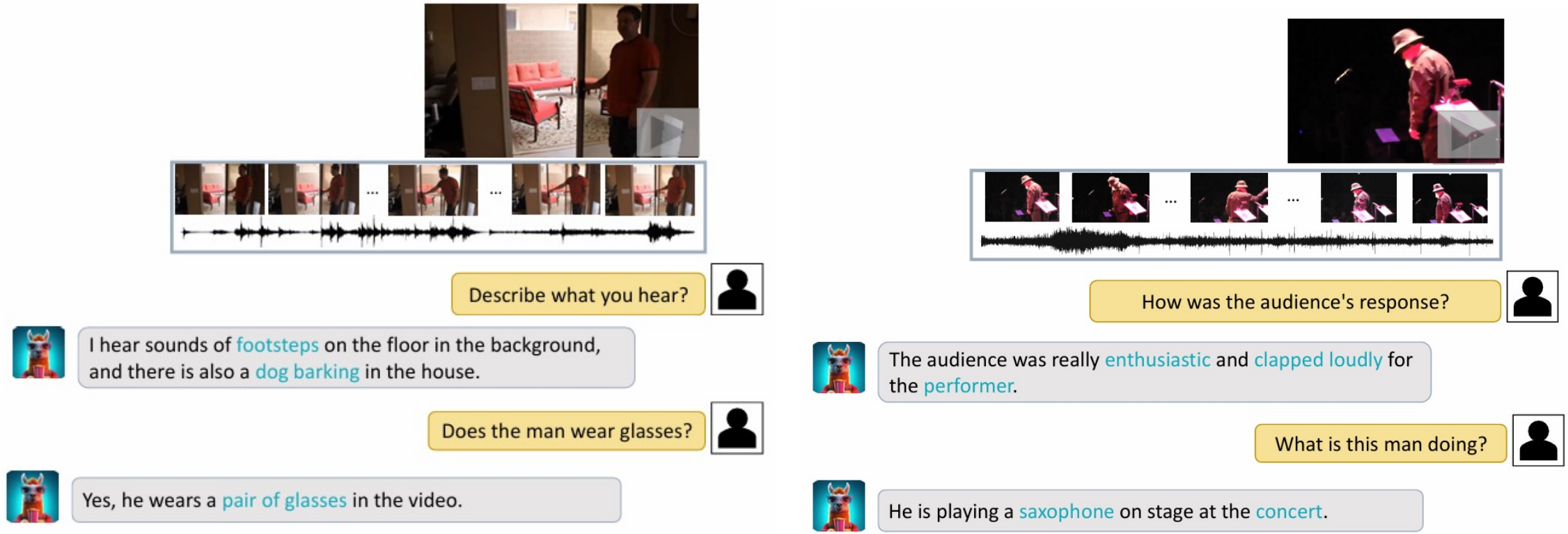
- **视觉-语言分支**：引入 Video Q-Former，专门捕获视频场景中的时序动态变化
- **音频-语言分支**：引入 Audio Q-Former，提取并整合听觉信号

❄️ 冻结：视觉/音频编码器、顶层 LLM，保留预训练知识

🔥 可训练：Q-Former 及 Linear 投影层，实现跨模态特征对齐至 LLM 语义空间

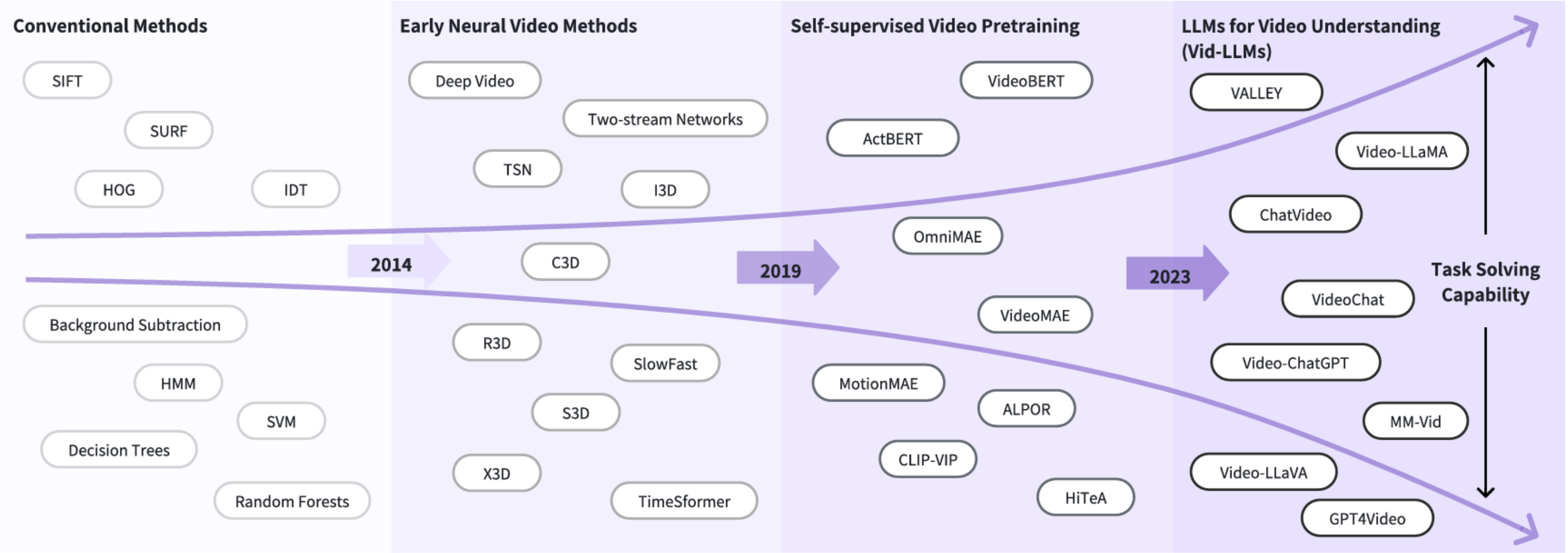
Video-LLaMA 框架图

MLLM 是如何“听”视频的?



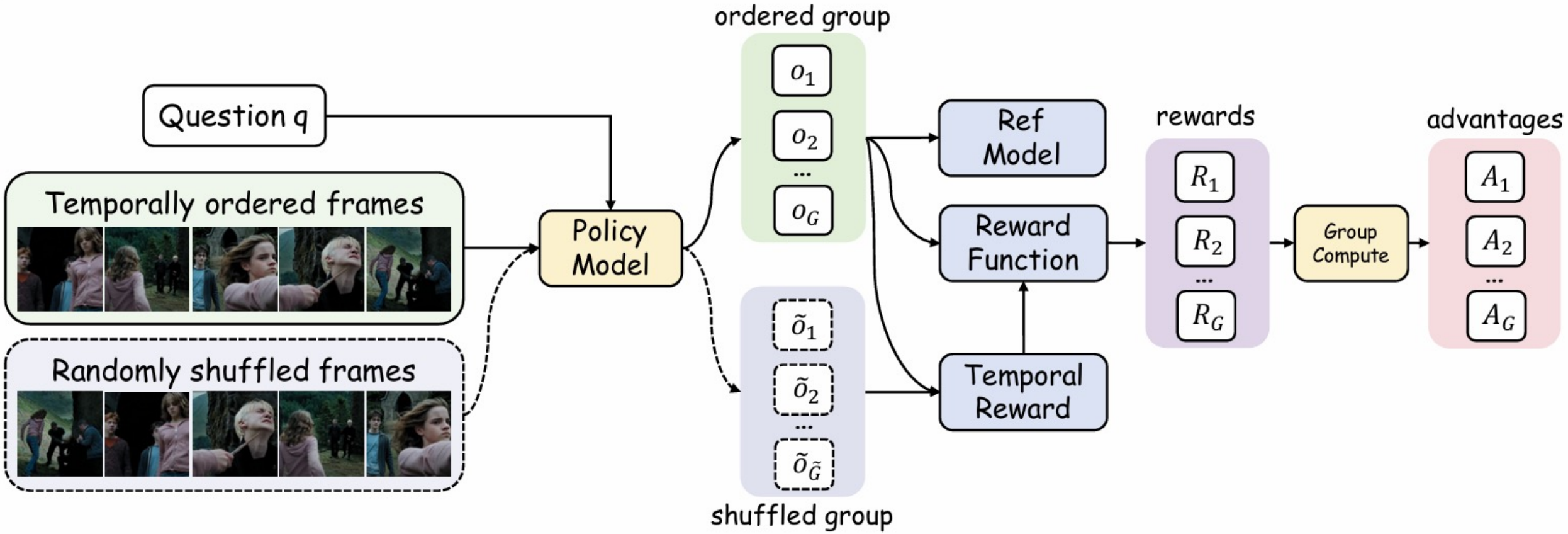
Video-LLaMA “听”视频内容的示例

演进脉络：视频理解的“进化史”



- **传统方法 (2014前)：** 依赖人工设计特征（如 SIFT/HOG）与传统机器学习算法
- **早期神经网络 (2014-2019)：** 引入 3D 卷积与双流网络，实现视觉特征的自动提取
- **自监督预训练 (2019-2023)：** 基于 Transformer 架构，利用海量无标注数据学习通用表征
- **视频大模型 (2023 至今)：** 融合 LLM，实现跨模态时空推理与开放域任务求解

多模态视频理解前沿：Video-R1



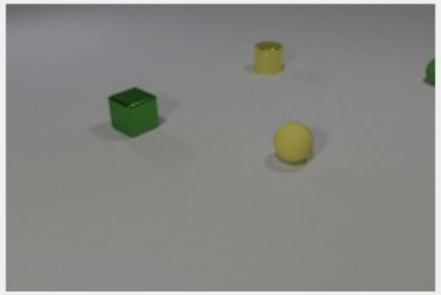
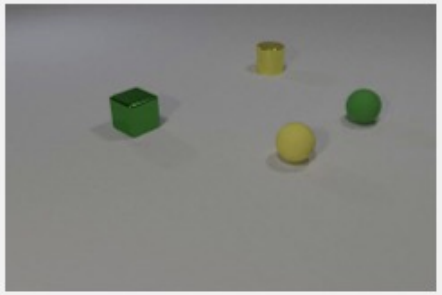
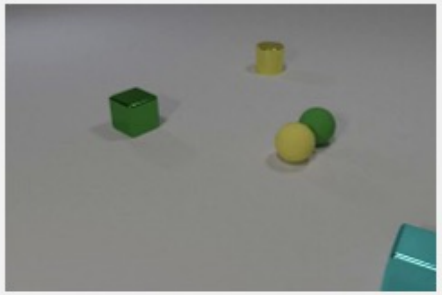
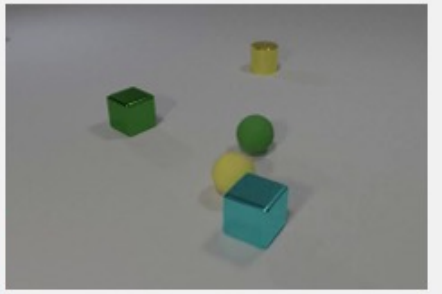
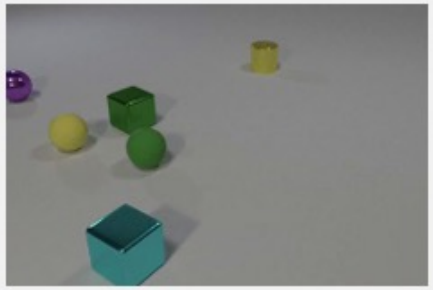
Video-R1 中的核心创新点： T-GRPO 算法

如何逼迫模型看懂时间线？研究人员让模型同时看两个版本的视频：一个是**正常按时间播放**的，一个是**帧顺序被打乱**的

- **奖励机制**：只有当模型在看“正常视频”时的答题正确率 **大于** 看“打乱视频”的正确率时，系统才给予正向奖励
- **效果**：彻底杜绝了模型单凭一张静态截图就“蒙混过关”的走捷径行为，强制其建立真正的**时序逻辑推理**

多模态视频理解前沿：Video-R1

Video:

Question: Are there any cyan cubes that enter the scene? A. not sure B.no C.yes

GRPO: <think>Let me think about this carefully……The question specifically asks about cyan cubes. I see a cyan cube in the scene (shortcut)……it was already present. Hmm, this is a bit tricky. The question is asking if any cyan cubes enter the scene, but the cyan cube was already there……. break it down one more time to be sure: No new cyan cubes enter the scene……</think>\n<answer>B</answer>

T-GRPO: <think>Let me think about this carefully……Now, let's focus on the cyan cube ……it is not there at the beginning of the scene. It appears to be moving into the scene from the bottom right corner. This indicates that the cyan cube is indeed entering the scene. Hmm, it seems that the cyan cube is a new object that appears in the scene. It is not present in the initial state……but it is clearly visible in the final state…… This confirms that the cyan cube enters the scene……After this thorough analysis, I am confident that the answer is yes……</think>\n<answer>C</answer>

Video-R1 的 T-GRPO 与 GRPO 对比示例

多模态视频理解前沿：Video-R1

Video: 

Question: Which move motion in the video lose the system energy?

Video-R1-7B: <think>Let me think about this for a moment. The question is asking about which move in the video loses kinetic energy. To answer this, I need to carefully analyze the video and identify the different actions and their outcomes.

First, let's break down the video content. The video shows a series of actions involving a pool table, a bowling alley, and a car crash. In the pool table scene, a player takes a shot, and the balls move around. In the bowling alley scene, a ball rolls down the lane and hits the pins. In the car crash scene, two cars collide.

Now, let's consider the energy loss in each scene:
1. In the pool table scene, the player takes a shot, and the balls move. The kinetic energy is transferred from the cue stick to the balls, and the balls continue to move. This does not seem to lose kinetic energy.
2. In the bowling alley scene, the ball rolls down the lane and hits the pins. The ball loses some kinetic energy upon hitting the pins, but it still has some energy left as it rolls away.
3. In the car crash scene, the two cars collide. The kinetic energy is transferred between the cars, and some energy is lost as heat and deformation of the cars. This is a significant loss of kinetic energy.
Hmm, it seems that the car crash scene is the one where the most kinetic energy is lost. The cars collide, and the energy is dissipated in the form of heat and deformation, which is a significant loss compared to the other scenes. (Perception and Reasoning)

Oh, I see. The car crash scene is the one where the most kinetic energy is lost. The other scenes involve transfers of energy, but the car crash involves a significant loss.

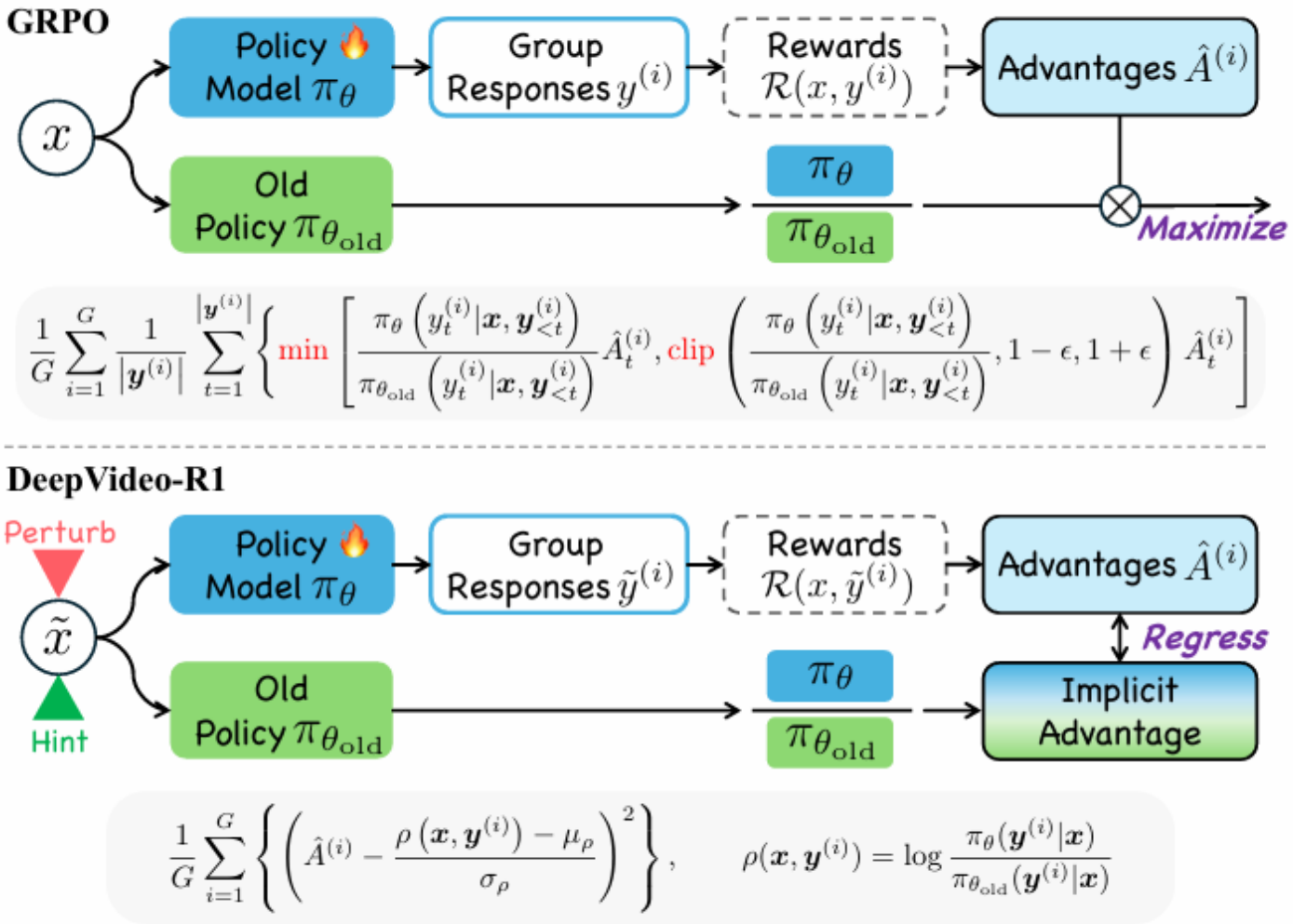
Let's verify this. The car crash scene is indeed the one where the most kinetic energy is lost, as the cars collide and the energy is dissipated in a dramatic way. (Reflection)

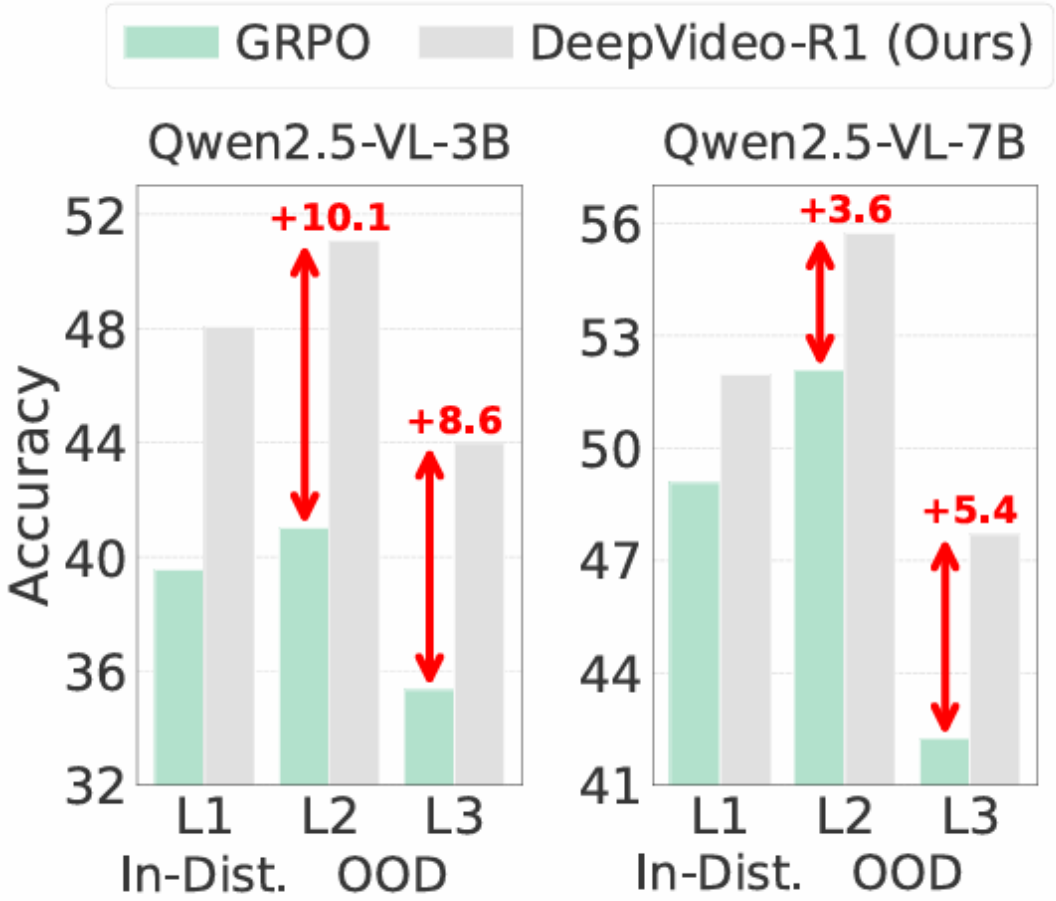
So, the move that loses the most kinetic energy in the video is the car crash.</think>

<answer>The car crash in the video loses the system energy.</answer>

Video-R1-7B 在 MMVU 数据集上的推理示例

多模态视频理解前沿：DeepVideo-R1





DeepVideo-R1显著提升了视频大语言模型的推理能力

- DeepVideo-R1 通过回归 GRPO 损失进行训练，以显式预测优势值 $\widehat{A^{(i)}}$
- 模型训练变得极其高效，并且与传统的 GRPO 相比，实现了 10.1 的性能提升

多模态视频理解前沿：DeepVideo-R1



Question: Considering the progress shown in the video and my current observation shown in the image, what action should I take next in order to prepare berries?

A. move tray B. close tray C. open container **D. move berries**

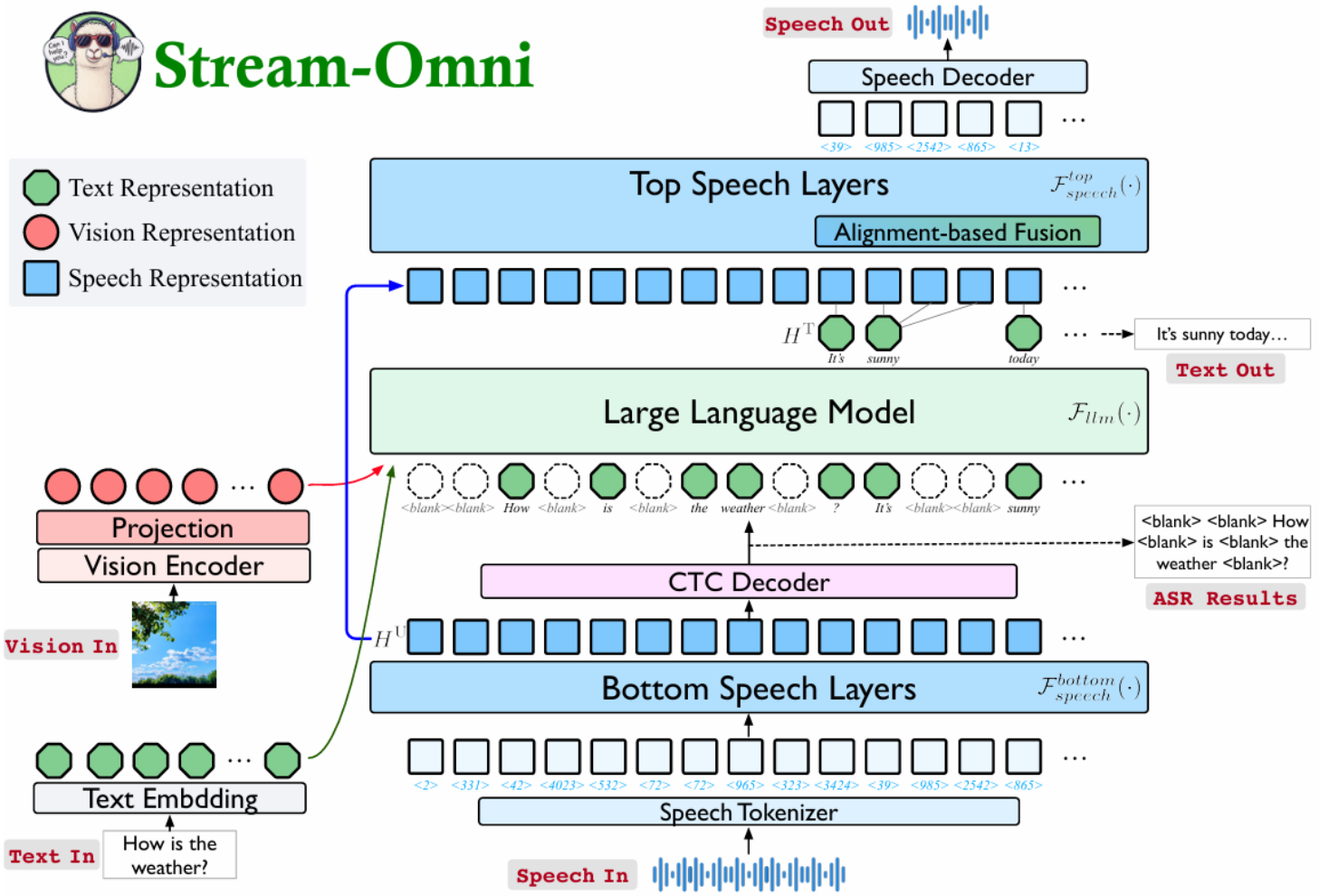
Output the thinking process in <think> </think> and final answer in <answer> </answer> tags, i.e., <think> reasoning process here </think><answer> answer here </answer>.

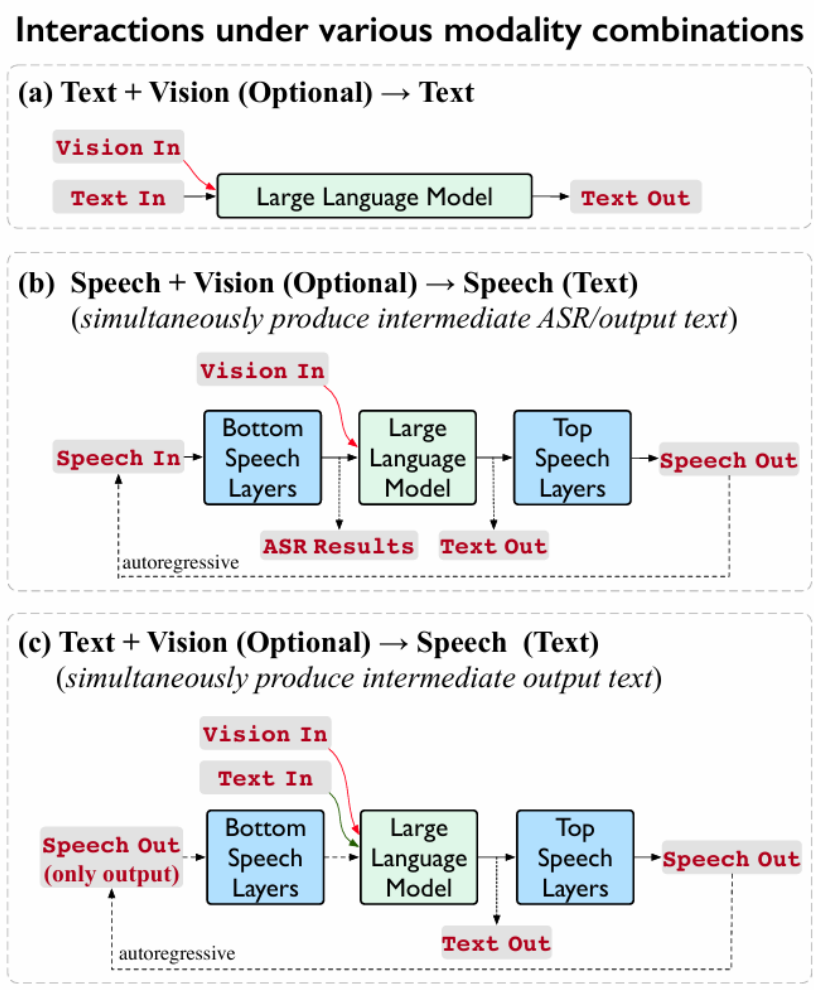
GRPO: <think> The image shows a person preparing food, specifically making a dish with vegetables and cheese on a flatbread. **There are no berries or trays visible in the image**, so it is not possible to determine the next steps for preparing berries. The options provided are related to trays and containers, which are not relevant to the current task of preparing berries. </think> <answer> **C. open container** </answer>

DeepVideo-R1(Ours): <think> The person in the video is currently preparing a dish with berries. The next logical step would be **to add more berries to the dish or to arrange the existing berries in a specific way**. Since the image shows the person already handling the berries, the next action should be to **add more berries or to arrange the existing ones.** </think> <answer> **D. move berries**</answer>

DeepVideo-R1-7B 与 Qwen2.5-VL-7B+GRPO 的定性结果对比

多模态视频理解前沿：Stream-Omni





Stream-Omni 的架构 以及 各种模态组合下的交互

多模态视频理解前沿：Stream-Omni



Question: Does the stairway from the basement allow traveling to the second floor without moving through another room first?

Speech:

Vision + Text → Text

GPT-4V: Yes, the stairway from the basement allows traveling to the second floor without moving through another room first, as it is in the same position in the plans for all three levels.

VITA-1.5: Based on the image, the stairway from the basement does not allow traveling to the second floor without moving through another room first. The stairway is shown as a direct vertical connection between the basement and the second floor, but there are no rooms depicted between these two levels.

Stream-Omni: Based on the image, the stairway from the basement leads directly to the second floor, bypassing the first floor. This suggests that one can travel from the basement to the second floor without moving through another room first.

mismatch

Vision + Speech → Text

VITA-1.5: Yes, the staircase from the basement leads directly to the second floor without requiring passage through another room.

Stream-Omni: Yes, the stairway from the basement leads directly to the second floor, allowing for travel between the basement and the second floor without passing through another room first.

Vision + Speech → Speech

Stream-Omni:

10s speech

*(**Whisper Transcription**: Yes, the stairway from the basement leads directly to the second floor, allowing for travel between the basement and the second floor without passing through another room.)*



Question: How does this device help you decide where to hang a picture?

Speech:

Vision + Text → Text

GPT-4V: This device helps you decide where to hang a picture by locating the wall studs behind the wallpaper, ensuring a secure and stable spot to anchor the picture frame.

VITA-1.5: This device, known as a stud finder, is designed to detect the presence of wall studs behind the surface. These studs are crucial for hanging heavy items like pictures because they provide structural support. By using this tool, you can identify the best spots to hang your picture frame where it will be secure and less likely to fall off the wall.

Stream-Omni: The device in the image is a stud finder, which is used to locate the studs behind a wall. To decide where to hang a picture, you would use this device to scan the wall at the spot where you want to hang the picture. The stud finder will detect the presence of a stud, indicated by a beep or a signal, and mark the location. This helps ensure that the picture frame is hung on a sturdy part of the wall, providing support and reducing the risk of the frame falling or the wall damaging.

Vision + Speech → Text

VITA-1.5: This device, known as a spirit level, helps ensure that pictures and other items are hung straight by providing visual feedback on the horizontal or vertical alignment. It's particularly useful for achieving precise and aesthetically pleasing results when hanging artwork or mirrors.

Stream-Omni: The device in the image is a stud finder, which helps in determining the location of wooden studs behind a wall. This is useful when hanging a picture because you want to find a spot where the picture will be securely supported by a stud. If you hang a picture on a spot without a stud, it may not be well-supported and could lead to the picture falling. By using a stud finder, you can locate a spot where a stud is present, ensuring that the picture will be securely mounted and safe from falling.

Vision + Speech → Speech

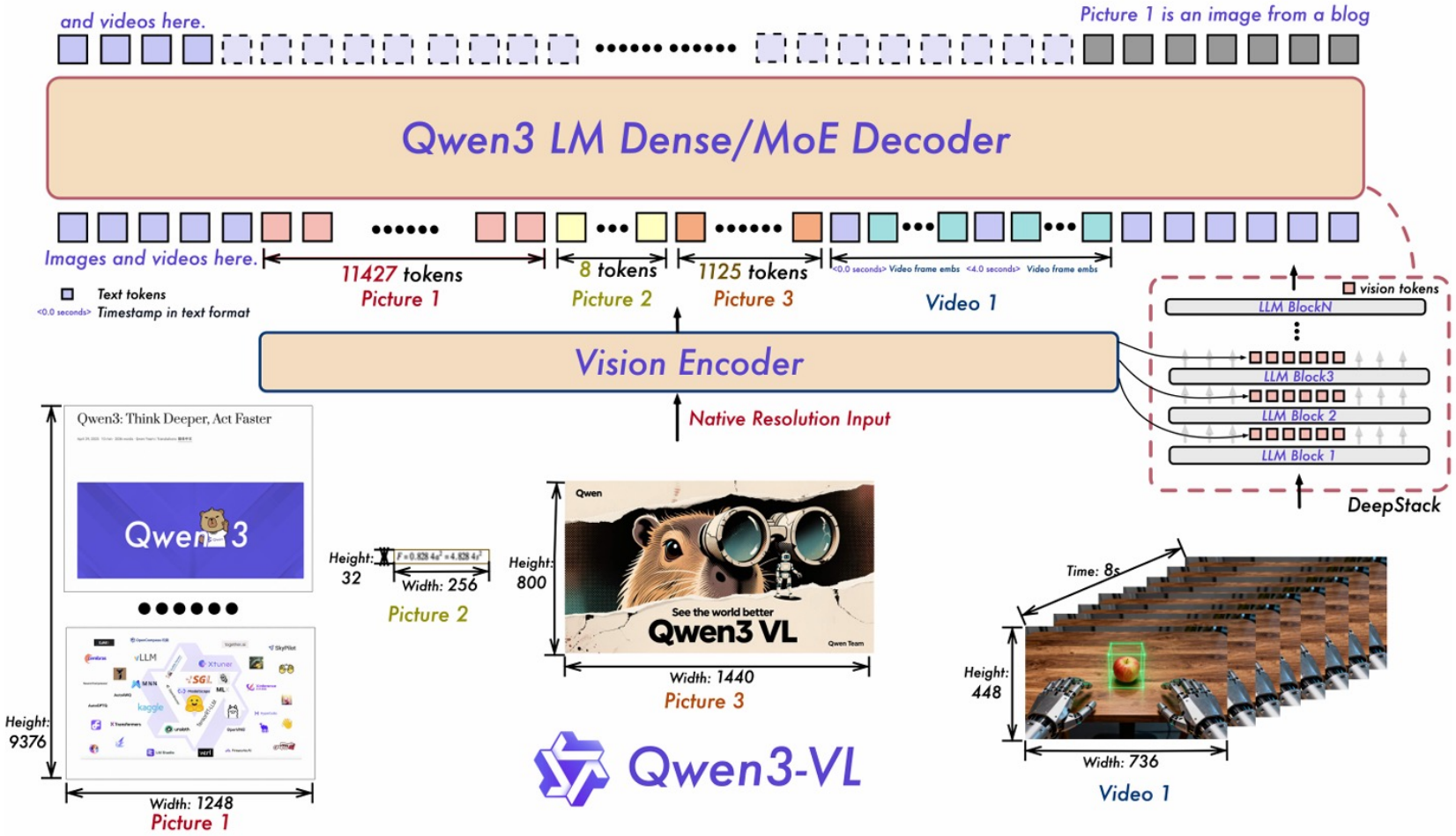
Stream-Omni:

30s speech

*(**Whisper Transcription**: The device in the image is a stud finder which helps to determining the location of the wooden studs behind a wall. This is useful when hanging picture because you want to find a spot where the picture will be securely supported by a stud. If you hang a picture on a spot without a stud, it may not be well-being and could lead to the picture falling. By using a stud finder, you can locate a spot where a stud is present, ensuring that the picture will be securely mounted and safe from falling.)*

Stream-Omni 在视觉细节理解（左）和长语音回复（右）场景下的示例

多模态视频理解近期SOTA：Qwen3-VL



		Qwen3-VL 235B-A22B Instruct	Gemini2.5-Pro Thinkingbudget 128	GPT5 Minimal or Without thinking	Claude-Opus-4.1 Without thinking	Other Best Without thinking
STEM&Puzzle	$MMMU_{VAL}$	**78.7**	80.9	74.4*	77.2	73.6* (Seed1.5VL)
	MMMU_Pro	**68.1**	71.2	62.7*	60.7	59.9* (Seed1.5VL)
	$MathVista_{mini}$	**84.9**	77.7	50.9	74.5	83.0* (Seed1.5VL)
	MathVision	**66.5**	66.0	45.8	57.7	65.5* (Seed1.5VL)
	$MathVerse_{mini}$	**72.5**	65.9	43.0	68.1	65.4* (GLM4.5V)
	VisuLogic	**29.9**	26.9	27.2	27.2	33.0* (Seed1.5VL)
General VQA	$MMBench_EN_V1.1_{dev}$	**89.9**	86.6	81.4	84.1	89.0* (InternVL3)
	RealWorldQA	**79.3**	76.0	77.3	68.5	78.0* (InternVL3)
	MMStar	**78.4**	78.5	65.2	71.0	76.2* (Seed1.5VL)
	SimpleVQA	**63.0**	66.9	56.7	55.7	63.1* (Seed1.5VL)
Subjective Experience and Instruction Following	HallusionBench	**63.2**	60.9	53.7	55.1	60.0* (Seed1.5VL)
	MM_MT_Bench	**8.5**	7.6	7.5	7.9	7.6* (Qwen2.5VL)
	MIABench	**91.3**	91.3	92.6	90.0	90.8 (Seed1.5VL)
Text Recognition and Chart/Document Understanding	MMLongBench-Doc	**57.0**	51.2	42.4	48.1	50.1 (Seed1.5VL)
	$DocVQA_{TEST}$	**97.1**	94.0	89.6	89.2	96.7* (Seed1.5VL)
	$InfoVQA_{TEST}$	**89.2**	82.9	69.9	60.9	87.3* (Qwen2.5VL)
	$AI2D_{TEST}$	**89.7**	90.0	84.1	84.4	89.7* (InternVL3)
	OCRBench	**920.0**	872.0	787.0	750.0	906* (InternVL3)
	$OCRBenchV2_{(en/ch)}$	**67.1 / 61.8**	55.2 / 53.1	48.2 / 37.7	47.2 / 38.0	61.5 / 63.7* (Qwen2.5VL)
	CC_OCR	**82.2**	76.8	66.1	66.0	79.8* (Qwen2.5VL)
	CharXiv(RQ)	**62.1**	62.9	57.8*	60.2	59.8* (Seed1.5VL)
2D/3D Grounding	$RefCOCO_{avg}$	**91.9**	–	–	–	91.6* (Seed1.5VL)
	CountBench	**93.0**	91.0	87.8	91.9	93.6* (Qwen2.5VL)
	ODinW13	**48.6**	34.5	–	–	40.6 (Seed1.5VL)
	ARKitScenes	**56.9**	–	–	–	27.5 (Seed1.5VL)
	Hypersim	**13.0**	–	–	–	9.6 (Seed1.5VL)
	SUNRGBD	**39.4**	–	–	–	33.5* (Seed1.5VL)
Multi-Image	BLINK	**70.7**	70.0	62.8	62.9	70.2* (Seed1.5VL)
	MUIRBENCH	**72.8**	74.0	66.5	–	71.1* (GLM4.5V)
Embodied and Spatial Understanding	ERQA	**51.3**	50.3	42.0*	26.0	46.5* (GLM4.5V)
	VSI-Bench	**62.6**	31.4	–	–	48.4* (InternVL3)
	EmbSpatialBench	**83.1**	73.3	75.1	66.0	78.6* (RoboBrain 2.0)
	RefSpatialBench	**65.5**	35.4	23.1	–	54.0* (RoboBrain 2.0)
	RoboSpatialHome	**69.5**	49.2	43.6	–	72.4* (RoboBrain 2.0)
Video	VideoMME(w/o sub)	**79.2**	80.6	77.3	73.3	77.6* (Seed1.5VL)
	MLVU	**84.3**	81.2	78.3	71.2	81.8* (Seed1.5VL)
	LVBench	**67.7**	69.0	–	–	64.0* (Seed1.5VL)
	CharadesSTA	**64.8**	–	–	–	64.7* (Seed1.5VL)
	VideoMMMU	**74.7**	79.4	61.6*	70.1	72.1* (Seed1.5VL)
Agent	ScreenSpot	**95.4**	–	–	–	88.7* (InternVL3)
	ScreenSpot Pro	**62.0**	–	–	–	60.9* (Seed1.5VL)
	OSWorldG	**66.7**	–	–	–	53.2* (InternVL3)
	AndroidWorld	**63.7**	–	–	–	62.1* (Seed1.5VL)
Coding	Design2Code	**92.0**	90.3	88.9	85.3	84.5* (GLM4.5V)
	ChartMimic_v2_Direct	**80.5**	79.9*	–	–	–
	UniSvg	**69.3**	67.9	–	–	–

Note: The default evaluation is conducted through API calls, and * indicates scores from the report. Results on video understanding are measured using a 256k-token context, handling up to 2048 frames.

Qwen3-VL 架构以及性能

- 原生高分辨率与超长上下文
- 细粒度图像与空间感知
- 精准的长视频时空推理
- 全面对标顶尖闭源大模型

课程提纲

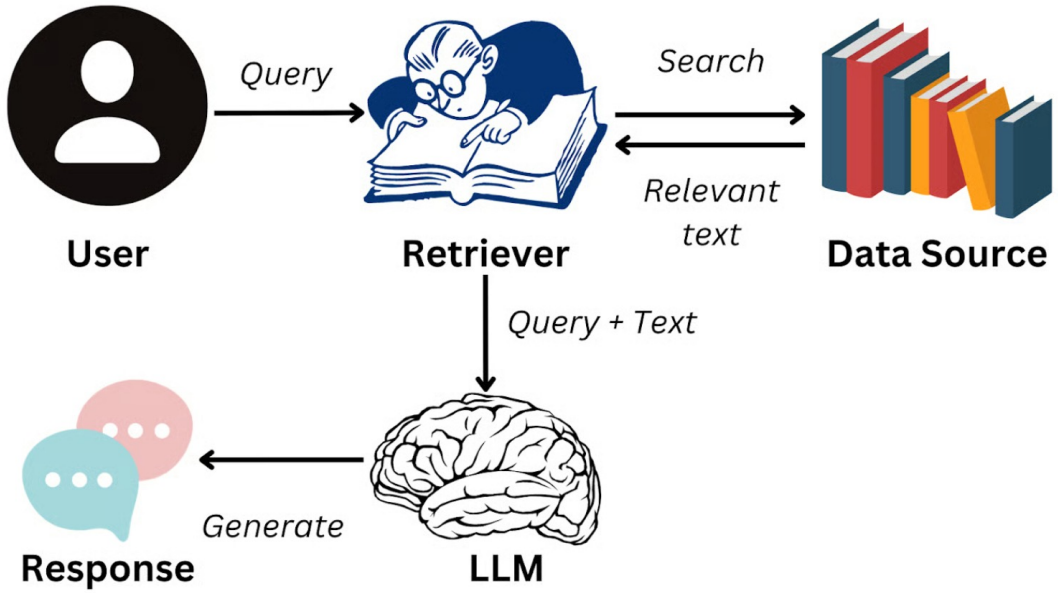
CONTENTS

一、 LLM/VLM/MLLM
二、多模态思维链推理
三、多模态视频理解
四、检索增强生成
五、总结

RAG 是什么?

RAG是什么?

RAG（Retrieval-Augmented Generation，检索增强生成）是当前人工智能领域，尤其是 LLM 应用中最重要、最流行的架构范式之一，它巧妙地结合了**信息检索和文本生成**的优势，旨在解决大模型固有的痛点



一个简单的比喻：

如果把大语言模型比作一个博学但记忆可能模糊、知识可能过时的“大师”，那么 RAG 就是给这位大师配了一个超级图书管理员，当用户提出问题时：

这位图书管理员**（检索器）**会飞速跑进图书馆**（知识库）**，找到最相关的几本参考书**（相关文档片段）**

然后把这几页关键内容递给大师**（生成器）**

大师基于这些最新的、准确的参考内容，组织语言，给出一个高质量的答案

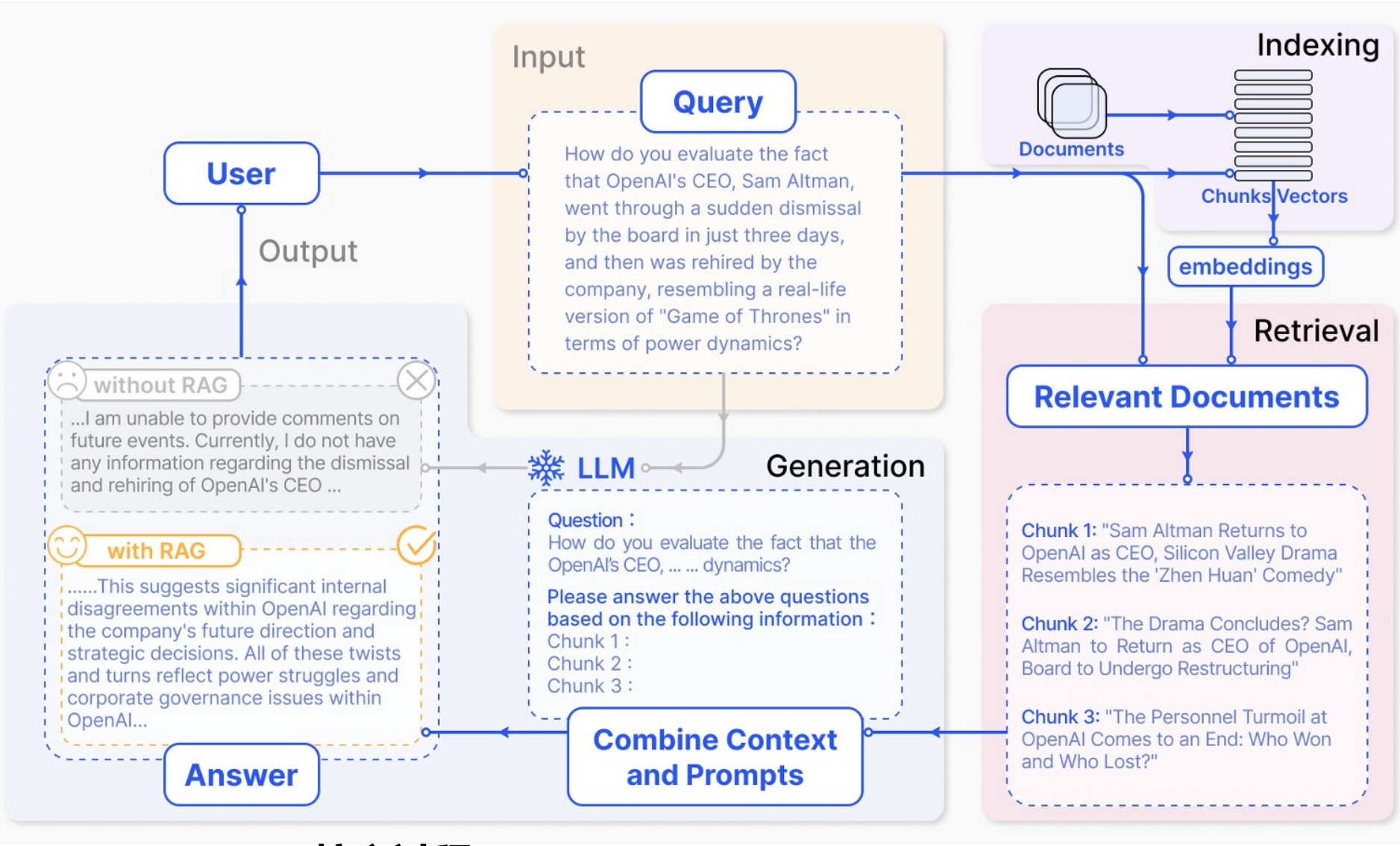
为什么需要 RAG?





涉及世界知识的文本到图像生成

RAG 基础与概念引入



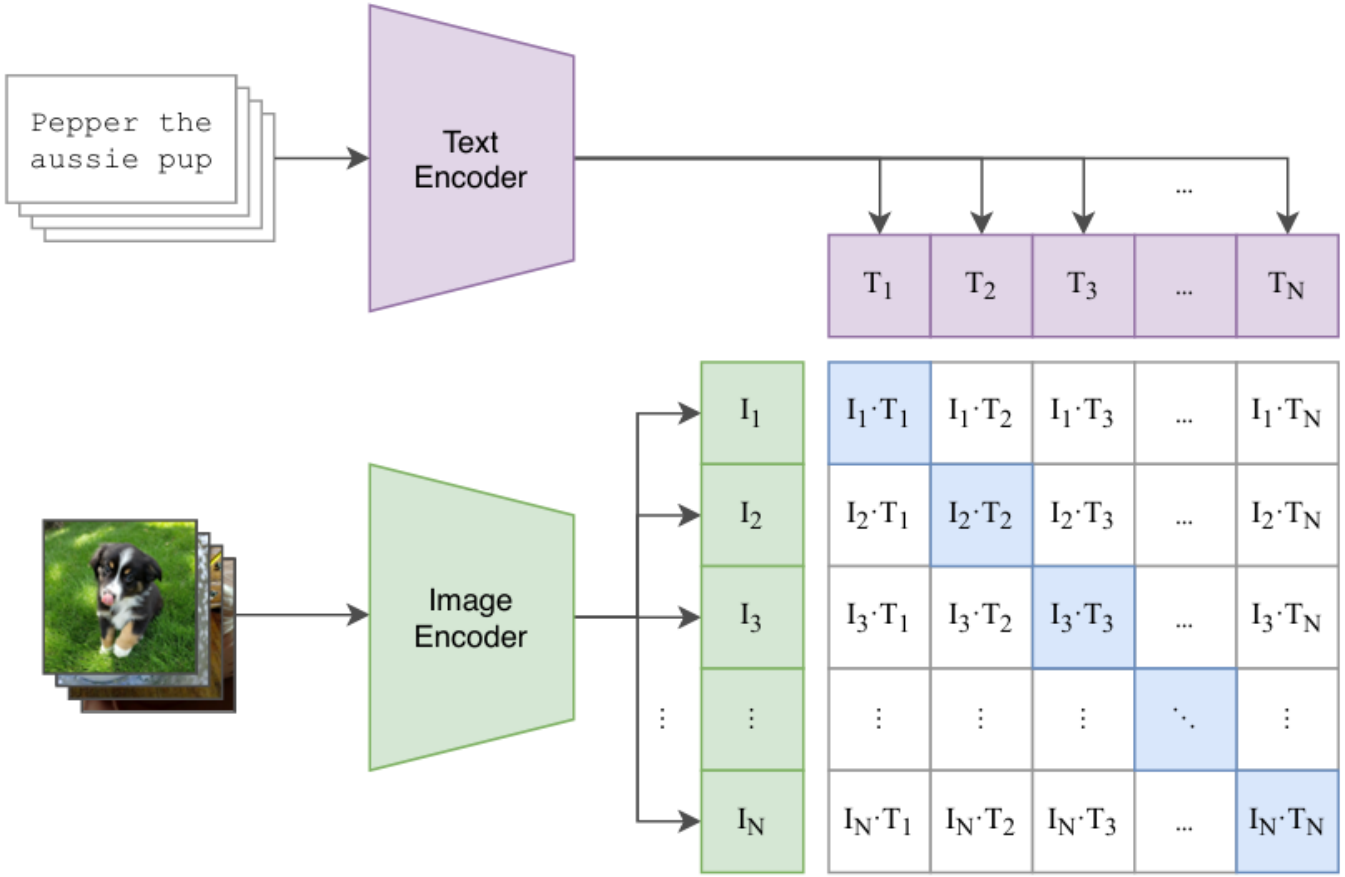
RAG 核心过程： Indexing, Retrieval, Generation

1) 索引（Indexing）将文档分割成块（chunks），编码为向量，并存储在**向量数据库**中

2) 检索（Retrieval）基于**语义相似度**，检索出与问题最相关的前 k 个块

3) 生成（Generation）将原始问题与检索到的块一起输入到**大语言模型**中，以生成最终答案

向量数据库：Embedding Space



什么是 Embedding?

图中的 $I_1, I_2 \ldots$ 和 $T_1, T_2 \ldots$ 就是向量（Embeddings）

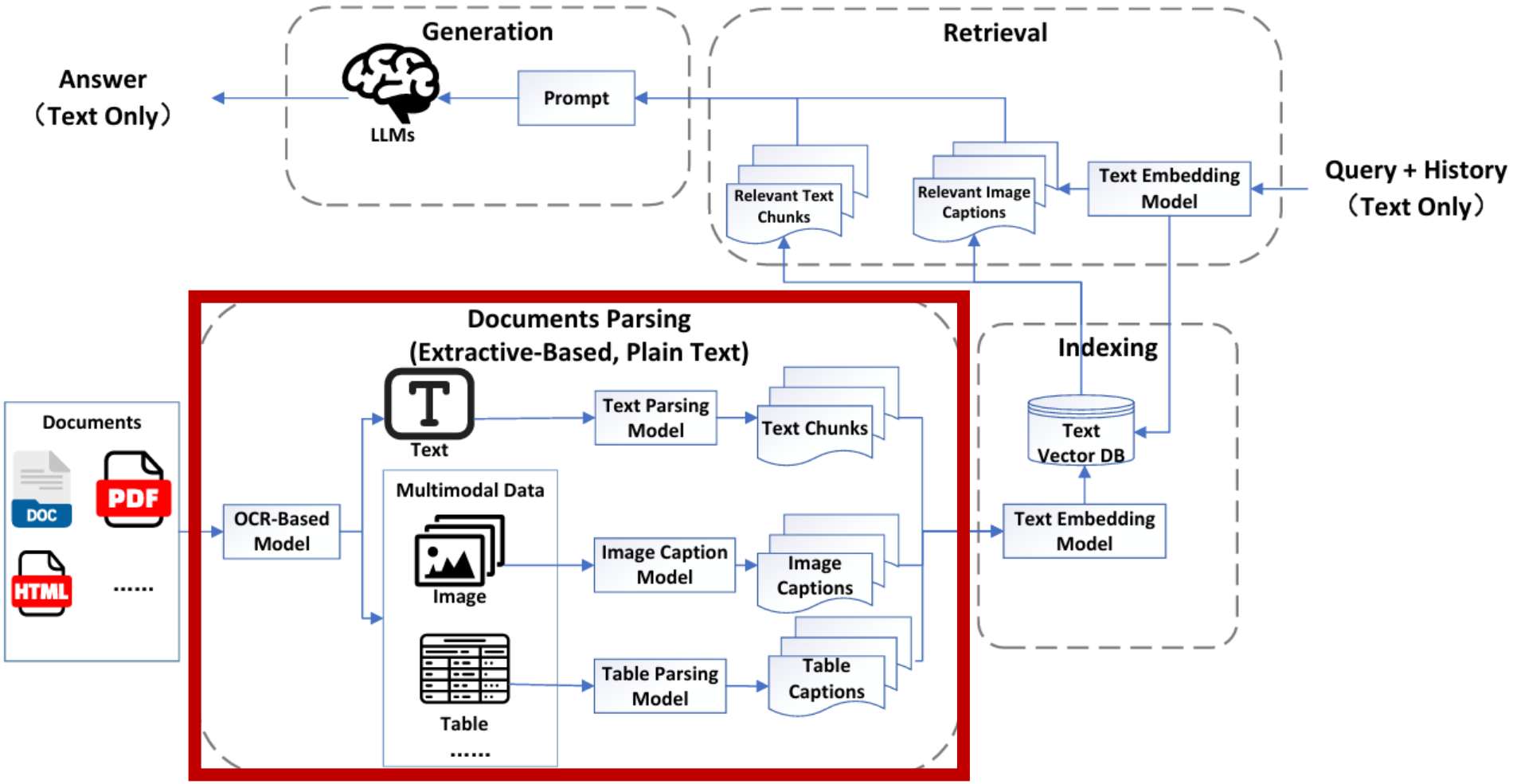
什么是向量距离?

在这个空间里，“模式识别”变成了简单的“算距离”

距离越近，相似度越高，说明文本和图像在语义模式上越匹配

向量数据库的核心工作，就是在海量数据中快速寻找这种距离最近的向量对

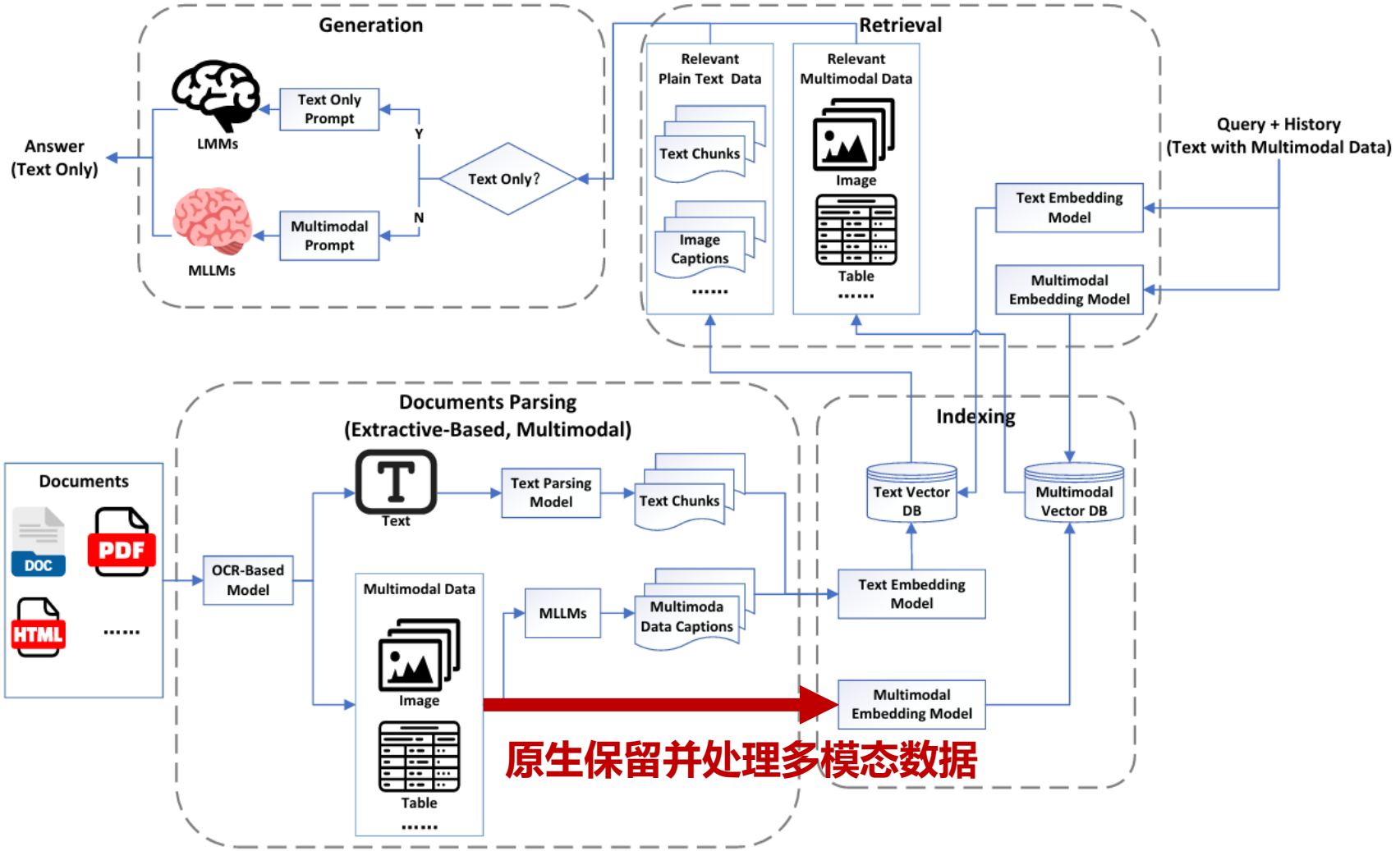
从纯文本RAG到多模态RAG （MRAG）



把所有非文本的模态，都“翻译”成了纯文本

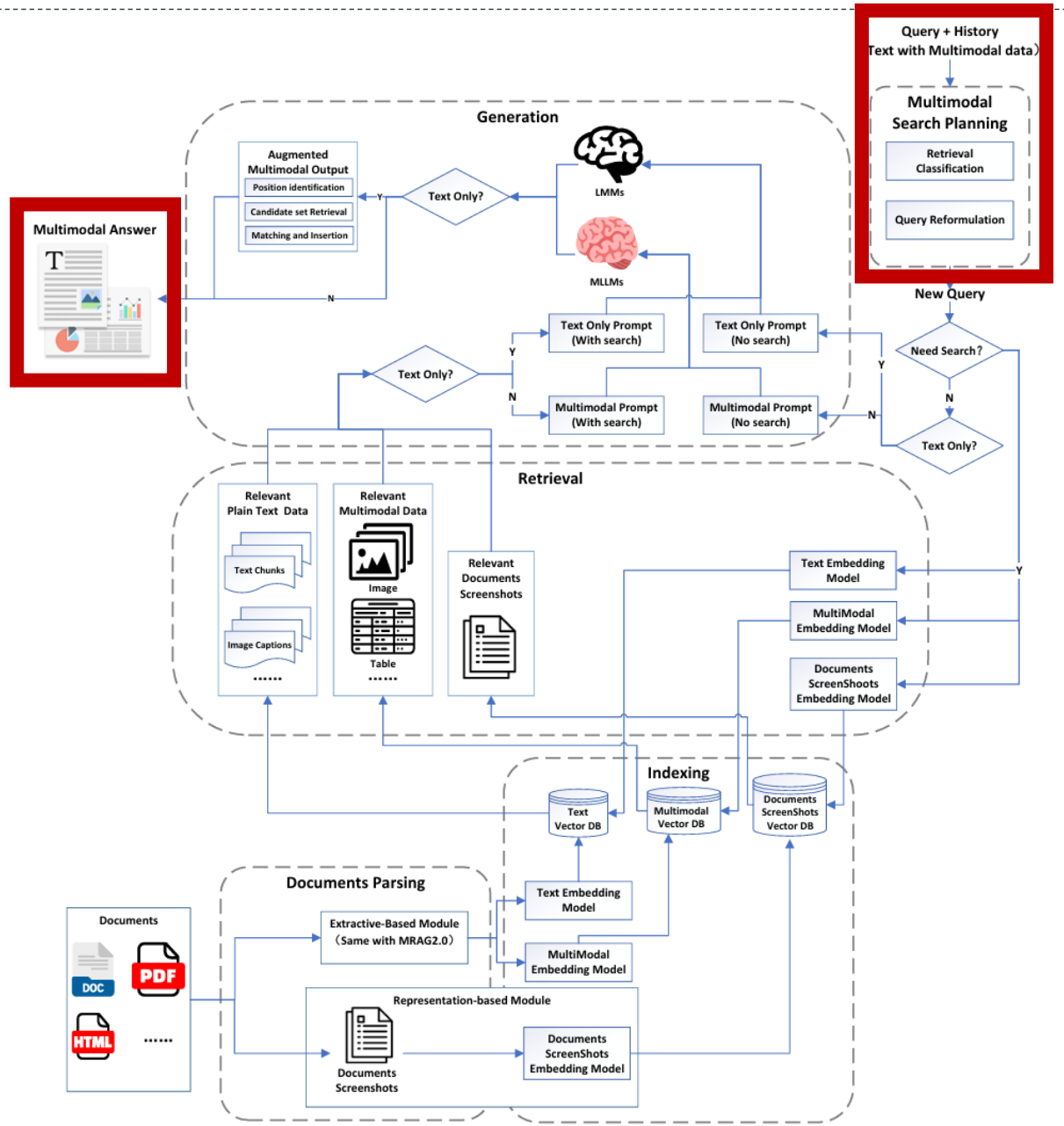
MRAG 1.0 架构

从纯文本RAG到多模态RAG（MRAG）



MRAG 2.0 架构

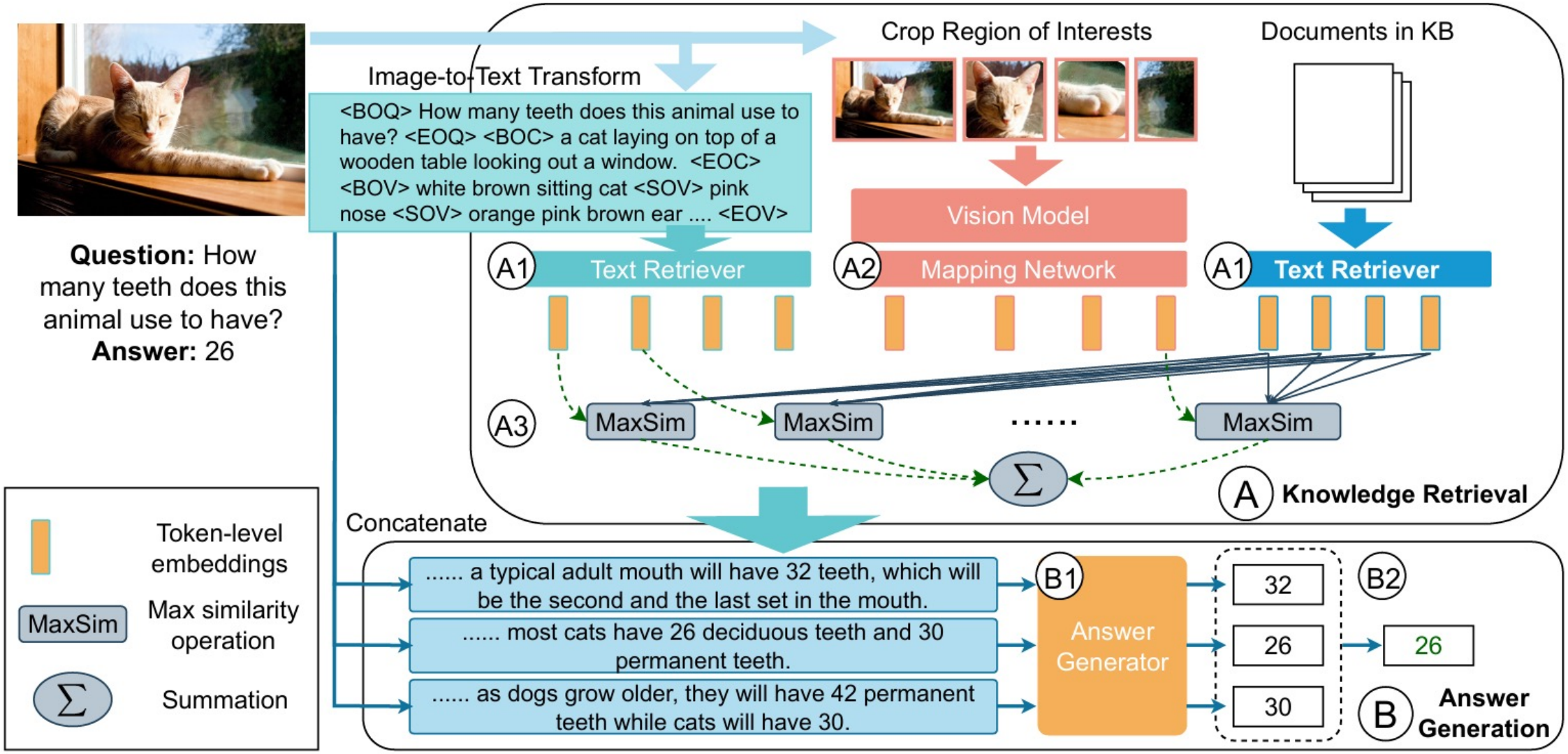
从纯文本RAG到多模态RAG（MRAG）



端到端多模态

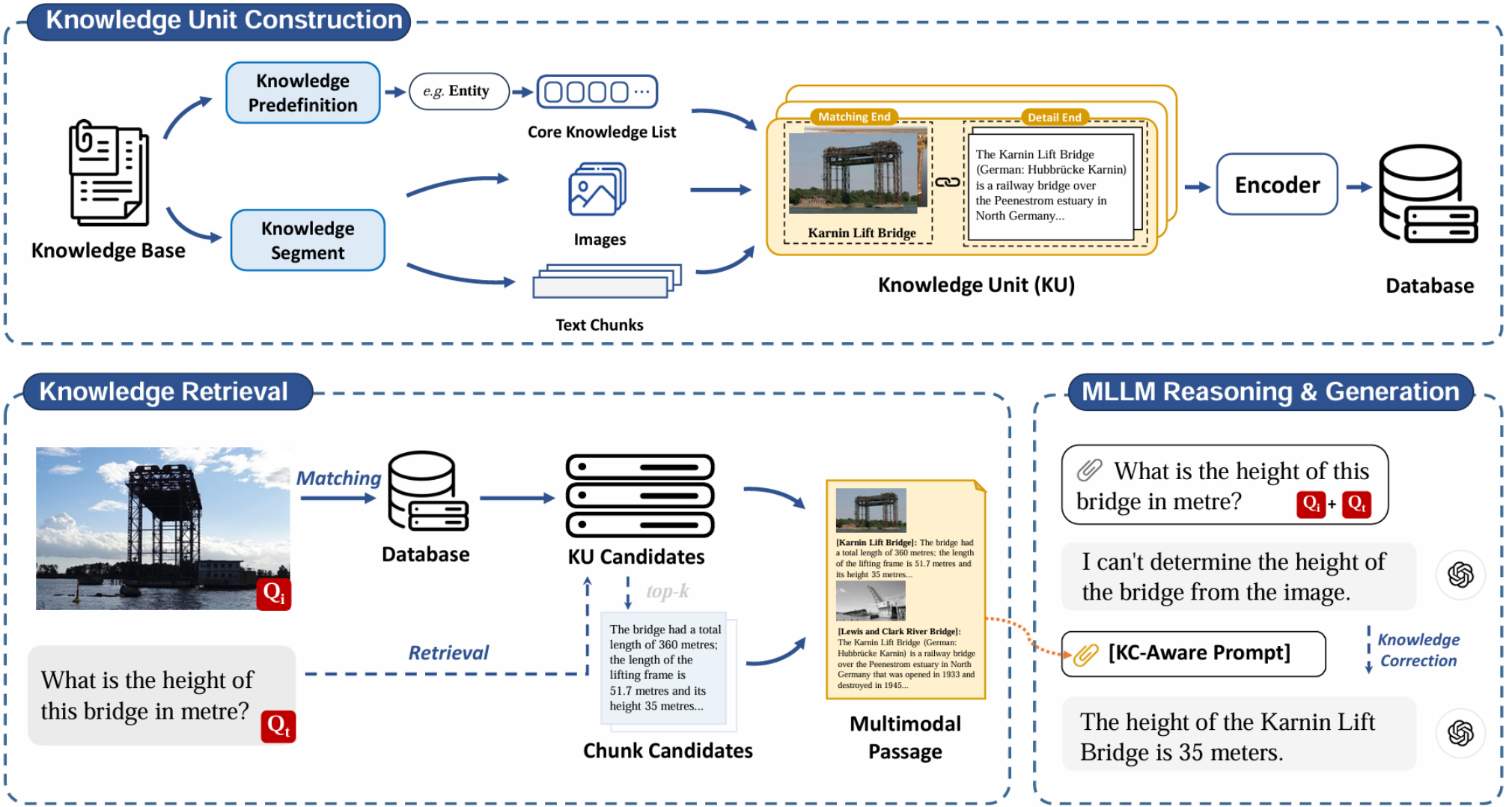
MRAG 3.0 架构

RAG在模式识别的应用： RA-VQA-v2



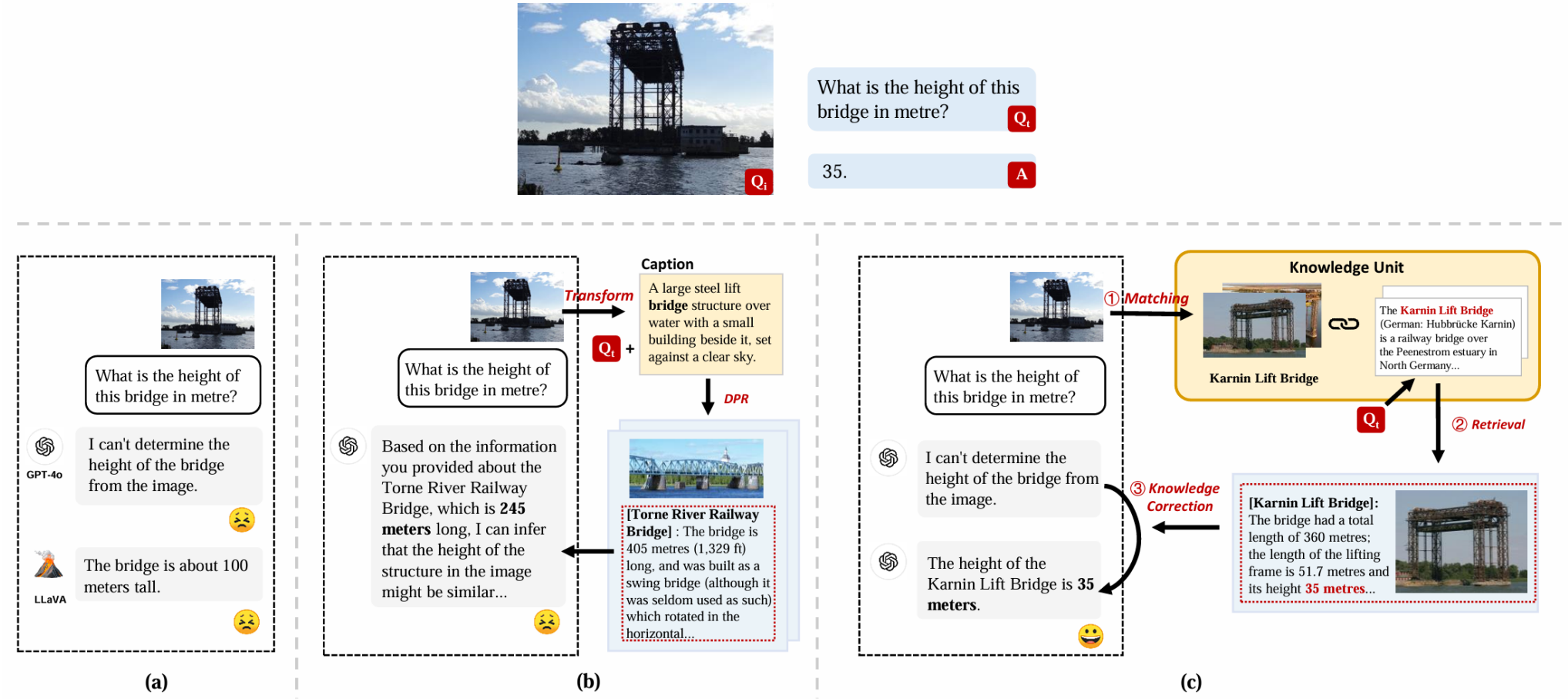
RA-VQA-v2 的框架

RAG在模式识别的应用：KU-RAG



KU-RAG 的框架

RAG在模式识别的应用： KU-RAG



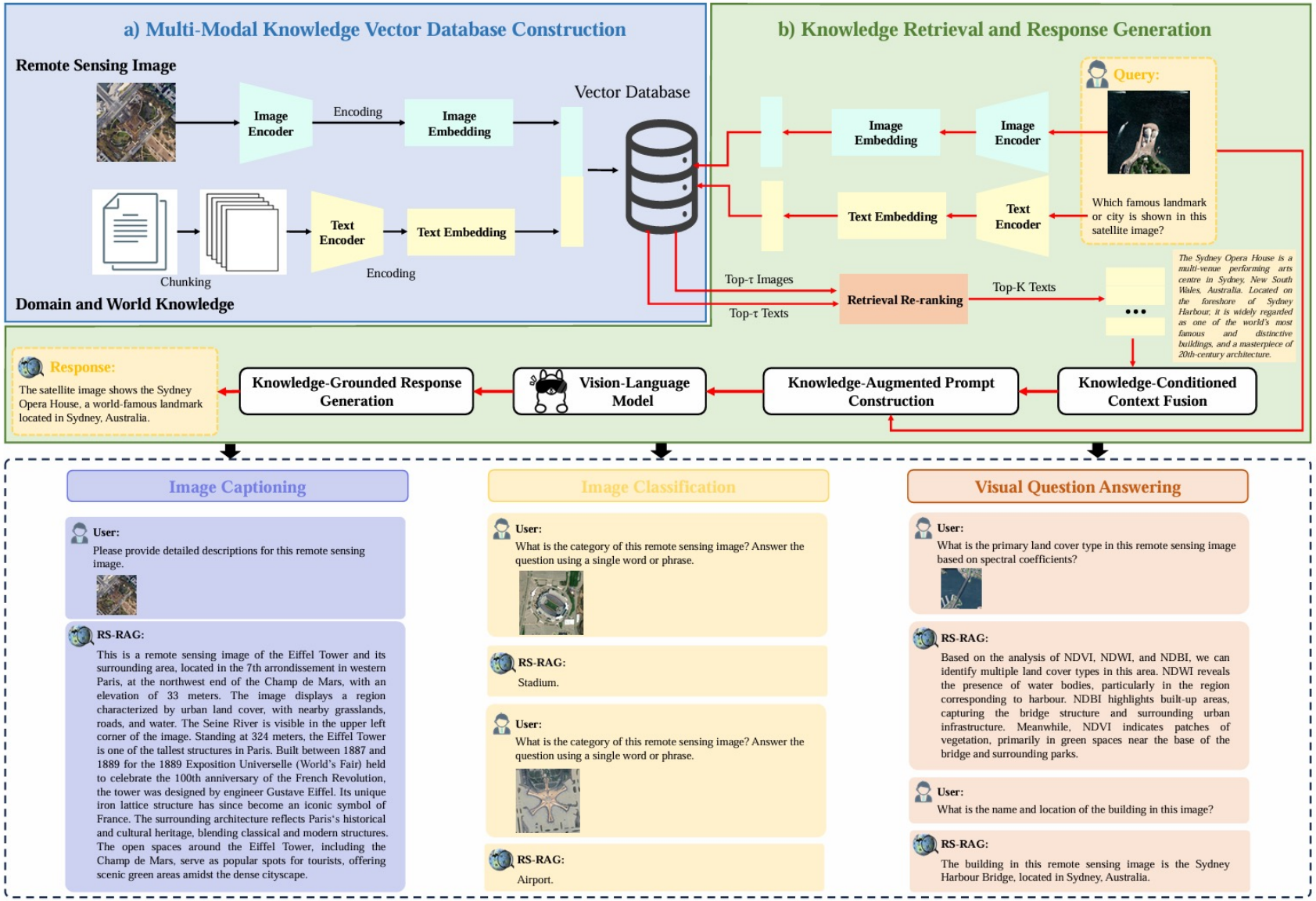
MLLMs 在视觉问答（VQA）任务中的不同解决方案对比

(a) 无外部知识的直接回答（易产生幻觉）

(b) 单模态粗粒度检索增强生成（图像转文本易丢失视觉细节）

(c) 提出的 KU-RAG 方法（基于细粒度知识单元的精准多模态检索与生成）

RAG在模式识别的应用： RS-RAG



RS-RAG 的框架

RAG在模式识别的应用： RS-RAG

Question:
Please provide detailed descriptions for this remote sensing image.



Ground truth:
Massachusetts Institute of Technology.

InternVL2.5: **This is a remote sensing image of Boston University.** The image shows a dense urban area with numerous buildings of varying sizes, shapes, and roof colors, interspersed with roads and parking lots...University located in Boston, Massachusetts, United States, at Location: 42° 21′52″N 71° 4′33″W with the Address: 404 Bay State Road, Boston, Massachusetts 02215, U.S., having a Physical Area of 107 acres (0.43 km²) and a Construction Period from 1867... Remote sensing expert knowledge includes an ALBEDO of 0.1, ALBNIRDF of 0.11, ALBNIRDR of 0.15, ALBVISDF of 0.05, ALBVISDR of 0.08, EMIS of 0.99, EVLAND of 845.53, GWETPROF of 0.6, LC_Type1 of 13, which represents Urban and Built-up Lands...

Janus-Pro: **This is a remote sensing image of University of Bucharest.** Numerous buildings with gray, brown, and white rooftops are arranged around a central plaza, with roads and green spaces interspersed...categorized as a University, is located in Bucharest, Romania, at coordinates 44° 25′41″N 26° 09′20″E, with the specific address being Str. Spiru Haret no. 6, Bucharest, Romania....Remote sensing expert knowledge for the university includes: ALBEDO (0.17), ALBNIRDF (0.23), ALBNIRDR (0.27), ALBVISDF (0.07), ALBVISDR (0.08), EMIS (0.98), EVLAND (512.47), GWETPROF (0.61), LC_Type1 (13, representing barren or sparsely vegetated areas, indicating minimal vegetation cover...

Qwen2.5-VL: **This is a remote sensing image of University of Illinois at Urbana–Champaign.** The image shows a dense collection of buildings with varying roof colors and shapes, interspersed with roads and patches of green vegetation...categorized as a University, is located in Champaign-Urbana, Illinois, United States at 40.28306° N 87.72472. Mathews Ave. Champaign, IL 61820. The campus spans 1,900 acres (770 ha) and was constructed between 1867 and 1907... Remote sensing expert knowledge reveals an albedo of 0.19, near-infrared albedo of diffuse radiation at 0.26, near-infrared albedo of direct radiation at 0.31, visible albedo of diffuse radiation at 0.09...

Llama-3.2: **This is a remote sensing image of University of Maryland, College Park.** A large, rectangular, dark-roofed building with a central courtyard is prominent in the center, surrounded by smaller, light-roofed structures, roads, and green spaces... located in Prince George's County, Maryland, at coordinates 39° 01′18″N 76° 56′16″W, is a prominent university with a physical area of 1,382 acres (5.59 km²)... Remote sensing expert knowledge for the site includes an albedo of 0.14, NIR diffuse albedo (ALBNIRDF) of 0.2, NIR direct albedo (ALBNIRDR) of 0.24, visible diffuse albedo (ALBVISDF) of 0.06, visible direct albedo (ALBVISDR) of 0.07, emissivity (EMIS) of 0.98, evapotranspiration (EVLAND) of 1009.56 mm, and groundwater profile (GWETPROF) of 0.62...

Geochat: **This is a remote sensing image of Cathedral Church of the Holy Trinity, Guildford.** The cathedral, with its distinctive cruciform shape and grey roof, is situated amidst a dense urban area of predominantly red-roofed buildings...categorized as a Church, is located in Guildford, Surrey, England at 51.24794° N, 0.59944° W (51.2499984741, -0.600000012), with the address High Street, Guildford, Surrey, England. The cathedral, situated at the junction of High Street and Stag Lane, occupies a physical area of 2 acres (0.81 ha)... Remote sensing expert knowledge collected shows an albedo of 0.18, ALBNIRDF of 0.25, ALBNIRDR of 0.32, ALBVISDF of 0.06, and ALBVISDR of 0.07. The land cover classification reveals LC_Type1 as 13, indicating barren or sparsely vegetated land, typical of deserts or rocky areas, suggesting limited vegetation cover in the immediate vicinity...

RSGPT: **This is a remote sensing image of University of Caldas.** The image shows a dense collection of buildings with varying roof colors, interspersed with green spaces, roads, and parking areas...located in Manizales, Colombia, at coordinates 4° 37′10″N 75° 26′48″W, is a public university established in 1954, originally founded as the Escuela Normal Superior de Manizales in 1930. It was renamed Universidad de Caldas in 1954 and has since grown into a significant educational institution, playing a pivotal role in the region's cultural and academic development...Remote sensing expert knowledge for the site reveals an albedo of 0.14, with diffuse and direct near-infrared albedo (ALBNIRDF and ALBNIRDR) values of 0.22 and 0.26, respectively, and visible albedo (ALBVISDF and ALBVISDR) values of 0.06 and 0.07. The visible diffuse albedo (ALBVISDR) is 0.07, and the near-infrared diffuse albedo (ALBNIRDR) is 0.26...

RS-RAG: **This is a remote sensing image of Massachusetts Institute of Technology.** The image displays a dense urban environment characterized by numerous buildings with varied roof shapes and colors, interspersed with roads, parking areas, and patches of green vegetation... categorized as a University located in Cambridge, Massachusetts, United States, specifically at coordinates 42° 21′35″N 71° 5′31″W or 42.3600006104 latitude and -71.09000092 longitude, with an address of 77 Massachusetts Avenue, Cambridge, MA 02139, USA, covering a physical area of 166 acres (67.2 ha); its construction period includes being established in 1861 and relocating to its current campus in 1916... ALBEDO: 0.1, ALBNIRDF: 0.11, ALBNIRDR: 0.15, ALBVISDF: 0.06, ALBVISDR: 0.1, EMIS: 0.99, EVLAND: 845.53mm, GWETPROF: 0.6, LC_Type1: 13, LC_Type2: 13.0, LC_Type3: 10.0, LC_Type4: 8.0, LC_Type5: 9.0, LST_Day_1km: 20.51° C, LST_Night_1km: 7.24° C, PRECTOTLAND: 1025.91mm, SAA: 158.14° , SLP: 101616.54Pa, SMLAND: 131.75mm, SPEED: 5.15m/s...

RS-RAG 和其他方法对比

麻省理工学院遥感影像的图像描述定性结果

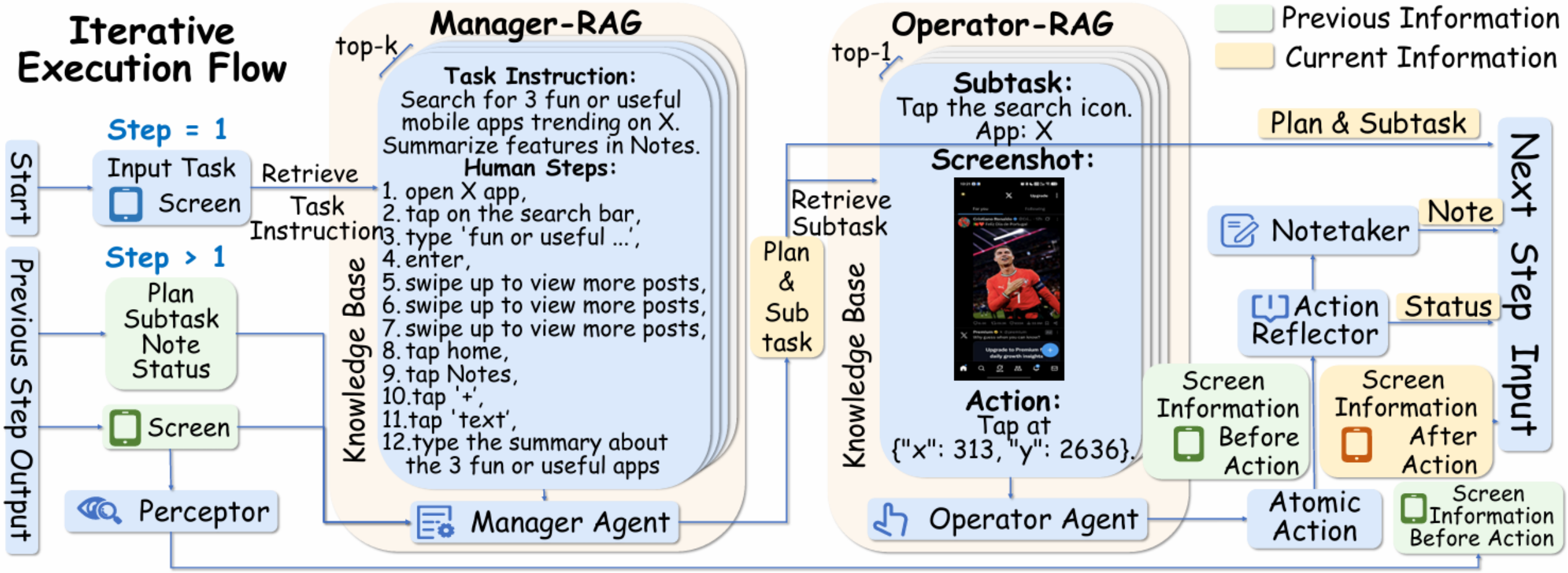
红色文本表示识别出的地标名称

蓝色突出显示从图像中可直接观察到的视觉内容

紫色表示检索到的世界知识（如历史、地理或文化事实）

绿色表示遥感知识（包括光谱指数、土地覆盖和反照率）

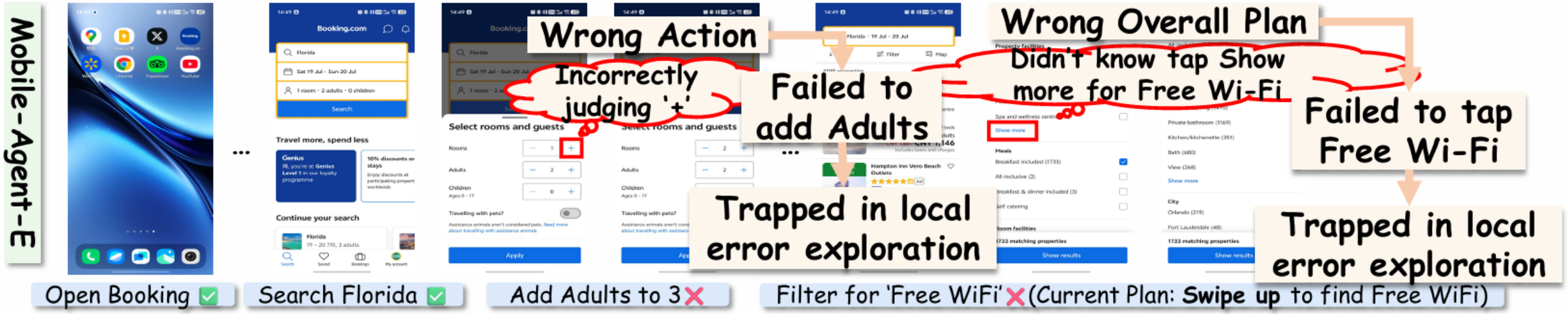
RAG在模式识别的应用：Mobile-Agent-RAG



Mobile-Agent-RAG 架构

RAG在模式识别的应用： Mobile-Agent-RAG

Task Instruction Find a breakfast and Free-WiFi including hotel in Florida under ¥1750/night for 3 people. Summarize final choice in Notes.



Mobile-Agent-RAG 与 w/o RAG 的对比

课程提纲

CONTENTS

一、 LLM/VLM/MLLM
二、多模态思维链推理
三、多模态视频理解
四、检索增强生成
五、总结

总结与回顾

这一章我们讲解了：

- **模态壁垒的打破：** 模式识别从传统的标签映射与单一文本处理，全面演进为整合文本、图像、音频、视频的 MLLM **全模态语义理解**
- **推理机制的进化：** 多模态思维链结合视觉锚定与动态自我纠错（如 T-GRPO 等算法），有效缓解了视觉幻觉，大幅提升了模型处理**复杂几何与逻辑任务**的上限
- **时空感知的跃迁：** 多模态视频理解使得模型从识别“静态快照”跨越到理解“动态因果”，实现了**原生高分辨率**与**超长序列的时空推理**
- **外部知识的赋能：** 从纯文本走向端到端多模态的 RAG 架构，为遥感分析、视觉问答及自动化智能体（如 Mobile-Agent-RAG）等模式识别下游任务提供了**精准的外部知识支撑**

核心启发与未来展望

- **“感知”与“推理”的深度闭环，从视觉映射走向世界模型**

模式识别已超越单纯的特征提取。通过多模态思维链（MCoT）的视觉锚定与自我纠正，结合 GRPO 等强化学习算法驱动的**自我进化**，模型正逐步建立物理世界的因果法则与**前瞻性推理**能力

- **“内生能力”与“外脑增强”的融合，跨越长周期任务的落地鸿沟**

强大的基础模型叠加多模态检索增强生成（MRAG），打破了参数记忆的物理瓶颈，通过动态上下文知识赋能，为**复杂场景下的长序列推理**提供精准支撑，是构建可信赖 AI 的必然路径

- **从“静态旁观”到“动态干预”，重塑多智能体协同的人机边界**

随着多模态大模型在流式感知与时空时序理解上的突破，AI 正加速向具身智能与 GUI Agent 等交互式应用演进，通过**多智能体协同**彻底改变**终端设备的自动化**形态

感谢倾听！



李冠彬 http://guanbinli.com